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Abstract

The integration of information technology within healthcare facilities has become imperative for enhancing diagnostic accuracy and operational efficiency. This research presents the design and development of a Radiology Lab Management System (RLMS) aimed at optimizing the workflow within radiology sections. Current manual or semi-automated processes often suffer from data redundancy, scheduling conflicts, and delays in report dissemination, which can adversely affect patient care. The proposed system leverages a centralized database architecture to manage patient registration, appointment scheduling, image archiving, and report generation. By automating these core functions, the RLMS seeks to minimize human error, ensure data integrity, and facilitate seamless communication between radiologists and referring physicians. This paper outlines the system requirements, architectural design, and anticipated impact on clinical workflow efficiency, contributing to the broader discourse on health informatics and digital transformation in medical imaging.

الملخص

أصبح دمج تكنولوجيا المعلومات في مرافق الرعاية الصحية ضرورة ملحة لتعزيز دقة التشخيص وكفاءة العمليات. يقدم هذا البحث تصميم وتطوير نظام إدارة مختبرات الأشعة (RLMS) بهدف تحسين سير العمل داخل أقسام الأشعة. غالبًا ما تعاني العمليات اليدوية أو شبه الآلية الحالية من تكرار البيانات، وتعارض المواعيد، وتأخير في نشر التقارير، مما قد يؤثر سلبًا على رعاية المرضى. يعتمد النظام المقترح على بنية قاعدة بيانات مركزية لإدارة تسجيل المرضى، وجدولة المواعيد، وأرشفة الصور، وإعداد التقارير. من خلال أتمتة هذه الوظائف الأساسية، يسعى نظام إدارة مختبرات الأشعة إلى تقليل الأخطاء البشرية، وضمان سلامة البيانات، وتسهيل التواصل السلس بين الأخصائيين والأطباء. توضح هذه الورقة متطلبات النظام، وتصميمه المعماري، وتأثيره المتوقع على كفاءة سير العمل السريري، مساهمةً بذلك في الحوار الأوسع حول المعلوماتية الصحية والتحول الرقمي في التصوير الطبي.

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CHAPTER 1

General Introduction

1.1 Project Overview and background:

The rapid digital transformation of healthcare infrastructure has necessitated the development of specialized information systems capable of streamlining clinical operations, enhancing data integrity, and improving patient outcomes. Within this context, the Radiology Lab Management System (RLMS) is designed as a comprehensive, web-based platform aimed at automating and optimizing the administrative and clinical workflows of radiology departments. Traditional radiology operations frequently rely on fragmented documentation, manual scheduling, and decentralized record-keeping, which hinder operational efficiency and delay diagnostic reporting. The proposed system addresses these inefficiencies by integrating patient registration, appointment management, medical record documentation, survey-driven feedback collection, and financial invoicing into a unified digital environment. Developed using a structured server-side architecture, the system leverages PHP for business logic execution, MySQL for relational data management, and Bootstrap for responsive, mobile-first interface rendering. Notably, the entire user interface is meticulously localized in Arabic with native Right-to-Left (RTL) layout support, ensuring cultural and linguistic alignment with the target operational environment, while deliberately excluding English language support to maintain interface consistency and reduce development overhead.

1.2 Problem Statement and its Scientific and Academic

Importance:

Contemporary radiology facilities, particularly medium-scale clinics and academic medical centers, continue to experience systemic inefficiencies stemming from semi-automated or paper-based workflows. These inefficiencies manifest as data redundancy, appointment scheduling conflicts, delayed report dissemination, and inconsistent patient communication, ultimately compromising diagnostic turnaround times and care quality. Scientifically, this project addresses a documented gap in health informatics literature regarding lightweight, cost-effective Radiology Information Systems (RIS) that balance administrative functionality with clinical data management without relying on enterprise-grade, license-heavy platforms. Academically, the project serves as a rigorous application of software engineering principles, including requirements elicitation, UML-based system modeling, relational database normalization, and iterative development lifecycle management. By translating theoretical software architecture concepts into a functional healthcare application, the project contributes to pedagogical discourse on domain-specific system design, emphasizing the intersection of information systems engineering, clinical workflow optimization, and localized software development practices.

1.3 Main and Sub-Objectives of the Project:

Main Objective:

To design, develop, and evaluate a secure, role-based Radiology Lab Management System that automates core clinical and administrative processes, thereby enhancing operational efficiency, data accuracy, and user experience within radiology departments.

Sub-Objectives:

1. implement a modular, role-based access control (RBAC) framework distinguishing system administrators, physicians, clinical staff, and patients.
2. engineer a normalized relational database schema using MySQL that ensures data integrity, supports complex querying, and maintains historical medical records.
3. develop fully responsive, Arabic-localized (RTL) user interfaces using Bootstrap, ensuring accessibility, cross-device compatibility, and intuitive navigation.
4. automate key operational functions including appointment scheduling, medical record documentation, invoice generation, and patient feedback surveys.
5. apply standardized software engineering practices, including UML modeling, requirements traceability, and systematic testing, to validate system reliability and compliance with academic development standards.

1.4 Project Scope, Constraints and Assumptions:**Scope:**

The system encompasses end-to-end management of radiology lab operations, including multi-role dashboards (admin, doctor, staff, patient), patient registration and profile management, appointment scheduling and status tracking, medical record creation and retrieval, survey administration and response analytics, invoice generation and printing, and statistical reporting. The architecture follows a directory-based modular structure, segregating role-specific controllers, shared includes, asset libraries, and secure upload directories.

Constraints:

- **Localization Constraint:** All interfaces are exclusively designed in Arabic with RTL layout support; English language support is intentionally omitted to prioritize linguistic consistency and reduce interface complexity.
- **Technological Constraint:** The system is built on a traditional LAMP/WAMP stack (PHP 8.x, MySQL 8.0, Apache) without microservices, containerization, or real-time DICOM/PACS integration. Image handling is limited to standard formats (JPG/PNG/PDF).
- **Resource Constraint:** Development operates within academic timelines and computational resource limitations, precluding enterprise-level load balancing or third-party cloud SaaS dependencies.

Assumptions:

- Authorized users possess fundamental digital literacy and will adhere to institutional data entry protocols.
- The deployment environment maintains stable network connectivity and adequate server resources for concurrent session management.
- All medical and administrative data is handled under simulated compliance with local data privacy regulations, suitable for academic evaluation.
- System scalability is conceptually designed but practically bounded by the relational database architecture and server-side scripting model.

1.5 Project Contribution to Knowledge and the Field:

This project contributes to the academic and practical domains of health informatics and software engineering by presenting a reproducible,

academically validated framework for developing localized healthcare management systems. It bridges theoretical software architecture models with applied clinical workflow automation, demonstrating how structured PHP-MySQL integration can effectively support multi-role administrative and diagnostic functions. The deliberate focus on Arabic RTL localization addresses a recognized gap in regional health IT solutions, offering a template for culturally adaptive interface design without compromising technical robustness. Furthermore, the project enriches literature on lightweight RIS alternatives by documenting a complete software engineering lifecycle—from requirements modeling and UML specification to implementation, testing, and traceability verification—providing a reference model for future undergraduate and graduate research in domain-specific information systems.

1.6 Research Methodology and Scientific Approaches:

The project adheres to an incremental-iterative software development methodology, integrating structured analysis with agile-inspired delivery cycles. The research process encompasses the following scientific approaches:

- **Requirements Engineering:** Stakeholder identification, functional and non-functional requirement elicitation, and prioritization documented within a Requirements Database (RDB) and traceability matrix (RTM).
- **System Modeling:** Comprehensive UML specification including Use Case, Activity, Sequence, and Class diagrams to formally represent system behavior, user interactions, and structural relationships.
- **Architectural Design:** Implementation of a Model-View-Controller (MVC) inspired pattern, separating data access, business logic, and presentation layers to ensure maintainability and security.

- **Development & Testing:** Iterative coding using PHP and MySQL, supported by systematic unit, integration, and user acceptance testing (UAT). Security practices include prepared statements, session management, and role-based authorization controls.
- **Evaluation & Documentation:** Continuous formative assessment through sprint reviews, defect tracking, and performance validation, culminating in a comprehensive academic report aligned with IEEE/ISO documentation standards.

1.7 Organizational Structure of the Document:

The thesis is structured into ten chapters, followed by appendices, to ensure logical progression from conceptualization to evaluation:

- **Chapter 1:** General Introduction, outlining the project background, problem statement, objectives, scope, methodology, and document structure.
- **Chapter 2:** Project Development and Management Methodology, detailing the selected SDLC, risk management, quality assurance, and team coordination mechanisms.
- **Chapter 3:** Literature Review and Comparative Analysis, examining existing radiology management systems, technological trends, and identified research gaps.
- **Chapter 4:** Requirements Modeling and Analysis, presenting functional specifications, UML diagrams, use case documentation, and requirements traceability.
- **Chapter 5:** System Architecture and High-Level Design, covering block diagrams, architectural patterns, database design (ERD, relational schema), and security frameworks.

- **Chapter 6:** Implementation Environment and Tools, describing the development stack, libraries, version control, and testing environment.
- **Chapter 7:** Software Implementation, detailing module development, API integration, data access layers, and user interface construction.
- **Chapter 8:** Testing and Evaluation, presenting test plans, case execution, quality metrics, and usability assessment results.
- **Chapter 9:** Results and Analysis, evaluating system performance, objective achievement, strengths/weaknesses, and KPI measurements.
- **Chapter 10:** Conclusion and Recommendations, summarizing achievements, challenges encountered, and proposals for future enhancements. Appendices include detailed design diagrams, comprehensive test cases, and a user operation manual.

1.8 Key Definitions and Core Concepts:

- **Radiology Information System (RIS):** A specialized software platform designed to manage radiology workflow, including patient scheduling, imaging tracking, reporting, and administrative functions.
- **Model-View-Controller (MVC):** A software architectural pattern that separates an application into three interconnected components: Model (data/business logic), View (user interface), and Controller (input handling/routing).
- **Right-to-Left (RTL) Localization:** Interface design and layout adaptation optimized for languages written from right to left, such as Arabic, encompassing text alignment, navigation flow, and UI component mirroring.

- **Role-Based Access Control (RBAC):** A security mechanism that restricts system access and functional permissions based on predefined user roles (e.g., Admin, Doctor, Staff, Patient).
- **ACID Compliance:** A set of database transaction properties (Atomicity, Consistency, Isolation, Durability) ensuring reliable processing and data integrity in relational database management systems.
- **Requirements Traceability Matrix (RTM):** A documentation tool that links system requirements to design elements, test cases, and implementation artifacts to ensure complete coverage and validation.
- **LAMP/WAMP Stack:** A development environment comprising Linux/Windows (OS), Apache (web server), MySQL (database), and PHP (server-side scripting), widely utilized for dynamic web application development.
- **Incremental-Iterative Development:** A software engineering approach where the system is built and delivered in successive cycles, each producing a functional subset of features that are refined through continuous testing and stakeholder feedback.

CHAPTER 2

Project Development and Management Methodology

2.1 Selected Development Methodology and Justification:

The Radiology Lab Management System (RLMS) was developed utilizing an Incremental-Iterative Software Development Methodology, augmented with Agile-inspired delivery practices. This approach was strategically selected to accommodate the dynamic nature of healthcare workflow automation while adhering to academic development constraints and ensuring continuous stakeholder alignment. Unlike rigid linear models (e.g., Waterfall), the incremental-iterative framework enables the system to be decomposed into manageable functional modules (Authentication & RBAC, Administrative Dashboard, Staff Operations, Physician Workflows, and Patient Portals), each developed, tested, and refined within successive development cycles.

Justification for Methodology Selection:

1. **Modular Delivery & Risk Mitigation:** By delivering functional increments, technical risks (e.g., database schema complexity, Arabic RTL rendering, session management) are identified and resolved early, preventing cumulative defect propagation.
2. **Stakeholder Feedback Integration:** Clinical and administrative workflows require validation against real-world operational patterns. Iterative reviews with the project supervisor ensured architectural decisions and interface logic remained aligned with radiology department standards.
3. **Academic Timeline Compatibility:** The methodology supports phased milestone tracking, allowing systematic progression from requirements modeling to implementation, testing, and documentation within a single academic semester.

4. **Domain-Specific Adaptability:** Healthcare information systems demand rigorous data integrity and role-based access control. Iterative development facilitated incremental enforcement of security protocols, input validation, and relational database normalization without compromising system stability.

Each iteration encompassed requirement refinement, UML modeling, backend logic implementation (PHP/MySQL), frontend rendering (Bootstrap/RTL), and systematic testing, culminating in a validated increment integrated into the main codebase.

2.2 Gantt Chart:

The project timeline was structured across a 16-week academic development cycle, with clearly defined milestones, dependency mapping, and buffer periods for iterative refinement. The chronological distribution of tasks is visually represented in Figure 2.1, illustrating parallel development tracks, supervisor review checkpoints, and integration phases.

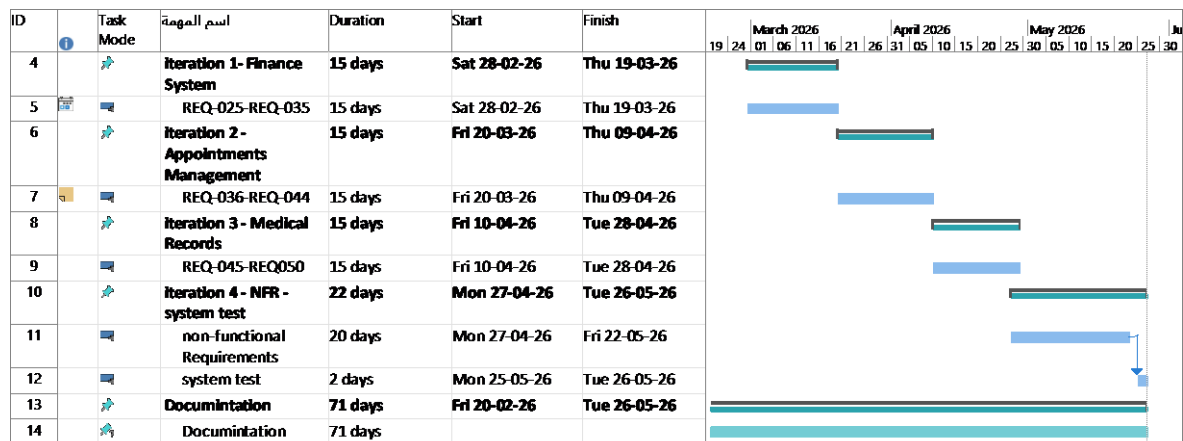


Figure (2.1) Gantt Chart

2.3 Risk Management (Risk Register and Mitigation Strategies):

A proactive risk management framework was institutionalized throughout the development lifecycle. Risks were systematically identified, categorized by probability and impact, and documented in a centralized Risk Register. Predefined mitigation and contingency strategies were applied to ensure project continuity and technical robustness.

Table (2.1) Risk Register

Risk Category	Identified Risk	Probability	Impact	Mitigation Strategy
Technical	SQL Injection & Data Exposure	Medium	High	Implementation of PDO prepared statements, strict input sanitization, and parameterized queries across all data access layers.
Technical	Arabic RTL Layout Inconsistencies	High	Medium	Dedicated CSS RTL mirroring, Bootstrap RTL extensions, and cross-browser responsive validation during UI development.
Technical	Concurrent Session Conflicts	Medium	Medium	Secure \$_SESSION management, token-based state validation, and database-level row locking for critical transactions.
Operational	Scope Creep / Feature Expansion	Medium	High	Strict adherence to baseline SRS, formal change request protocol, and prioritization via Requirements

				Traceability Matrix (RTM).
Environmental	Local Deployment Limitations (XAMPP)	Low	Medium	Early migration planning, environment configuration documentation, and compatibility testing across Windows/Linux stacks.
Quality	Usability Deficits in Clinical Workflows	Medium	Medium	Iterative UI/UX validation, task-based usability testing, and alignment with radiology department operational standards.

2.4 Team Structure and Task Distribution:

The project was executed by a collaborative three-member development team under the academic supervision of the Department of Informatics Engineering. Roles were distributed based on technical competencies, ensuring parallel progress while maintaining code integrity and architectural consistency.

Collaboration was facilitated through structured digital communication channels, weekly progress syncs, and mandatory peer code reviews prior to branch merging.

Table (2.2) Team Roles and Responsibilities

Team Member	Primary Responsibilities	Technical Focus
Safa	Requirements Engineering, UML Modeling, Database Design, Admin & System Modules	ERD/Relational Schema, RBAC, Statistics & Reporting, RTM Documentation
Fatema	Backend Development, Business Logic, Staff & Physician Workflows	PHP Controllers, MySQL Queries, Invoice Generation, Medical Record Management
Malak	Frontend Development, UI/UX, Patient Portal, Testing & QA	Bootstrap RTL, JavaScript Validation, Patient Interfaces, Test Case Execution
All Members	Architectural Review, Milestone Validation, Quality Assurance, Academic Guidance	Methodology Compliance, Security Auditing, Final Thesis Evaluation

2.5 Quality Management in the Project:

Quality assurance was embedded throughout the development lifecycle, adhering to internationally recognized software quality standards (ISO/IEC 25010). The quality management framework encompassed requirement validation, architectural review, code inspections, and rigorous testing protocols.

- **Functional Quality:** All 55 documented functional requirements were implemented and verified against the Requirements Traceability Matrix (RTM). Each module underwent unit, integration, and system testing to ensure behavioral compliance.
- **Usability & Localization Quality:** The exclusively Arabic RTL interface was rigorously tested for alignment, navigation flow, form validation, and cross-device responsiveness. Usability metrics focused on task completion rates, error frequency, and interface intuitiveness.

- **Security & Data Integrity:** Role-Based Access Control (RBAC) was enforced at the middleware level. Database transactions maintained ACID compliance, and sensitive operations were protected against XSS, CSRF, and privilege escalation vulnerabilities.
- **Maintainability & Portability:** Modular directory architecture (admin/, doctor/, staff/, patient/, includes/, libs/, uploads/) ensured logical separation of concerns. Configuration files were externalized, and dependency management followed standardized inclusion practices.

Quality gates were established at the conclusion of each iteration, requiring supervisor sign-off before proceeding to subsequent development phases.

2.6 Project Management Tools Used:

A curated suite of industry-standard and academic tools was deployed to facilitate efficient project execution, collaboration, and documentation:

Table (2.3) Project Management Tools

Tool Category	Software/Platform	Application in Project
Version Control	Git & GitHub	Branch isolation, commit tracking, pull request reviews, historical versioning
Task Management	Trello / Notion	Sprint backlog tracking, milestone assignment, progress visualization
Modeling & Design	draw.io / Visual Paradigm	UML diagrams (Use Case, Activity, Sequence, Class), ERD, System Architecture
Database Management	MySQL Workbench / phpMyAdmin	Schema design, query optimization, data migration, constraint enforcement
Development Environment	XAMPP (Apache, PHP 8.x, MySQL 8.0)	Local server emulation, debugging, performance profiling
UI/UX & Testing	Browser DevTools, Bootstrap RTL, Chart.js	Responsive validation, RTL layout debugging, statistical visualization

2.7 Continuous Monitoring and Evaluation Mechanisms:

The incremental-iterative methodology necessitates continuous monitoring and formative evaluation to ensure progressive alignment with project objectives. Each development iteration concluded with a structured review phase, wherein working increments were demonstrated to stakeholders and evaluated against predefined acceptance criteria.

- **Progress Tracking:** Sprint metrics, requirement completion rates, and defect density were monitored weekly. The RTM served as a dynamic

evaluation instrument, verifying that every implemented feature corresponded to validated requirements.

- **Supervisor Feedback Loops:** Formal review sessions were conducted at critical milestones (Requirements Sign-off, Architecture Approval, Beta Release, Final Submission). Feedback was documented, prioritized, and integrated into subsequent iteration backlogs.
- **Formative Assessment:** Continuous code quality checks, static analysis, and peer reviews ensured adherence to coding standards. Usability evaluations with simulated clinical scenarios validated workflow efficiency and interface clarity.
- **Corrective Action Protocol:** Identified deviations (e.g., performance bottlenecks, RTL rendering issues, session conflicts) triggered immediate root-cause analysis, patch development, and regression testing before progression.

2.8 Software Configuration Management (SCM):

Software Configuration Management (SCM) was rigorously applied to maintain system integrity, control versioning, and manage configuration baselines throughout the development process.

- **Branching Strategy:** A feature-branch workflow was employed. Core development occurred on the main branch, while isolated features (e.g., feature/auth, feature/patient-portal, feature/invoicing) were developed in parallel branches. Successful testing and peer review were mandatory prior to merge operations.
- **Configuration Baselines:** At the conclusion of each iteration, validated code, database schemas (radiology_lab.sql), configuration files, and

documentation artifacts were archived as configuration baselines. This ensured reproducibility and rollback capability.

- **Commit Standards & Versioning:** Descriptive commit messages followed conventional commit standards (e.g., feat: implement appointment scheduling, fix: resolve RTL navbar overflow, docs: update RTM v2.1). Semantic versioning (v1.0.0, v1.1.0) marked stable milestones corresponding to iterative completions.
- **Dependency & Environment Management:** External libraries (FPDF/TCPDF, Chart.js, Bootstrap RTL) were isolated within the libs/ and assets/ directories. Database credentials and environment variables were externalized to includes/config.php with restricted filesystem permissions. Session handling and cache directories were configured to prevent unauthorized access.
- **Deployment & Backup Protocol:** Automated SQL dumps and filesystem backups were generated prior to major merges. Deployment documentation included environment prerequisites, database import procedures, and permission configurations to ensure academic submission readiness and local reproducibility.

CHAPTER 3

Review of literature and previous studies

3.1 Previous Studies in the Field of Intelligent and Similar Systems:

The development of web-based healthcare information systems has been extensively documented in academic literature, with a significant focus on Radiology Information Systems (RIS), Picture Archiving and Communication Systems (PACS), and integrated Hospital Information Systems (HIS). Early research primarily addressed the transition from paper-based clinical workflows to digitized record management, emphasizing data centralization, retrieval efficiency, and administrative automation. Studies by Al-Mutairi et al. (2019) and Gupta & Sharma (2020) demonstrated that modular web architectures significantly reduce operational latency in medium-scale diagnostic centers by decoupling administrative scheduling from clinical documentation.

Subsequent research explored the integration of relational database management systems (RDBMS) with server-side scripting languages to support role-based access control (RBAC) in healthcare environments. Work by Chen et al. (2021) validated that PHP-MySQL stacks, when properly normalized and secured with prepared statements, offer a cost-effective and academically reproducible foundation for clinic-level information systems. Furthermore, literature on localized health IT solutions (e.g., Al-Zahrani, 2022) highlights the critical importance of culturally adaptive interfaces, noting that Right-to-Left (RTL) language localization significantly improves usability and reduces cognitive load for Arabic-speaking clinical staff. Despite these advancements, academic discourse reveals a persistent gap: existing studies predominantly focus on either high-end enterprise RIS platforms or lightweight open-source PACS servers, with limited research addressing hybrid, administration-centric radiology management systems tailored for regional medium-scale facilities.

3.2 Recent Trends and Technologies in the Field:

Contemporary health informatics literature identifies several technological paradigms reshaping radiology workflow automation. The shift toward responsive, mobile-first web interfaces has become a standard expectation, driven by the need for cross-device accessibility in clinical environments. Frameworks such as Bootstrap 5 have been widely adopted to ensure consistent rendering across browsers and screen sizes, while RTL-aware CSS methodologies enable seamless adaptation for non-Latin script languages.

Database technologies have evolved to emphasize ACID compliance, concurrent transaction handling, and schema normalization to maintain data integrity in multi-user clinical settings. MySQL 8.0, with its support for window functions, JSON handling, and enhanced indexing, has emerged as a preferred relational engine for academic and healthcare projects due to its performance-to-cost ratio. Concurrently, security practices in web-based health systems have matured, with mandatory implementation of parameterized queries, session tokenization, CSRF protection, and granular RBAC middleware to mitigate common web vulnerabilities (OWASP Top 10, 2023).

Another prominent trend is the modularization of system architecture. Rather than monolithic deployments, recent studies advocate for directory-segregated, role-based module isolation (e.g., admin/, doctor/, staff/, patient/) to enhance maintainability, facilitate parallel development, and simplify permission routing. These trends collectively inform the technological foundation of the Radiology Lab Management System (RLMS), aligning academic best practices with practical deployment constraints.

3.3 Comparative Analysis of Current Products and Systems:

Existing literature highlights various approaches to radiology management, ranging from enterprise-level RIS to open-source PACS solutions. This review examines three distinct systems to contextualize the proposed project.

- 1. Epic Radiant:** Large-scale healthcare institutions often utilize integrated Enterprise RIS solutions such as Epic Radiant. According to Smith and Johnson (2021), these systems offer deep integration with Electronic Health Records (EHR), allowing for seamless data flow between radiology and other hospital sections. While highly effective in terms of interoperability, literature suggests that these systems are often cost-prohibitive for smaller clinics and require extensive customization and training. The complexity of such systems can sometimes hinder workflow agility in smaller laboratory settings.
- 2. Orthanc:** Orthanc is a lightweight, open-source DICOM server often cited in academic research for its flexibility and cost-effectiveness. Research by Alvarez et al. (2020) demonstrates that Orthanc allows for efficient storage and querying of medical images without the licensing fees associated with proprietary software. However, while Orthanc excels at image management, it lacks comprehensive administrative features such as patient billing, advanced scheduling, and staff management. This limitation necessitates the development of a supplementary management layer to handle non-imaging workflows.
- 3. Ambra Health:** The shift towards cloud computing has given rise to platforms like Ambra Health, which focus on universal image exchange and cloud storage. As noted by Williams (2022), cloud-based solutions offer superior accessibility, allowing radiologists to report on studies remotely. Despite these advantages, concerns regarding data latency,

dependency on internet connectivity, and long-term subscription costs remain prevalent in the literature. Furthermore, data sovereignty issues can arise when patient data is stored on third-party servers, posing compliance challenges for certain jurisdictions.

While the aforementioned systems offer robust features, there remains a gap for a localized, cost-effective management system that combines the administrative capabilities of an RIS with the image handling of a PACS, specifically tailored for medium-scale facilities. This project aims to address this gap by developing a hybrid solution that balances functionality, cost, and data security.

The following is a table to compare the referenced applications compared to our application, as shown in Table (3.1).

Table (1.1) Literature Review

Feature	Epic Radiant	Orthanc	Ambra Health	Proposed RLMS
Core Purpose	Enterprise EHR-integrated Radiology Information System (RIS)	Open-source DICOM Server & Lightweight PACS	Cloud-based Imaging Suite & Medical Data Exchange	Lightweight Practice Management with Workflow Automation
Data Handling	Comprehensive EHR (Clinical history, billing, scheduling)	DICOM-native (Medical imaging files focus)	Cloud PACS (Focus on image sharing and mobility)	Relational SQL (Patient records, appointments, invoices)
Imaging Support	High-end DICOM integration with advanced viewers	Full DICOM/PACS support & RESTful API	Enterprise Cloud PACS with HTML5 DICOM Viewer	Basic Document/Image Upload (JPG/PNG/PDF)

Artificial Intelligence	AI-driven workflow optimization & clinical decision support	Extensible via plugins for AI image analysis	AI for automated triage and image recognition	Predictive Analytics for Patient Volume (Statistical Modeling)
Billing & Invoices	Deeply integrated Revenue Cycle Management (RCM)	None (Requires external integration)	Cloud billing and claims management	Built-in Invoice Generation & PDF Export
Patient Engagement	MyChart Integration (Full portal access)	Minimal (Focuses on server-side)	Patient-facing portal for image sharing	Appointment Booking, Survey Feedback, Status Tracking
Technology Stack	Proprietary (InterSystems IRIS/Caché)	C++ / REST API	Cloud-native / Web-based	PHP 8.x / MySQL 8.0 / Bootstrap 5
Deployment	Enterprise Cloud or Large On-premise	Lightweight (Raspberry Pi to Cloud)	SaaS (Software as a Service)	Local Server (On-premise / XAMPP/WAMP)
Interface Localization	Multi-language (Enterprise-grade)	English/Community patches	Multi-language	Arabic-Only (Native RTL Layout)

The comparative analysis underscores that while enterprise solutions offer deep clinical integration, they impose significant financial and infrastructural barriers. Open-source PACS servers excel in image handling but lack administrative workflow automation. Cloud platforms provide accessibility but introduce dependency, latency, and data sovereignty concerns. The proposed RLMS strategically occupies a niche focused on operational efficiency, cost-effective deployment, and culturally aligned interface design, making it suitable for medium-scale radiology departments and academic healthcare training environments.

3.4 Proposed Innovation and Research Gap:

Despite the proliferation of digital health platforms, a documented research gap persists in the development of lightweight, administration-centric radiology management systems that balance clinical documentation, scheduling, and financial tracking without requiring enterprise-grade imaging infrastructure. Existing academic and commercial solutions either over-engineer clinical workflows (neglecting administrative simplicity) or under-deliver on operational automation (focusing solely on image storage).

The Radiology Lab Management System addresses this gap through three key innovations:

1. **Modular Role-Based Architecture:** Implementation of a strictly segregated directory structure (admin/, doctor/, staff/, patient/) with middleware-enforced RBAC, ensuring secure, context-aware access without complex microservices overhead.
2. **Localized Usability Engineering:** Deliberate exclusion of English interface support in favor of a fully Arabic-native RTL layout, reducing interface complexity, eliminating translation inconsistencies, and aligning with regional operational standards.
3. **Hybrid Workflow Automation:** Integration of appointment management, medical record documentation, survey-driven feedback collection, and invoice generation within a single relational framework, eliminating the need for disparate third-party tools.

This innovation bridges the divide between theoretical software architecture models and practical, deployable health IT solutions, offering a reproducible academic reference for lightweight RIS development.

3.5 Theoretical Framework and Reference Studies:

The architectural and developmental foundation of the RLMS is grounded in established software engineering and health informatics theories:

- **Model-View-Controller (MVC) Pattern:** Provides separation of concerns between data access (Model), user interaction (View), and request routing (Controller), ensuring maintainability and scalable business logic execution.
- **Relational Database Normalization Theory:** Application of 1NF, 2NF, and 3NF principles to eliminate data redundancy, enforce referential integrity, and optimize query performance in multi-entity relationships (patients, appointments, records, invoices).
- **Role-Based Access Control (RBAC) Security Model:** Formalizes permission assignment based on user roles rather than individual identities, minimizing privilege escalation risks and ensuring compliance with academic security standards.
- **Usability Engineering Principles (ISO 9241-210):** Guides the design of intuitive, task-oriented interfaces with consistent navigation, error prevention mechanisms, and RTL-compliant visual hierarchy.

Reference studies by Sommerville (2016) on software engineering methodologies, Pressman & Maxim (2020) on web application architecture, and El-Sappagh et al. (2021) on Arabic health IT localization provide the theoretical scaffolding for system design, validation, and deployment strategies.

3.6 Recent and Future Trends in the Field:

The trajectory of radiology information systems is increasingly shaped by interoperability standards, predictive analytics, and edge-cloud hybrid architectures. Recent literature emphasizes the adoption of HL7 FHIR (Fast Healthcare Interoperability Resources) to enable seamless data exchange between RIS, EHR, and laboratory systems. Concurrently, machine learning models are being integrated for automated triage, anomaly detection, and predictive patient volume forecasting, enhancing operational planning and resource allocation.

Future developments in the field are expected to prioritize:

- **DICOM/PACS Integration:** Standardized medical imaging ingestion and web-based viewer embedding for comprehensive clinical reporting.
- **API-First Interoperability:** RESTful/GraphQL endpoints enabling third-party integrations with telemedicine platforms and national health registries.
- **Cloud-Native Deployment:** Containerized architectures (Docker/Kubernetes) with auto-scaling and encrypted data lakes for enterprise readiness.
- **Multi-Language & Accessibility Expansion:** Dynamic localization engines supporting bidirectional text rendering and WCAG 2.1 compliance for inclusive healthcare access.

For the RLMS, future work will focus on implementing DICOM metadata handling, integrating predictive scheduling algorithms, developing secure API gateways, and evaluating cloud migration feasibility while preserving the system's core advantages: modularity, cost efficiency, and localized usability.

CHAPTER 4

Requirements Modeling and Analysis

4.1 Feasibility Study (Technical and Financial Viability):

A comprehensive feasibility assessment was conducted to validate the practicality of developing the Radiology Lab Management System (RLMS) within academic and operational constraints. The study evaluated technical, financial, and operational dimensions:

- **Technical Feasibility:** The system architecture leverages a mature, open-source technology stack (PHP 8.x, MySQL 8.0, Bootstrap 5, Apache) that ensures robust backend processing, relational data integrity, and responsive frontend rendering. The modular directory structure (admin/, doctor/, staff/, patient/, includes/, libs/) facilitates parallel development and simplifies role-based routing. Arabic RTL localization is natively supported through CSS logical properties and Bootstrap RTL extensions, eliminating third-party dependency risks. Database normalization (up to 3NF) and ACID compliance guarantee data consistency for clinical and administrative transactions. The deployment environment (XAMPP/WAMP) is widely supported across academic and institutional infrastructures, ensuring reproducibility and low maintenance overhead.
- **Financial Feasibility:** The project operates entirely on open-source software and freemium development tools, resulting in zero licensing costs. Computational requirements are minimal, allowing deployment on standard local servers or institutional intranets without specialized hardware. This cost-efficient model aligns with the budgetary constraints of medium-scale radiology departments and academic training environments, ensuring long-term sustainability without recurring subscription fees.

- **Operational Feasibility:** The system directly addresses documented workflow inefficiencies in radiology facilities, including appointment scheduling conflicts, manual record-keeping, and fragmented billing processes. By digitizing and automating core operations, the RLMS reduces administrative workload, minimizes human error, and accelerates report dissemination. The intuitive, Arabic-native interface lowers the learning curve for non-technical staff, ensuring high adoption rates and operational continuity.

4.2 Requirements Gathering Methodology:

Requirements were systematically elicited using a multi-method approach aligned with IEEE 830 standards and software engineering best practices:

1. **Domain Analysis & Literature Review:** Examination of existing Radiology Information Systems (RIS), healthcare informatics standards, and academic benchmarks to identify functional gaps and best practices.
2. **Stakeholder Simulation & Workflow Mapping:** Role-based process mapping was conducted to simulate real-world radiology operations, capturing task sequences, decision points, and data exchange patterns across administrative, clinical, and patient-facing workflows.
3. **Comparative System Analysis:** Evaluation of commercial and open-source RIS/PACS platforms to extract industry-standard features, usability patterns, and security protocols.
4. **Iterative Refinement:** Elicited requirements were documented, validated against the project scope, and refined through supervisor review cycles to ensure academic rigor and technical feasibility.

All gathered requirements were cataloged in a centralized Requirements Database (RDB), assigned unique identifiers (REQ-XXX), and classified by type, priority, and role ownership.

4.3 Stakeholder Identification and Needs Analysis:

Four primary stakeholder groups were identified, each with distinct operational objectives and system interaction patterns:

Table (4.1) Stakeholder identification and needs analysis

Stakeholder	Primary Role	Core Needs & Expectations
System Administrator	Platform governance, security, configuration	User/role management, section/dept configuration, system statistics, report generation, global settings management, audit trails
Clinical Staff (Reception/Lab)	Daily operations, patient coordination	Patient registration, appointment scheduling, invoice generation, payment method management, survey administration, data search & filtering
Physician (Doctor)	Clinical documentation, diagnostic reporting	Patient history access, medical record creation/editing, report viewing, PDF export, appointment overview
Patient	Service access, self-management	Account creation, appointment booking & tracking, medical record viewing, survey participation, profile management

4.4 Function Requirements Specification:

Functional requirements were derived from stakeholder analysis and structured into a Requirements Database (RDB). A total of **55 functional requirements** were documented, categorized by user role, and assigned priority levels (1=Critical, 5=Deferred/Optional). The complete RDB is maintained in the project documentation; representative excerpts are presented below:

Table (4.2) Requirements Database RDB

Id	Title	Description	Type	Priority
REQ-001	Log in	Login for each user level and direct each user to their respective pages.	Functional	1
REQ-002	Edit personal account	Users can edit their personal data, such as email address and password.	Functional	1
REQ-003	View Profile	View user account information.	Functional	1
REQ-004	Change password	Change user password	Functional	1
REQ-005	View section list	View the list of sections listed in the system	Functional	1
REQ-006	Add section	Add a new section to the list of sections	Functional	1
REQ-007	Edit section	Edit an existing section in the database	Functional	1
REQ-008	Change section status	Change the status of an existing section in the database (Activate, Deactivate)	Functional	1
REQ-009	Add user	Add a new user to the system's user list	Functional	1
REQ-010	Edit user data	Edit the data of an existing user	Functional	1
REQ-011	Change user account status	Change the account status of an existing user in the database (Activate, Deactivate)	Functional	1
REQ-012	View user list	View the list of users listed in the system	Functional	1
REQ-013	View statistics	Manage the accounts of registered volunteer organization administrators in the system (Approve, Reject, Deactivate)	Functional	2
REQ-014	Export reports	Manage the accounts of registered volunteers in the system (Approve, Reject, Deactivate)	Functional	2
REQ-015	Manage system settings	Manage system settings such as (Daily appointment limit, Billing prefix)	Functional	2
REQ-016	View survey list	View the list of surveys listed in the system	Functional	2
REQ-017	Add survey	Add a new survey to the survey list	Functional	2
REQ-018	Edit survey	Edit an existing survey	Functional	2
REQ-019	Change survey status	Change the status of an existing survey in the database (Activate, Deactivate)	Functional	2
REQ-020	Delete survey	Delete an existing survey	Functional	2
REQ-021	View survey questions	View questions added to a survey	Functional	2
REQ-022	Add survey question	Add a new question to the survey's question list	Functional	2
REQ-023	Edit survey question	Edit an existing question added to a survey	Functional	2
REQ-024	Delete survey question	Delete an existing question added to a survey	Functional	2

REQ-025	View survey responses	View patient responses to the survey	Functional	3
REQ-026	View payment methods list	View the list of payment methods listed in the payment methods list	Functional	3
REQ-027	Add payment method	Add a payment method Add to Payment Method List	Functional	3
REQ-028	Edit payment method	Edit Existing Payment Method	Functional	3
REQ-029	Change payment method status	Change Status of Existing Payment Method (Activate, Deactivate)	Functional	3
REQ-030	Delete payment method	Delete Existing Payment Method	Functional	3
REQ-031	Add invoice	Add New Invoice to Invoice List	Functional	3
REQ-032	Edit invoice data	Edit Existing Invoice Details	Functional	3
REQ-033	Print invoice	Print Invoice from Invoice List	Functional	3
REQ-034	View invoice list	View List of Invoices Added by Staff	Functional	3
REQ-035	Search invoice	Search for Invoice by Patient Name or Invoice Number	Functional	3
REQ-036	Add new patient	Add New Patient to Patient List	Functional	3
REQ-037	Edit patient data	Edit Existing Patient Details	Functional	4
REQ-038	View patient list	View List of Patients Registered in the System	Functional	4
REQ-039	Search patient	Search for Patient Using Name or Phone Number	Functional	4
REQ-040	Add new appointment	Add New Appointment to Appointment List	Functional	4
REQ-041	Edit appointment data	Edit Existing Appointment Details	Functional	4
REQ-042	Change appointment status	Change Status of Existing Appointment (Accept, Cancel)	Functional	4
REQ-043	Search appointment	Search for Existing Appointment Using Patient Name or Doctor Name with Time and Section Filters	Functional	4
REQ-044	View appointment list	View List of Listed Appointments	Functional	4
REQ-045	View patient's medical record	View Patient's Medical Record	Functional	4
REQ-046	View medical records list	View List of Patient Medical Records	Functional	4
REQ-047	Add patient's medical record	Add New Medical Record for Patient After Doctor Review	Functional	4
REQ-048	Edit patient's medical record	Edit Existing Medical Record	Functional	4
REQ-049	Print patient's medical record	Print Patient Medical Record in PDF Format	Functional	5
REQ-050	Search medical record	Search for Medical Record Using Patient Name or Phone Number	Functional	5
REQ-051	Create an account	Create a new account so that the user can use the system	Functional	5
REQ-052	Request an appointment	Request New Appointment by Patient	Functional	5
REQ-053	View appointment history	View Patient's Appointment History	Functional	5

REQ-054	Track appointment status	View Appointment Status Appointment confirmed or canceled	Operational	5
REQ-055	Answer a survey	Patient questionnaire response	Operational	5

4.5 Non-Functional Requirements Specification:

Non-functional requirements define quality attributes constraining system operation. These were specified in tabular format per the methodology guideline:

Table (4.3) Non-Functional Requirements Matrix

Category	Requirement Specification	Metric/Standard
Performance	Page load time ≤ 2 seconds under concurrent user load; database query response ≤ 500 ms	Load testing, Apache Bench
Security	Prepared statements (PDO), session tokenization, RBAC middleware, CSRF protection, XSS prevention	OWASP Top 10 (2023) compliance
Usability	Fully Arabic RTL interface, consistent navigation, error prevention, WCAG 2.1 AA contrast standards	Task completion rate $\geq 90\%$
Maintainability	Modular directory structure, externalized configuration, standardized naming conventions, inline documentation	ISO/IEC 25010 Maintainability Index
Portability	Cross-browser compatibility (Chrome, Firefox, Edge, Safari), responsive mobile-first design	Bootstrap 5 responsive grid validation
Data Integrity	ACID compliance, foreign key constraints, cascade deletion rules, transaction rollback mechanisms	MySQL InnoDB engine standards

4.6 Requirements Prioritization and Management Mechanism:

Requirements were prioritized using a 1–5 numeric scale aligned with the MoSCoW framework:

- **Priority 1 (Must Have):** Core authentication, RBAC routing, patient/staff/doctor dashboards, appointment management, medical records, invoicing.
- **Priority 2–3 (Should Have):** Survey administration, payment methods, statistics, report generation, system settings.
- **Priority 4–5 (Could/Will Have):** Profile management, password reset, historical tracking, feedback surveys.

A formal Change Control Protocol was instituted to manage scope evolution. All modifications required impact analysis, supervisor approval, and RTM updates before integration. The Requirements Traceability Matrix (RTM) served as the primary governance instrument, ensuring complete coverage and preventing unauthorized feature expansion.

4.7 UML Modeling for Requirements Specification:

Unified Modeling Language (UML) was employed to formally represent system behavior, structural relationships, and dynamic interactions prior to implementation.

4.7.1 Main Use Case Diagrams:

Three hierarchical use case diagrams were developed to capture system functionality at varying granularities:

- **Figure 4.1 General Use Case Diagram:** Illustrates high-level actor-system interactions and core functional boundaries.

- Figure 4.2 Use Case Diagram for Project Phase 1: Details foundational modules (authentication, profile management, admin configuration).
- Figure 4.3 Use Case Diagram for Current Project: Expands to include clinical workflows, appointment scheduling, invoicing, surveys, and statistical reporting.

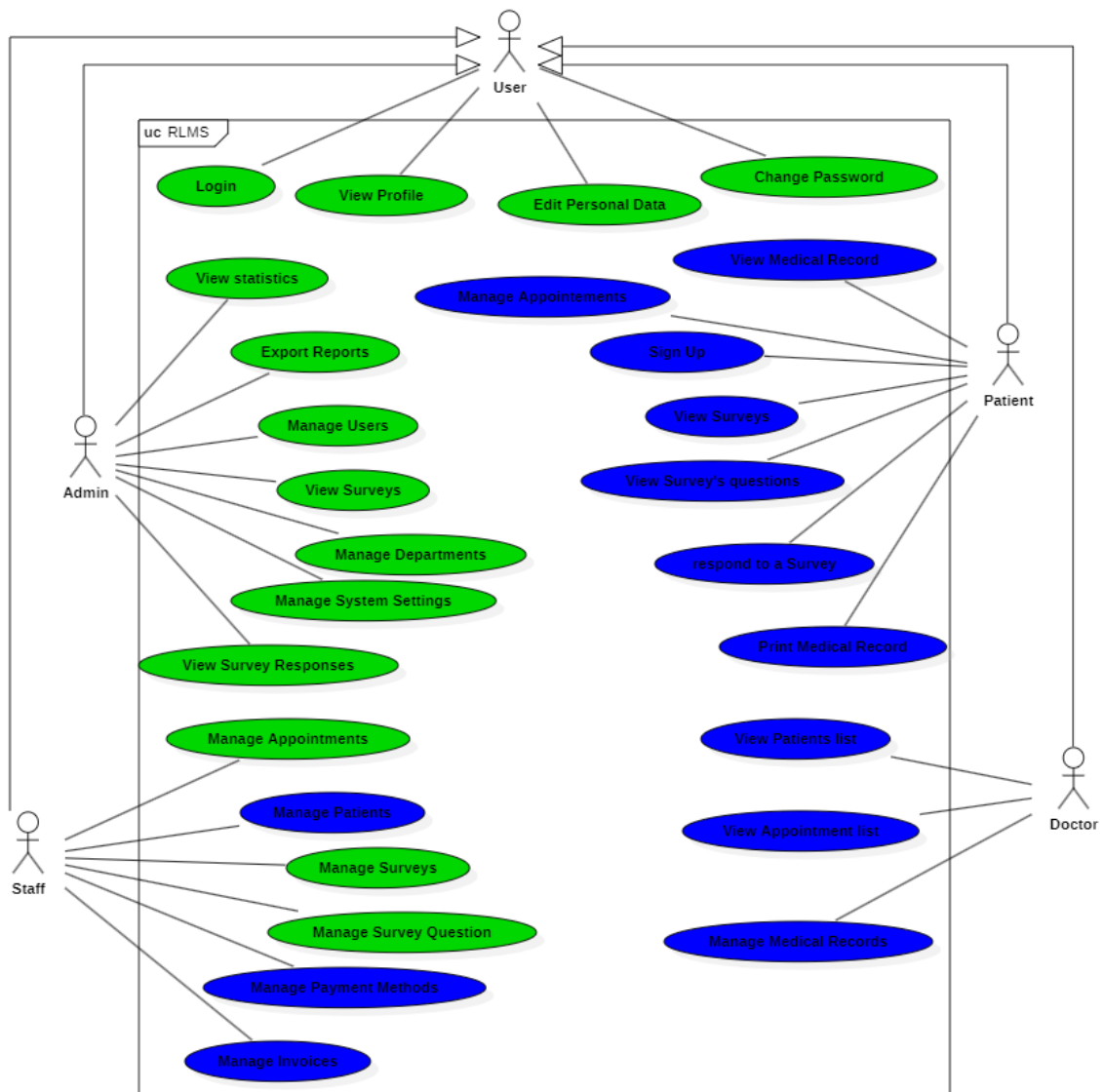


Figure (4.1) General Use Case Diagram

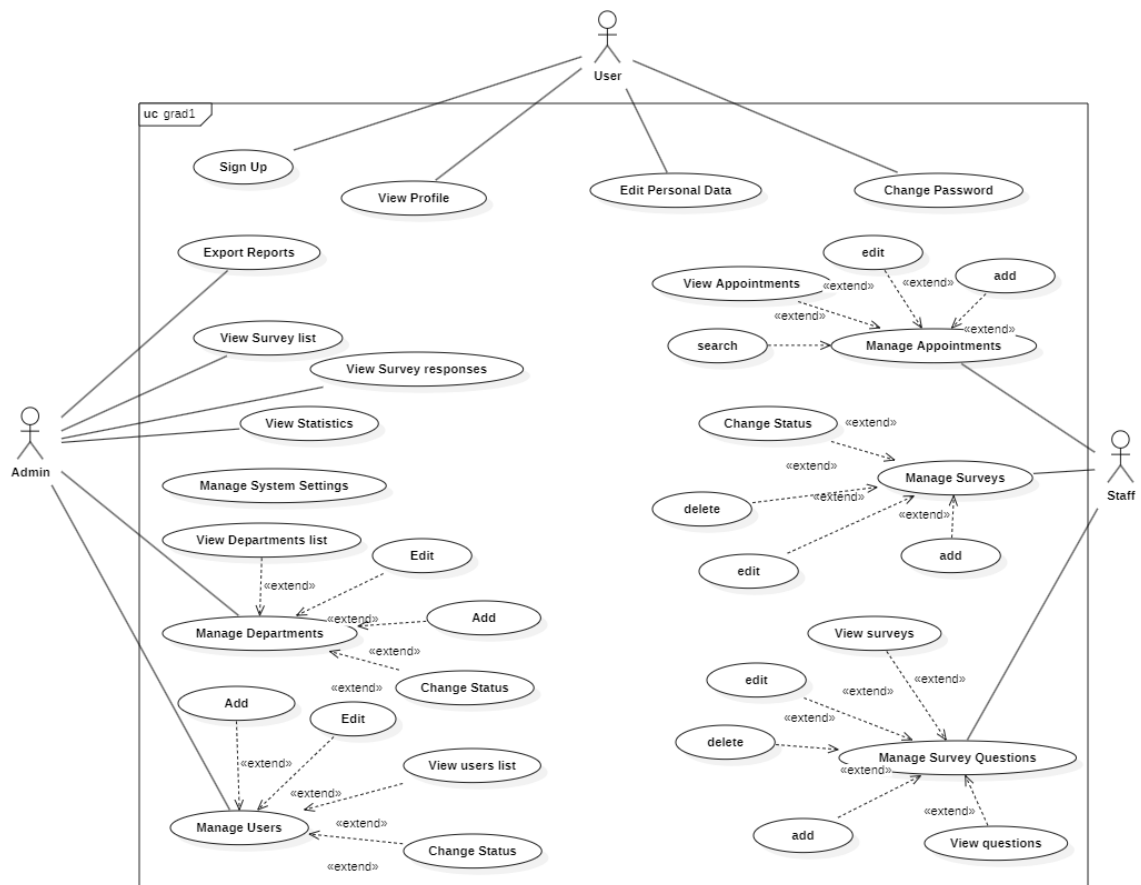


Figure (4.2) Use Case Diagram for Project 1

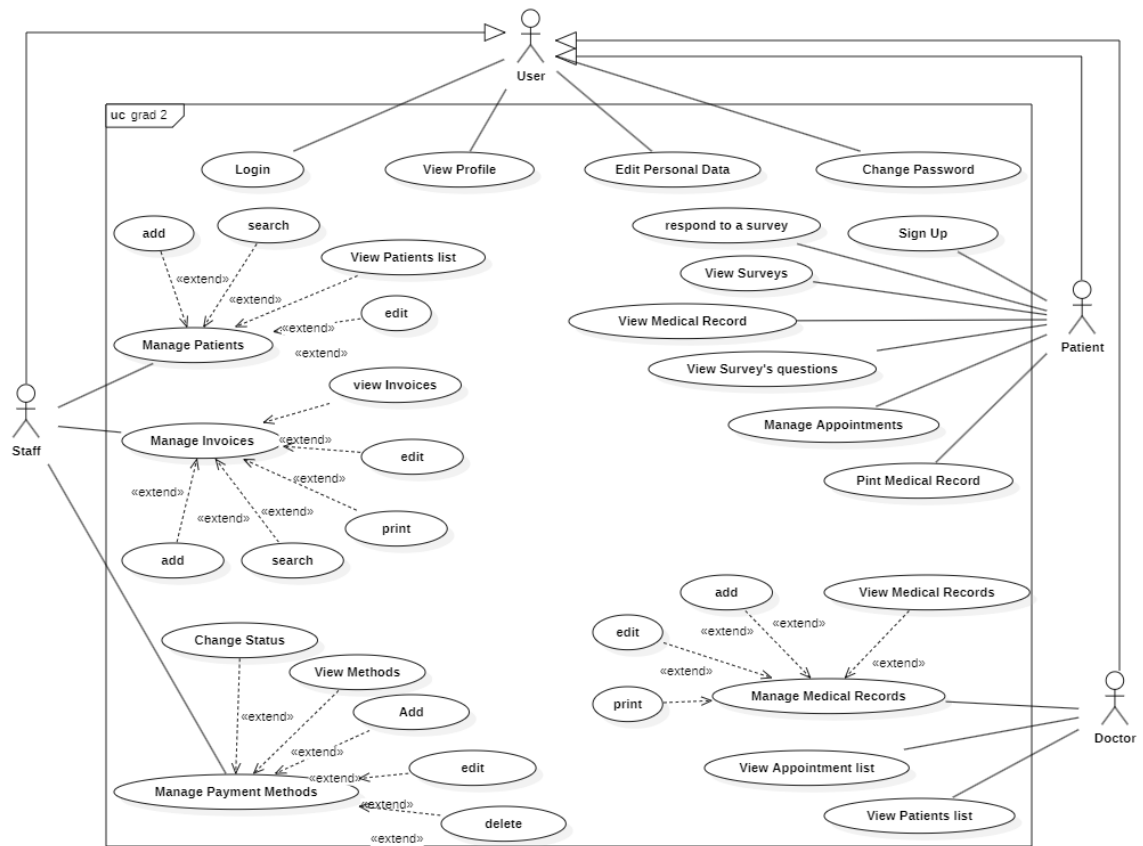


Figure (4.3) Use Case Diagram for Current Project

4.7.2 Use Case Description:

Each functional requirement was elaborated into a structured Use Case specification comprising: unique ID, descriptive title, actor(s), pre/post-conditions, main success scenario, and alternative flows.

Table (4.4) Login Use Case

ID	REQ-001
Use Case	Login
Description	Access system
Actors	User
Pre-Conditions	- Having a pre-existing user account
Subsequent-Conditions	- Access User's Dashboard
Main Flow	- When the user selects to log in, the system displays a login form. - The user fills in the required information (username, password). - When the user clicks the login button, the system verifies the entered data. - If the entered data is correct, the system redirects the user to their user interface.
Alternative Flow	- If the entered data is incorrect, an error message will appear requesting that the data be re-entered correctly.

Table (4.5) Edit Personal Account Use Case

ID	REQ-002
Use Case	Edit Personal Account
Description	Edit user's personal account data
Actors	User
Pre-Conditions	- Login
Subsequent-Conditions	- The process was completed successfully.
Main Flow	- When the user clicks the "Edit Personal Data" option, the system displays the user's data. - The user fills in the fields with the data they wish to modify (username, password). - When the "Save Changes" button is clicked, the system verifies the entered data. - If the entered data is correct (matches the pattern), the system saves the changes and updates the database.
Alternative Flow	- If there is an error in the entered data (pattern not matching), an error message is displayed asking the user to modify the fields indicated in red.

Table (4.6) View Personal Account

ID	REQ-003
Use Case	View Personal Account
Description	View User's Profile information
Actors	User
Pre-Conditions	- Login
Subsequent-Conditions	- The process was completed successfully.
Main Flow	- When the user clicks on the profile option, the system displays the user's profile data.

Table (4.7) Change Password

ID	REQ-004
Use Case	Change Password
Description	Change User's Password
Actors	User
Pre-Conditions	- Login
Subsequent-Conditions	- The process was completed successfully.
Main Flow	- When the user clicks the "Change Passwrd" option, the system displays the user's password. - The user fills in the fields with the new password and password confirmation. - When the "Save Changes" button is clicked, the system verifies the entered data. - If the entered data is correct (matches the pattern), the system saves the changes and updates the database.
Alternative Flow	- If there is an error in the entered data (pattern not matching), an error message is displayed asking the user to modify the fields indicated in red.

Table (4.8) View section list

ID	REQ-005
Use Case	View sections list
Description	View the list of sections already listed in the database
Actors	Admin
Pre-Conditions	- Log in - Existed sections listed in the database
Subsequent-Conditions	- Successful completion of the operation
Main Flow	- Admin selects "View Sections" - The system displays a table containing a list of sections registered on the platform
Alternative Flow	- If no sections are listed in the database, a "No Data" message is displayed

Table (4.9) Add Section

ID	REQ-006
Use Case	Add Section
Description	Adding a new Section to the sections list by the admin
Actors	Admin
Pre-Conditions	- Login
Subsequent-Conditions	- Successful completion of the operation
Main Flow	- When admin selects sections Management, a list of sections listed in the system appears with the option to add a new section. - When admin chooses to add a new section, the system displays the add section form - Admin fills in the data according to the form - When the “Save” button is clicked, the system verifies the entered data. - If the entered data is correct (matches the pattern), the system saves the changes and updates the database.
Alternative Flow	- If there is an error in the entered data (pattern not matching), an error message is displayed asking admin to modify the fields indicated in red.

Table (4.10) Edit Section

ID	REQ-007
Use Case	Edit Section
Description	Edit previously added section data
Actors	Admin
Pre-Conditions	- Login - The presence of sections in the list of sections
Subsequent-Conditions	- successful completion of the operation
Main Flow	- when admin selects sections management, the system displays the list of sections - Admin selects the section for which he wants to edit the data - Admin modifies the data he wishes to modify - When the Save Changes, button is pressed, the system checks the entered data - If the data is correct (matching the pattern), the system saves the changes and updates the database
Alternative Flow	- If there is an error in the entered data (non-conformance to the pattern), an error message is displayed asking admin to modify the fields indicated in red

Table (4.11) Change Section Status

ID	REQ-008
Use Case	Change section Status
Description	Change the status of a section previously listed in the database
Actors	Admin
Pre-Conditions	- Login - The presence of Sections in the list of Sections
Subsequent-Conditions	- successful completion of the operation
Main Flow	- When Admin selects sections management, the system shows a table containing the list of sections - Admin selects the section whose status he wants to modify - When the save modifications button is pressed, the system saves the data and updates the database

Table (4.12) Add User

ID	REQ-009
Use Case	Add User
Description	Adding a new User to the users list by the admin
Actors	Admin
Pre-Conditions	- Login
Subsequent-Conditions	- Successful completion of the operation
Main Flow	- When admin selects users Management, a list of users listed in the system appears with the option to add a new user. - When admin chooses to add a new user, the system displays the add user form - Admin fills in the data according to the form - When the “Save” button is clicked, the system verifies the entered data. - If the entered data is correct (matches the pattern), the system saves the changes and updates the database.
Alternative Flow	- If there is an error in the entered data (pattern not matching), an error message is displayed asking admin to modify the fields indicated in red.

Table (4.13) Edit User

ID	REQ-010
Use Case	Edit User
Description	Edit previously added user data
Actors	Admin
Pre-Conditions	- Login - The presence of users in the list of users
Subsequent-Conditions	- successful completion of the operation
Main Flow	- when admin selects users management, the system displays the list of users - Admin selects the user for which he wants to edit the data - Admin modifies the data he wishes to modify - When the Save Changes, button is pressed, the system checks the entered data - If the data is correct (matching the pattern), the system saves the changes and updates the database
Alternative Flow	- If there is an error in the entered data (non-conformance to the pattern), an error message is displayed asking admin to modify the fields indicated in red

Table (4.14) Change user account Status

ID	REQ-011
Use Case	Change user account Status
Description	Change the status of a section previously listed in the database
Actors	Admin
Pre-Conditions	- Login - The presence of users in the list of users
Subsequent-Conditions	- successful completion of the operation
Main Flow	- When Admin selects users management, the system shows a table containing the list of users - Admin selects the user whose status he wants to modify - When the save modifications button is pressed, the system saves the data and updates the database

Table (4.15) View users list

ID	REQ-012
Use Case	View users list
Description	View the list of users already listed in the database
Actors	Admin
Pre-Conditions	- Log in - Existed users listed in the database
Subsequent-Conditions	- Successful completion of the operation
Main Flow	- Admin selects "View Users" - The system displays a table containing a list of users registered on the platform
Alternative Flow	- If no users are listed in the database, a "No Data" message is displayed

Table (4.16) View Statistics

ID	REQ-013
Use Case	View Statistics
Description	View statistics on workflow within the platform
Actors	Admin
Pre-Conditions	- Log in
Subsequent-Conditions	- Successful completion of the operation
Main Flow	- When the system administrator logs in, the system directs him to his control panel. - In the system administrator's control panel, statistics are displayed in the form of cards.
Alternative Flow	- If no citizens are listed in the database, a "No Data" message is displayed

Table (4.17) Export Reports

ID	REQ-014
Use Case	Export reports
Description	Export system performance reports
Actors	Admin
Pre-Conditions	- Log in
Subsequent-Conditions	- The report file is generated and downloaded
Main Flow	- The admin selects the "Export Reports" option - The system generates a report based on system data. - The system triggers the file download

Table (4.18) Manage System Settings

ID	REQ-015
Use Case	Manage System Settings
Description	Manage system settings such as dark mode or changing invoice printing settings
Actors	Admin
Pre-Conditions	- Log in
Subsequent-Conditions	- Successful completion of the operation
Main Flow	- When Admin clicks on the System Settings option, the system displays a form for managing the allowed settings. - Admin fills in the required data according to the form. - When the Save button is clicked, the system verifies the entered data. - If the entered data is complete, the system updates the settings according to the new data.
Alternative Flow	- If there is an error in the entered data, an error message is displayed asking the administrator to correct the fields highlighted in red.

Table (4.19) View Surveys list

ID	REQ-016
Use Case	View surveys list
Description	View the list of surveys already listed in the database
Actors	Staff
Pre-Conditions	- Log in - Existed surveys listed in the database
Subsequent-Conditions	- Successful completion of the operation
Main Flow	- Staff selects "View surveys" - The system displays a table containing a list of surveys registered on the platform
Alternative Flow	- If no surveys are listed in the database, a "No Data" message is displayed

Table (4.20) Add Survey

ID	REQ-017
Use Case	Add Survey
Description	Adding a new Survey to the surveys list
Actors	Staff
Pre-Conditions	- Login
Subsequent-Conditions	- Successful completion of the operation
Main Flow	- When staff selects surveys Management, a list of surveys listed in the system appears with the option to add a new survey. - When staff chooses to add a new survey, the system displays the add survey form - Staff fills in the data according to the form - When the “Save” button is clicked, the system verifies the entered data. - If the entered data is correct (matches the pattern), the system saves the changes and updates the database.
Alternative Flow	- If there is an error in the entered data (pattern not matching), an error message is displayed asking staff to modify the fields indicated in red.

Table (4.21) Edit Survey

ID	REQ-018
Use Case	Edit survey
Description	Edit previously added survey data
Actors	Staff
Pre-Conditions	- Login - The presence of surveys in the list of surveys
Subsequent-Conditions	- successful completion of the operation
Main Flow	- When staff selects surveys management, the system displays the list of surveys - Staff selects the survey for which he wants to edit the data - Staff modifies the data he wishes to modify - When the Save Changes, button is pressed, the system checks the entered data - If the data is correct (matching the pattern), the system saves the changes and updates the database
Alternative Flow	- If there is an error in the entered data (non-conformance to the pattern), an error message is displayed asking staff to modify the fields indicated in red

Table (4.22) Change survey Status

ID	REQ-019
Use Case	Change survey Status
Description	Change the status of a survey previously listed in the database
Actors	Staff
Pre-Conditions	- Login - The presence of surveys in the list of surveys
Subsequent-Conditions	- successful completion of the operation
Main Flow	- When staff selects surveys management, the system shows a table containing the list of surveys - Staff selects the survey whose status he wants to modify - When the save modifications, button is pressed, the system saves the data and updates the database

Table (4.23) Delete survey

ID	REQ-020
Use Case	Delete survey
Description	Delete previously added survey data
Actors	Staff
Pre-Conditions	- Login - The presence of surveys in the list of surveys
Subsequent-Conditions	- successful completion of the operation
Main Flow	- When Staff selects surveys management, the system shows a table containing a list of survey - Staff selects the survey whose data he wants to delete - When the Delete button is pressed, the system requests confirmation of the deletion process - If staff confirms the deletion operation, the system saves the changes and updates the database
Alternative Flow	- If the deletion is not confirmed, the list of surveys is returned to the view

Table (4.24) View Survey Questions

ID	REQ-021
Use Case	View Survey Questions
Description	View the list of questions already added to a survey
Actors	Staff
Pre-Conditions	- Log in - Existed questions added to the survey
Subsequent-Conditions	- Successful completion of the operation
Main Flow	- Staff selects "View surveys" - Staff selects the survey for which he wants to view the added questions. - The system displays a table containing a list of questions added to survey
Alternative Flow	- If no questions were added in the database, a "No Data" message is displayed

Table (4.25) Add Survey question

ID	REQ-022
Use Case	Add Survey question
Description	Adding a new question to a survey
Actors	Staff
Pre-Conditions	- Login
Subsequent-Conditions	- Successful completion of the operation
Main Flow	- When staff selects surveys Management, a list of surveys listed in the system appears - Staff selects the survey for which he wants to view the added questions, The system displays a table containing a list of questions added to survey, with the option to add a new question - When staff chooses to add a new survey, the system displays the add question form - Staff fills in the data according to the form - When the “Save” button is clicked, the system verifies the entered data. - If the entered data is correct (matches the pattern), the system saves the changes and updates the database.
Alternative Flow	- If there is an error in the entered data (pattern not matching), an error message is displayed asking staff to modify the fields indicated in red.

Table (4.26) Edit Survey question

ID	REQ-023
Use Case	Edit Survey question
Description	Edit an existing question added to a survey
Actors	Staff
Pre-Conditions	- Login
Subsequent-Conditions	- Successful completion of the operation
Main Flow	- When staff selects surveys Management, a list of surveys listed in the system appears. - Staff selects the survey for which he wants to view the added questions, The system displays a table containing a list of questions added to survey - Staff choose the question he wants to edit - Staff fills in the data according to the form - When the “Save” button is clicked, the system verifies the entered data. - If the entered data is correct (matches the pattern), the system saves the changes and updates the database.
Alternative Flow	- If there is an error in the entered data (pattern not matching), an error message is displayed asking staff to modify the fields indicated in red.

Table (4.27) Delete Survey question

ID	REQ-024
Use Case	Delete Survey question
Description	Delete an existing question added to a survey
Actors	Staff
Pre-Conditions	- Login
Subsequent-Conditions	- Successful completion of the operation
Main Flow	- When staff selects surveys Management, a list of surveys listed in the system appears - Staff selects the survey for which he wants to view the added questions, The system displays a table containing a list of questions added to survey - Staff choose the question he wants to Delete - When the Delete button is pressed, the system requests confirmation of the deletion process - If staff confirms the deletion operation, the system saves the changes and updates the database
Alternative Flow	- If the deletion is not confirmed, the list of questions is returned to the view

Table (4.28) View Survey responses

ID	REQ-025
Use Case	View survey responses
Description	View the list of survey responses already listed in the database
Actors	Staff
Pre-Conditions	- Log in - Existed surveys listed in the database
Subsequent-Conditions	- Successful completion of the operation
Main Flow	- When staff selects surveys Management, a list of surveys listed in the system appears - Staff selects the survey for which he wants to view the responses, The system displays a table containing a list of responses added to survey
Alternative Flow	- If no responses are listed in the database, a "No Data" message is displayed

Table (4.29) View payment methods list

ID	REQ-026
Use Case	View payment methods list
Description	View the list of payment methods already listed in the database
Actors	Staff
Pre-Conditions	- Log in - Existed payment methods listed in the database
Subsequent-Conditions	- Successful completion of the operation
Main Flow	- Staff selects "View payment methods " - The system displays a table containing a list of payment methods registered on the platform
Alternative Flow	- If no payment methods are listed in the database, a "No Data" message is displayed

Table (4.30) Add Payment Method

ID	REQ-027
Use Case	Add payment method
Description	Adding a new payment method to the payment methods list
Actors	Staff
Pre-Conditions	- Login
Subsequent-Conditions	- Successful completion of the operation
Main Flow	- When staff selects payment methods Management, a list of payment methods listed in the system appears with the option to add a new payment method. - When staff chooses to add a new payment method, the system displays the add payment method form - Staff fills in the data according to the form - When the “Save” button is clicked, the system verifies the entered data. - If the entered data is correct (matches the pattern), the system saves the changes and updates the database.
Alternative Flow	- If there is an error in the entered data (pattern not matching), an error message is displayed asking staff to modify the fields indicated in red.

Table (4.31) Edit Payment Method

ID	REQ-028
Use Case	Edit payment method
Description	Edit previously added payment method data
Actors	Staff
Pre-Conditions	- Login - The presence of payment methods in the list of payment methods
Subsequent-Conditions	- successful completion of the operation
Main Flow	- When staff selects payment methods management, the system displays the list of payment methods - Staff selects the payment method for which he wants to edit the data - Staff modifies the data he wishes to modify - When the Save Changes, button is pressed, the system checks the entered data - If the data is correct (matching the pattern), the system saves the changes and updates the database
Alternative Flow	- If there is an error in the entered data (non-conformance to the pattern), an error message is displayed asking staff to modify the fields indicated in red

Table (4.32) Change Payment Method Status

ID	REQ-029
Use Case	Change payment method Status
Description	Change the status of a payment method previously listed in the database
Actors	Staff
Pre-Conditions	- Login - The presence of payment methods in the list of payment methods
Subsequent-Conditions	- successful completion of the operation
Main Flow	- When staff selects payment methods management, the system shows a table containing the list of payment methods - Staff selects the payment method whose status he wants to modify - When the save modifications, button is pressed, the system saves the data and updates the database

Table (4.33) Delete Payment Method

ID	REQ-030
Use Case	Delete payment method
Description	Delete previously added payment method data
Actors	Staff
Pre-Conditions	- Login - The presence of payment methods in the list of payment methods
Subsequent-Conditions	- successful completion of the operation
Main Flow	- When Staff selects payment methods management, the system shows a table containing a list of payment methods - Staff selects the payment method whose data he wants to delete - When the Delete button is pressed, the system requests confirmation of the deletion process - If staff confirms the deletion operation, the system saves the changes and updates the database
Alternative Flow	- If the deletion is not confirmed, the list of payment methods is returned to the view

Table (4.34) Add Invoice

ID	REQ-031
Use Case	Add Invoice
Description	Adding a new Invoice to the Invoices list
Actors	Staff
Pre-Conditions	- Login
Subsequent-Conditions	- Successful completion of the operation
Main Flow	- When staff selects Invoices Management, a list of Invoices listed in the system appears with the option to add a new Invoice. - When staff chooses to add a new Invoice, the system displays the add Invoice form - Staff fills in the data according to the form - When the “Save” button is clicked, the system verifies the entered data. - If the entered data is correct (matches the pattern), the system saves the changes and updates the database.
Alternative Flow	- If there is an error in the entered data (pattern not matching), an error message is displayed asking staff to modify the fields indicated in red.

Table (4.35) Edit Invoice

ID	REQ-032
Use Case	Edit Invoice
Description	Edit previously added Invoice data
Actors	Staff
Pre-Conditions	- Login - The presence of Invoices in the list of Invoices
Subsequent-Conditions	- successful completion of the operation
Main Flow	- When staff selects Invoices management, the system displays the list of Invoices - Staff selects the Invoice for which he wants to edit the data - Staff modifies the data he wishes to modify - When the Save Changes, button is pressed, the system checks the entered data - If the data is correct (matching the pattern), the system saves the changes and updates the database
Alternative Flow	- If there is an error in the entered data (non-conformance to the pattern), an error message is displayed asking staff to modify the fields indicated in red

Table (4.36) Print Invoice

ID	REQ-033
Use Case	Print Invoice
Description	Print Invoice from Invoices list
Actors	Staff
Pre-Conditions	- Login - The presence of Invoices in the list of Invoices
Subsequent-Conditions	- The invoice file is generated
Main Flow	- When staff selects Invoices management, the system displays the list of Invoices - Staff selects the Invoice wants to print - The system triggers the file print

Table (4.37) View Invoices list

ID	REQ-034
Use Case	View Invoices list
Description	View the list of Invoices already listed in the database
Actors	Staff
Pre-Conditions	- Log in - Existed Invoices listed in the database
Subsequent-Conditions	- Successful completion of the operation
Main Flow	- Staff selects "View Invoices " - The system displays a table containing a list of Invoices registered on the platform
Alternative Flow	- If no Invoices are listed in the database, a "No Data" message is displayed

Table (4.38) Search invoice

ID	REQ-035
Use Case	Search Invoice
Description	Search for invoice in the Invoices list
Actors	Staff
Pre-Conditions	- Login
Subsequent-Conditions	- Successful completion of the operation
Main Flow	- When staff selects Invoices Management, a list of Invoices listed in the system appears with the option to search for Invoice. - When staff chooses to search, fills in the data he wants to search for - If there are matches for entered data, the system shows the matches.
Alternative Flow	- If there are no matches, a "No Data" message is displayed

Table (4.39) Add New Patient

ID	REQ-036
Use Case	Add New Patient
Description	Adding a new patient to the patients list
Actors	Staff
Pre-Conditions	- Login
Subsequent-Conditions	- Successful completion of the operation
Main Flow	- When staff selects patients Management, a list of patients listed in the system appears with the option to add a new patient. - When staff chooses to add a new patient, the system displays the add patient form - Staff fills in the data according to the form - When the “Save” button is clicked, the system verifies the entered data. - If the entered data is correct (matches the pattern), the system saves the changes and updates the database.
Alternative Flow	- If there is an error in the entered data (pattern not matching), an error message is displayed asking staff to modify the fields indicated in red.

Table (4.40) Edit Patient Data

ID	REQ-037
Use Case	Edit patient data
Description	Edit previously added patient data
Actors	Staff
Pre-Conditions	- Login - The presence of Invoices in the list of Invoices
Subsequent-Conditions	- successful completion of the operation
Main Flow	- When staff selects patients management, the system displays the list of patients - Staff selects the patient for which he wants to edit the data - Staff modifies the data he wishes to modify - When the Save Changes, button is pressed, the system checks the entered data - If the data is correct (matching the pattern), the system saves the changes and updates the database
Alternative Flow	- If there is an error in the entered data (non-conformance to the pattern), an error message is displayed asking staff to modify the fields indicated in red

Table (4.41) View Patients list

ID	REQ-038
Use Case	View patients list
Description	View the list of patients already listed in the database
Actors	Staff
Pre-Conditions	- Log in - Existed patients listed in the database
Subsequent-Conditions	- Successful completion of the operation
Main Flow	- Staff selects "View patients " - The system displays a table containing a list of patients registered on the platform
Alternative Flow	- If no patients are listed in the database, a "No Data" message is displayed

Table (4.42) Search Patient

ID	REQ-039
Use Case	Search Patient
Description	Search for patient in the Patients list
Actors	Staff
Pre-Conditions	- Login
Subsequent-Conditions	- Successful completion of the operation
Main Flow	- When staff selects Patients Management, a list of patients listed in the system appears with the option to search for patient. - When staff chooses to search, fills in the data he wants to search for - If there are matches for entered data, the system shows the matches.
Alternative Flow	- If there are no matches, a "No Data" message is displayed

Table (4.43) Add New Appointment

ID	REQ-040
Use Case	Add New Appointment
Description	Adding a new appointment to the appointments list
Actors	Staff
Pre-Conditions	- Login
Subsequent-Conditions	- Successful completion of the operation
Main Flow	- When staff selects appointments Management, a list of appointments listed in the system appears with the option to add a new appointment. - When staff chooses to add a new appointment, the system displays the add appointment form - Staff fills in the data according to the form - When the “Save” button is clicked, the system verifies the entered data. - If the entered data is correct (matches the pattern), the system saves the changes and updates the database.
Alternative Flow	- If there is an error in the entered data (pattern not matching), an error message is displayed asking staff to modify the fields indicated in red.

Table (4.44) Edit Appointment Data

ID	REQ-041
Use Case	Edit Appointment data
Description	Edit previously added appointment data
Actors	Staff
Pre-Conditions	- Login - The presence of appointments in the list of appointments
Subsequent-Conditions	- successful completion of the operation
Main Flow	- When staff selects appointments management, the system displays the list of appointments - Staff selects the appointment for which he wants to edit the data - Staff modifies the data he wishes to modify - When the Save Changes, button is pressed, the system checks the entered data - If the data is correct (matching the pattern), the system saves the changes and updates the database
Alternative Flow	- If there is an error in the entered data (non-conformance to the pattern), an error message is displayed asking staff to modify the fields indicated in red

Table (4.45) Change Appointment Status

ID	REQ-042
Use Case	Change Appointment Status
Description	Change the status of a appointment previously listed in the database
Actors	Staff
Pre-Conditions	- Login - The presence of appointments in the list of appointments
Subsequent-Conditions	- successful completion of the operation
Main Flow	- When staff selects appointments management, the system shows a table containing the list of appointments - Staff selects the appointment whose status he wants to modify - When the save modifications, button is pressed, the system saves the data and updates the database

Table (4.46) Search Appointment

ID	REQ-043
Use Case	Search Appointment
Description	Search for appointment in the appointments list
Actors	Staff
Pre-Conditions	- Login
Subsequent-Conditions	- Successful completion of the operation
Main Flow	- When staff selects appointments Management, a list of appointments listed in the system appears with the option to search for appointment. - When staff chooses to search, fills in the data he wants to search for - If there are matches for entered data, the system shows the matches.
Alternative Flow	- If there are no matches, a "No Data" message is displayed

Table (4.47) View Appointments list

ID	REQ-044
Use Case	View appointments list
Description	View the list of appointments already listed in the database
Actors	Staff
Pre-Conditions	- Log in - Existed appointments listed in the database
Subsequent-Conditions	- Successful completion of the operation
Main Flow	- Staff selects "View appointments " - The system displays a table containing a list of appointments registered on the platform
Alternative Flow	- If no appointments are listed in the database, a "No Data" message is displayed

Table (4.48) View Patient's medical record

ID	REQ-045
Use Case	View Patient's Medical Record
Description	View Patient's Medical Record that added by doctor
Actors	Patient, Doctor
Pre-Conditions	- Log in - Existed records listed in the database
Subsequent-Conditions	- Successful completion of the operation
Main Flow	- User selects "View Medical Records" - The system displays a list of patient's medical records registered on the platform
Alternative Flow	- If no records are listed in the database, a "No Data" message is displayed

Table (4.49) View Medical Records list

ID	REQ-046
Use Case	View Medical Records list
Description	View the list of medical records already listed in the database
Actors	Doctor
Pre-Conditions	- Log in - Existed medical records listed in the database
Subsequent-Conditions	- Successful completion of the operation
Main Flow	- Doctor selects "View medical records" - The system displays a list of medical records registered on the platform
Alternative Flow	- If no records are listed in the database, a "No Data" message is displayed

Table (4.50) Add Patient's Medical Record

ID	REQ-047
Use Case	Add Patient's Medical Record
Description	Adding a new Patient's Medical Record
Actors	Doctor
Pre-Conditions	- Login
Subsequent-Conditions	- Successful completion of the operation
Main Flow	- When Doctor selects Medical Records Management, a list of medical records listed in the system appears with the option to add a new record. - When doctor chooses to add a new record, the system displays the add record form - Doctor fills in the data according to the form - When the "Save" button is clicked, the system verifies the entered data. - If the entered data is correct (matches the pattern), the system saves the changes and updates the database.
Alternative Flow	- If there is an error in the entered data (pattern not matching), an error message is displayed asking Doctor to modify the fields indicated in red.

Table (4.51) Edit Patient' s Medical Record

ID	REQ-048
Use Case	Edit Patient's Medical Record
Description	Edit previously added Patient's Medical Record
Actors	Doctor
Pre-Conditions	- Login - The presence of records in the list of Patient's Medical Record
Subsequent-Conditions	- successful completion of the operation
Main Flow	- When Doctor selects Patient's Medical Record management, the system displays the list of records - Doctor selects the record for which he wants to edit the data - Doctor modifies the data he wishes to modify - When the Save Changes, button is pressed, the system checks the entered data - If the data is correct (matching the pattern), the system saves the changes and updates the database
Alternative Flow	- If there is an error in the entered data (non-conformance to the pattern), an error message is displayed asking Doctor to modify the fields indicated in red

Table (4.52) Print Patient's Medical Record

ID	REQ-049
Use Case	Print Patient's Medical Record
Description	Print Medical Record from Patient's Medical Record list
Actors	Doctor, Patient
Pre-Conditions	- Login - The presence of records in the list of Patient's Medical Record
Subsequent-Conditions	- The record file is generated
Main Flow	- When User selects Medical Records management, the system displays the list of Patient's Medical Record - User selects the record he wants to print - The system triggers the file print

Table (4.53) Search Medical Record

ID	REQ-050
Use Case	Search Medical Record
Description	Search for record in the Medical Records list
Actors	Doctor
Pre-Conditions	- Login
Subsequent-Conditions	- Successful completion of the operation
Main Flow	- When doctor selects medical records Management, a list of medical records listed in the system appears with the option to search for record. - When doctor chooses to search, fills in the data he wants to search for - If there are matches for entered data, the system shows the matches.
Alternative Flow	- If there are no matches, a "No Data" message is displayed

Table (4.54) Create new Account

ID	REQ-051
Use Case	Create new Account
Description	Create a new account so that the user can use the system
Actors	Patient
Pre-Conditions	- No pre-created account
Subsequent-Conditions	- Successful completion of the account creation process - Redirection to the login interface
Main Flow	- When Patient chooses to create a new account, the system displays the account creation form - Patient fills in the data according to the form - When the account creation button is pressed, the system verifies the entered data - If the entered data is correct (matches the pattern), the system saves the data and updates the database
Alternative Flow	- If there is an error in the entered data (non-conformance to the pattern), an error message is displayed asking the Patient to modify the fields indicated in red

Table (4.55) Add Request an Appointment

ID	REQ-052
Use Case	Request an Appointment
Description	Request New Appointment by Patient
Actors	Patient
Pre-Conditions	- Login
Subsequent-Conditions	- Successful completion of the operation
Main Flow	- When Patient selects appointments Management, his history of appointments listed in the system appears with the option to request an appointment. - When patient chooses to request an appointment, the system displays the request appointment form - Patient fills in the data according to the form - When the “Save” button is clicked, the system verifies the entered data. - If the entered data is correct (matches the pattern), the system saves the changes and updates the database.
Alternative Flow	- If there is an error in the entered data (pattern not matching), an error message is displayed asking patient to modify the fields indicated in red.

Table (4.56) View Appointments history

ID	REQ-053
Use Case	View appointments history
Description	View the list of appointments already listed in the database
Actors	Patient
Pre-Conditions	- Log in - Existed appointments listed in the database
Subsequent-Conditions	- Successful completion of the operation
Main Flow	- Patient selects "My Appointments " - The system displays a table containing a history of appointments registered on the platform
Alternative Flow	- If no appointments are listed in the database, a "No Data" message is displayed

Table (4.57) Track Appointment Status

ID	REQ-054
Use Case	Track Appointment Status
Description	Tracking the status of appointment submitted by patient
Actors	Patient
Pre-Conditions	- Login
Subsequent-Conditions	- Successful completion of the operation
Main Flow	- When patient clicks on the appointments History option, the system displays the history of appointments submitted by patient
Alternative Flow	- If there are no appointments, a "No Data" message is displayed

Table (4.58) Answer a Survey

ID	REQ-055
Use Case	Answer a survey
Description	Patient survey response
Actors	Patient
Pre-Conditions	- Log in - Existed surveys listed in the database
Subsequent-Conditions	- Successful completion of the operation
Main Flow	- When Patient selects "View surveys", system displays a table containing a list of surveys registered on the platform - Patient choose a survey to response, system shows survey's questions - When patient finish all question, system save responses and update the database
Alternative Flow	- If no surveys are listed in the database, a "No Data" message is displayed

4.7.3 Activity Diagrams:

An activity diagram is defined as a sophisticated visual representation in software engineering and various fields. It excels at illustrating the smooth flow of activities, procedures, and processes within complex systems, business workflows, or any dynamic process. Activity diagrams provide a comprehensive view of the dynamic coordination of tasks, creating clear visualizations of workflows in software systems.

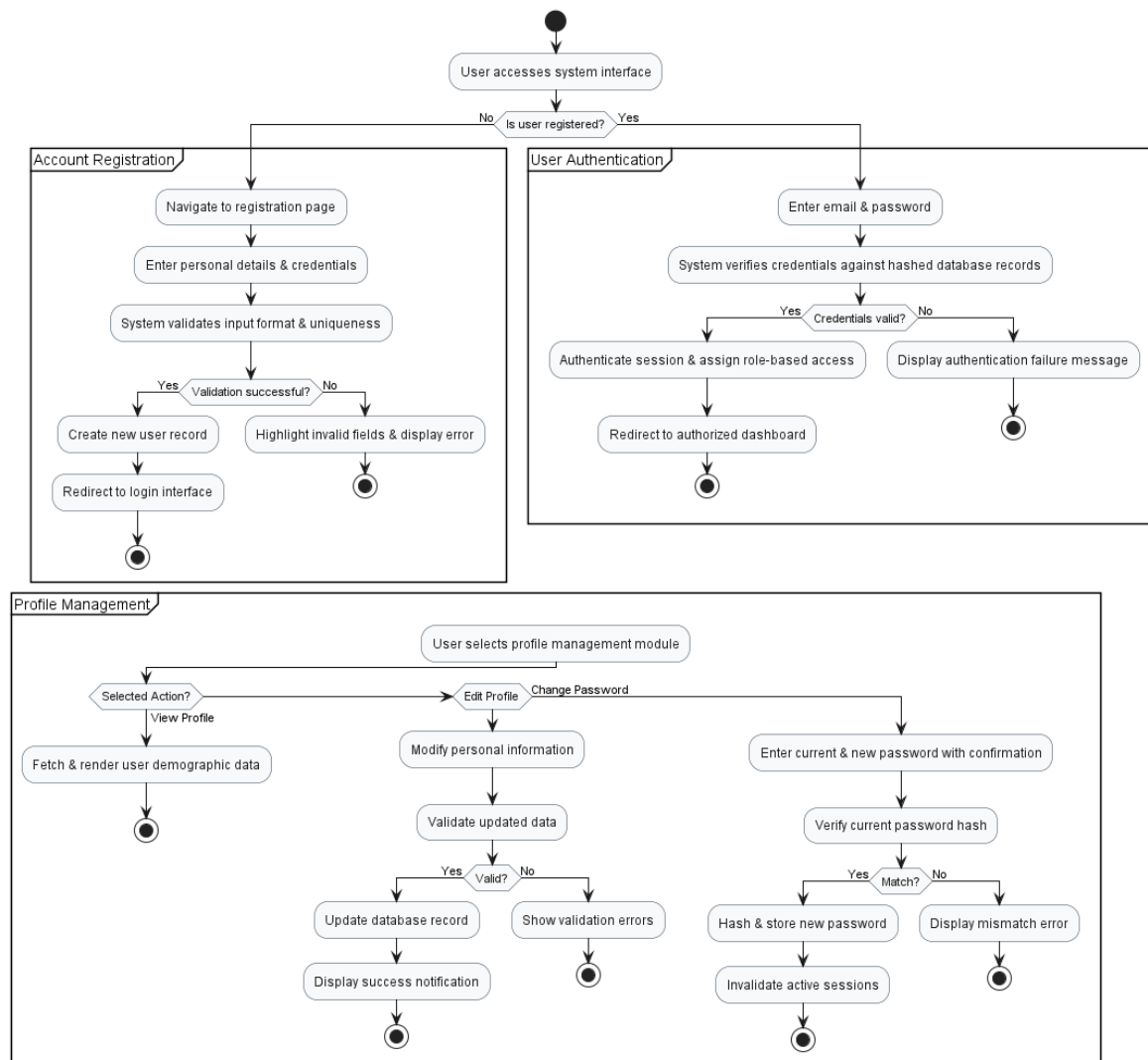


Figure (4.4) Activity Diagram for Authentication & Account Management

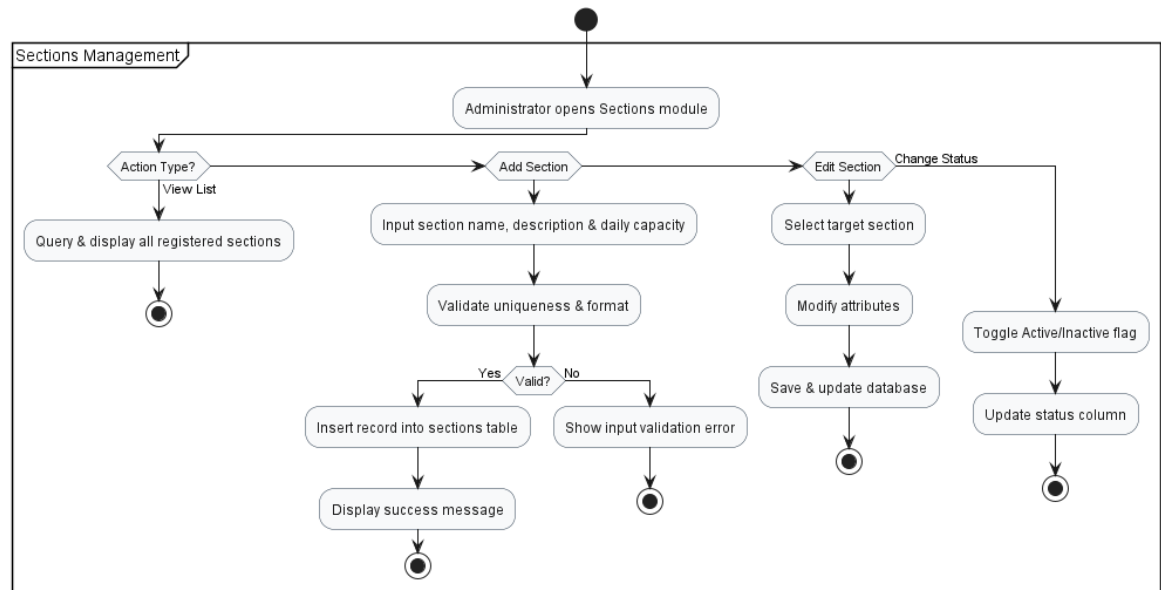


Figure (4.5) Activity Diagram for Sections Management

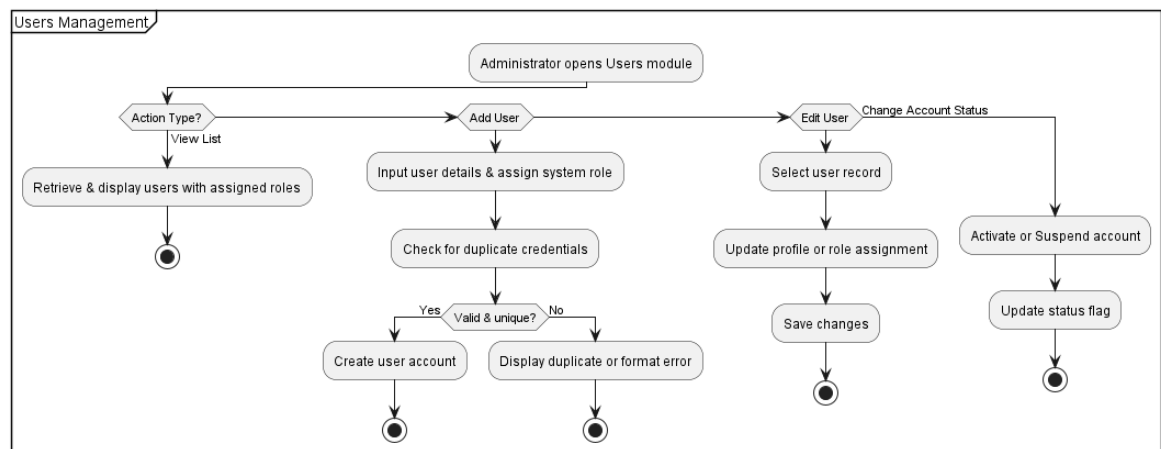


Figure (4.6) Activity Diagram for Users Management

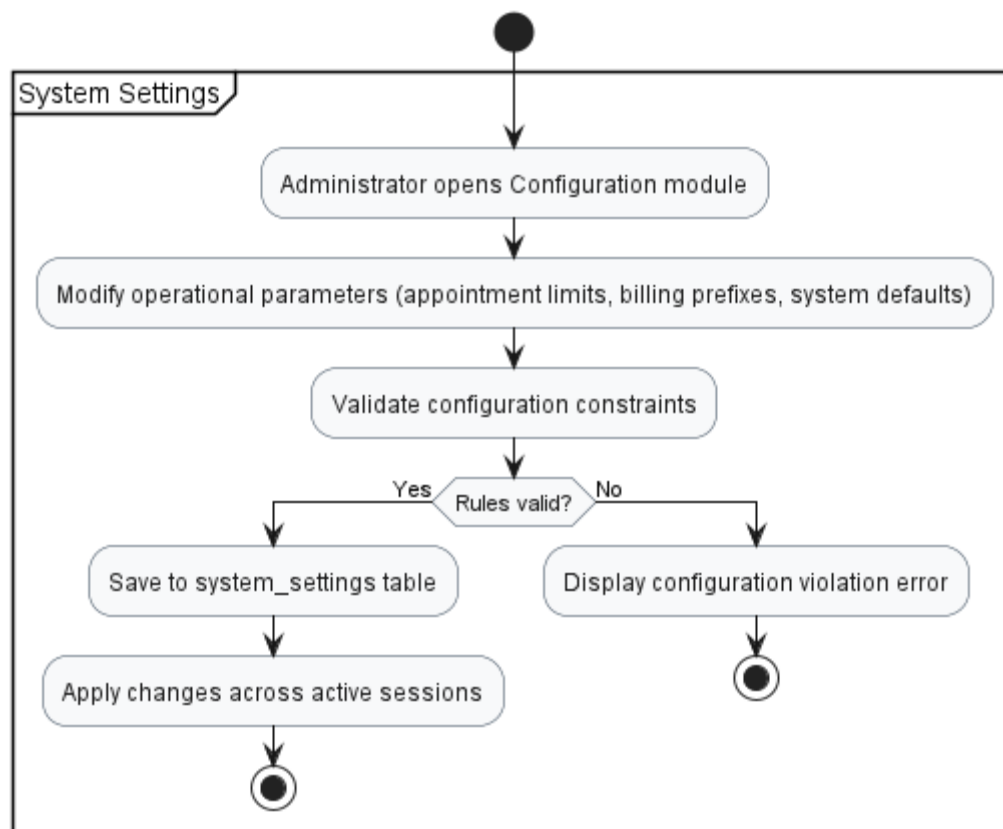


Figure (4.7) Activity Diagram for system settings

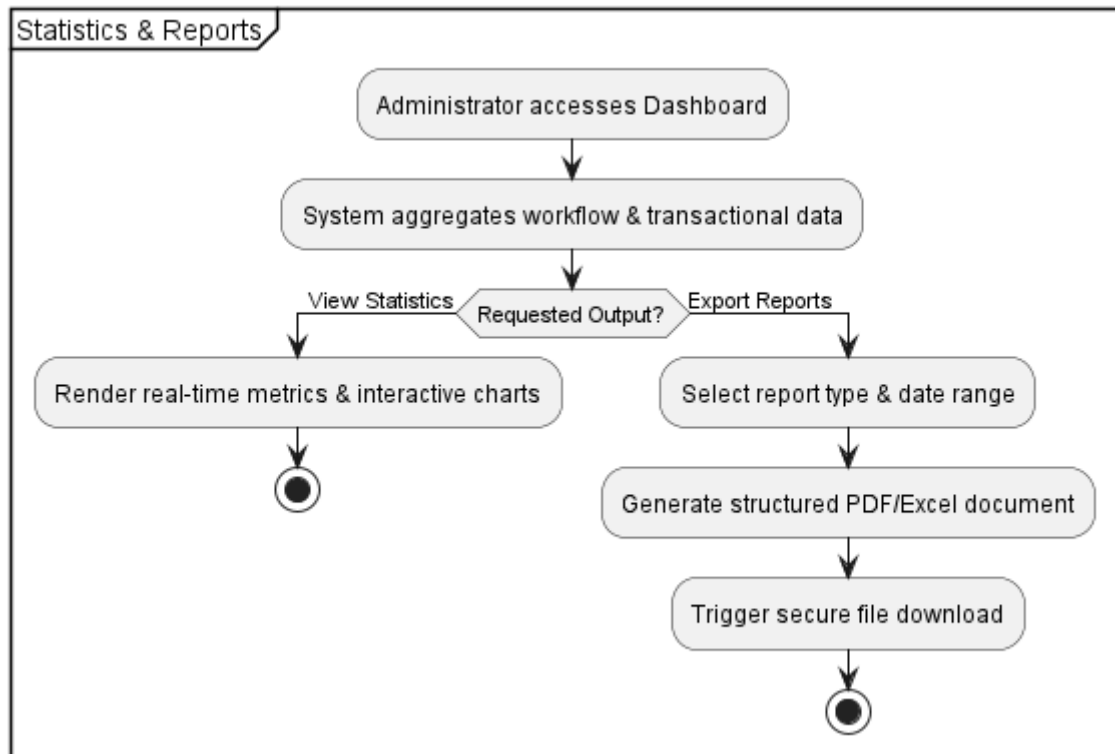


Figure (4.8) Activity Diagram for Statistics & Reports

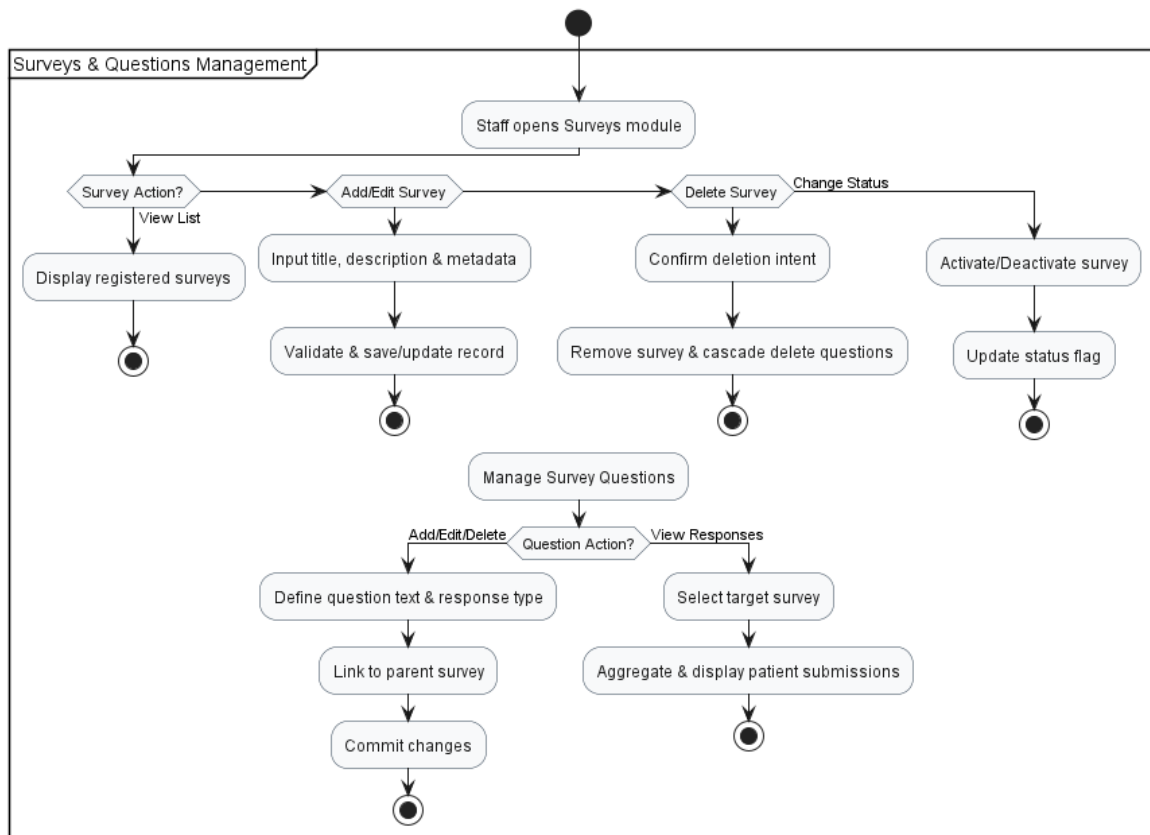


Figure (4.9) Activity Diagram for Surveys & Questions Management

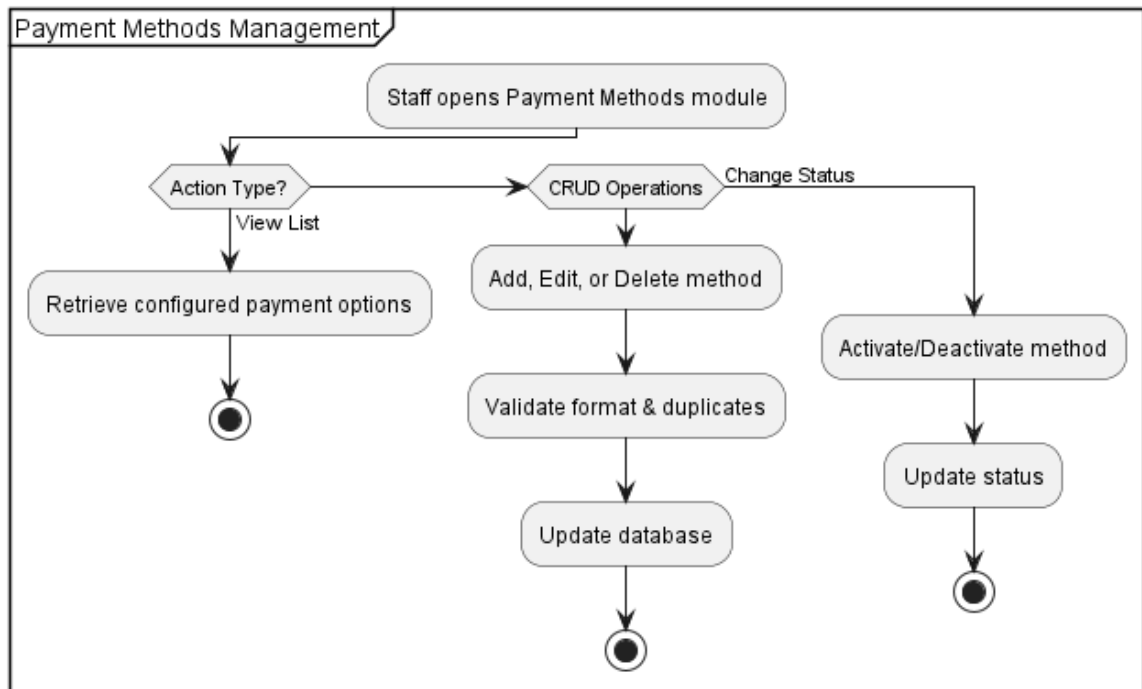


Figure (4.10) Activity Diagram for Payment Methods Management

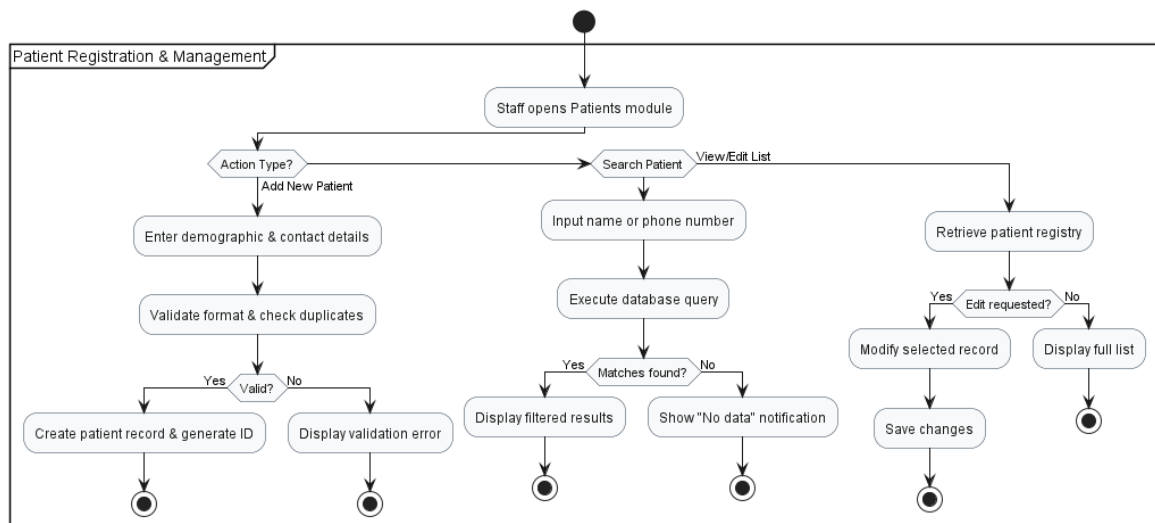


Figure (4.11) Activity Diagram for Patients Management

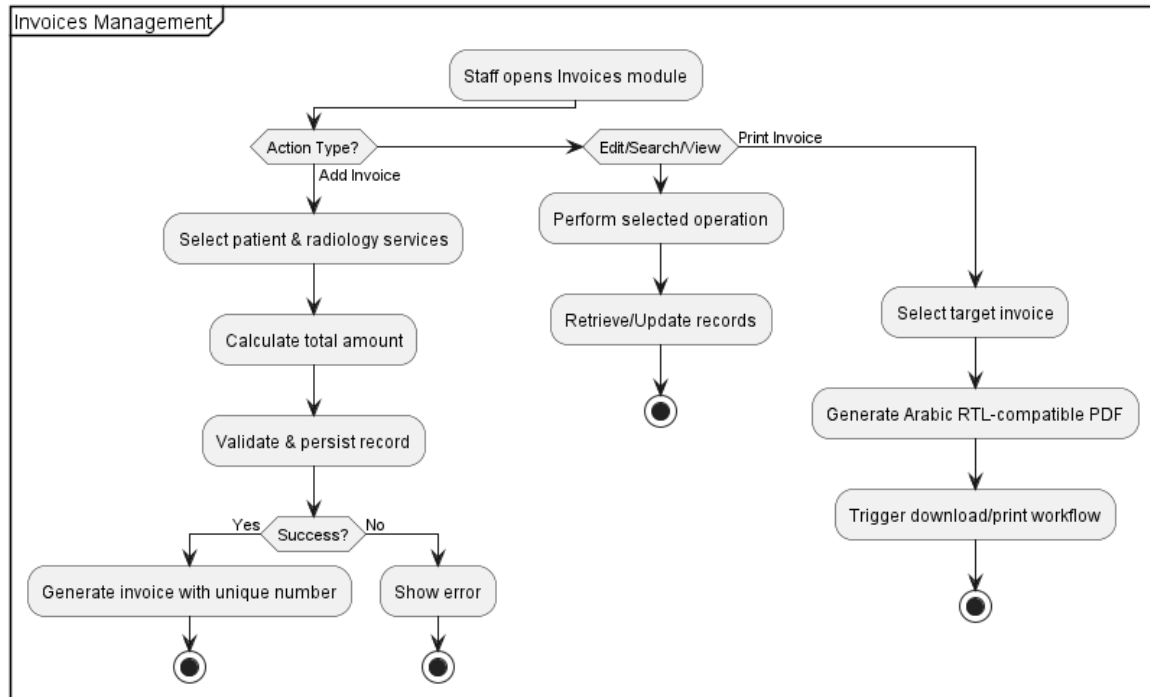


Figure (4.12) Activity Diagram for Invoices Management

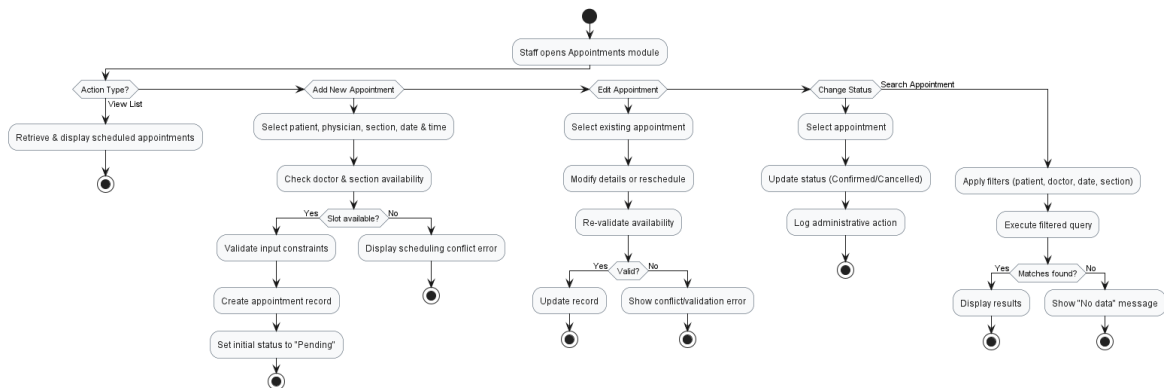


Figure (4.13) Activity Diagram for Appointments Scheduling

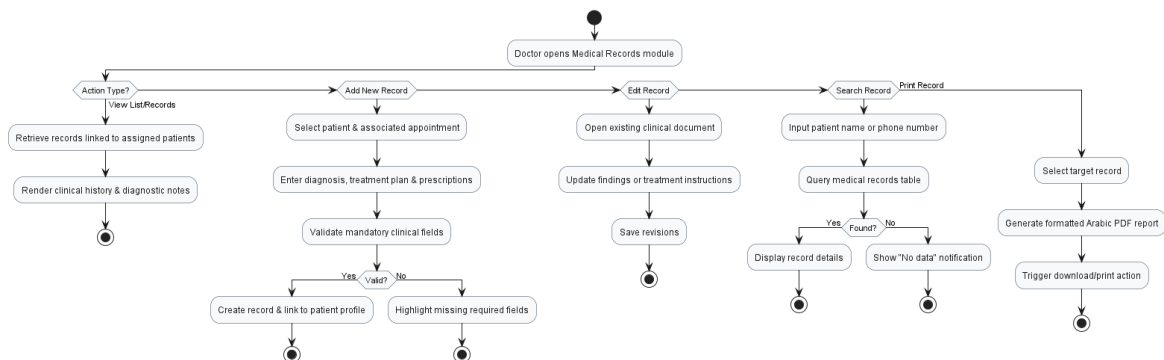


Figure (4.14) Activity Diagram for Medical Records Management

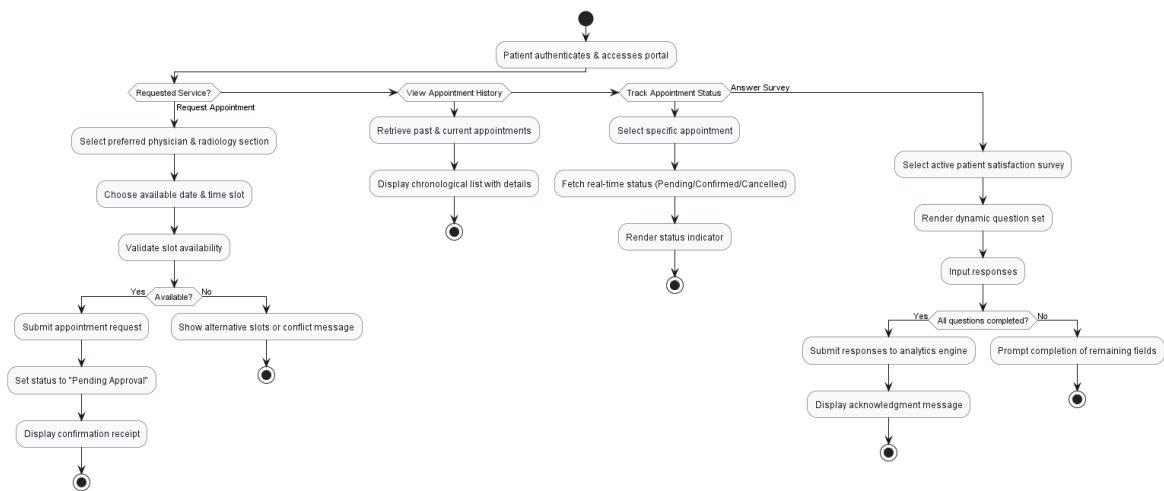


Figure (4.15) Activity Diagram for Patient self-service portal

4.7.4 Sequence Diagrams:

Sequence diagrams are defined as interactive diagrams that illustrate how a group of objects interact with each other and in what order. Sequence diagrams are particularly useful for depicting the dynamic behavior of a system and how different components cooperate to achieve a specific task or goal.

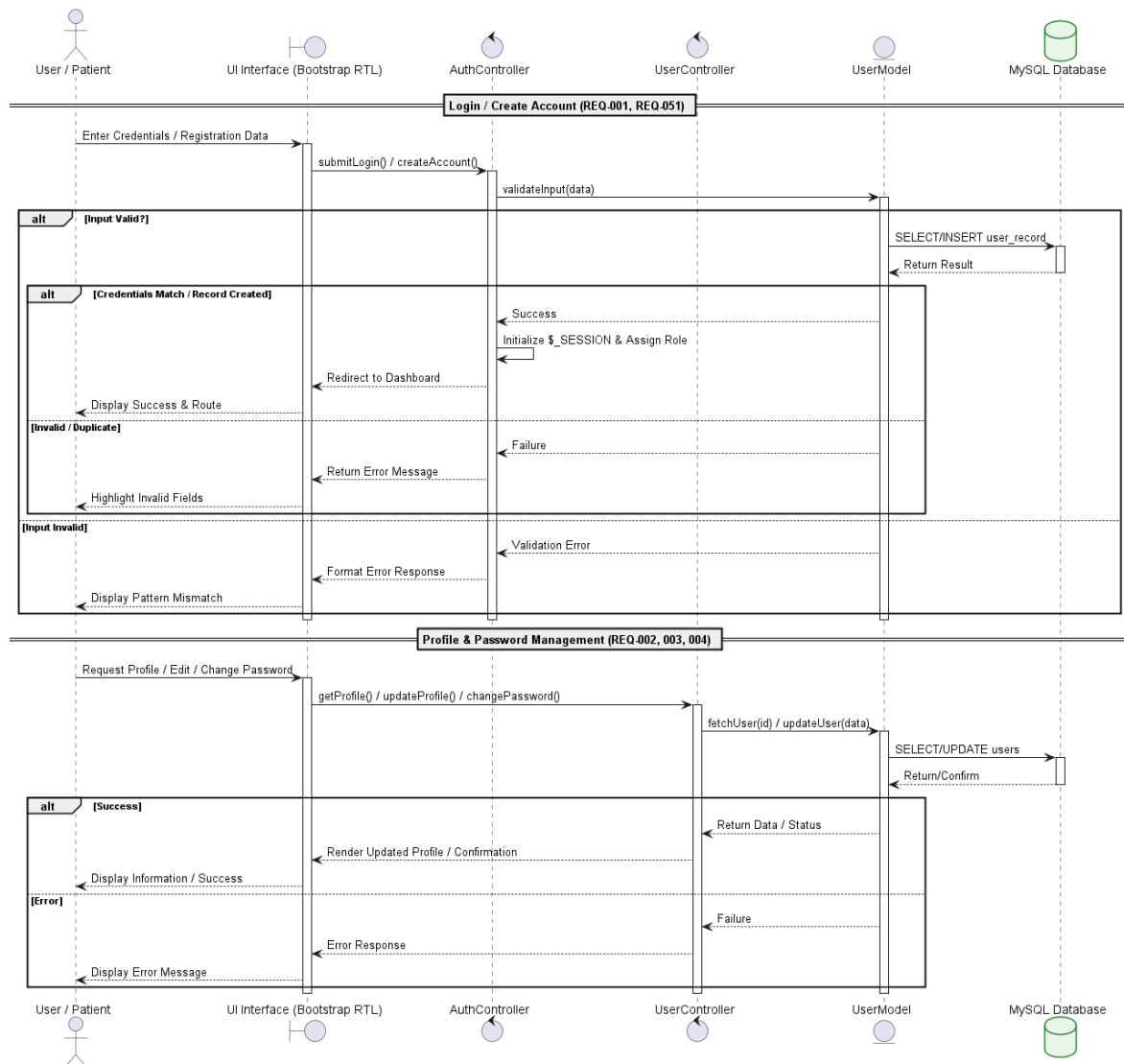


Figure (4.16) Sequence Diagram for Authentication & Account Management

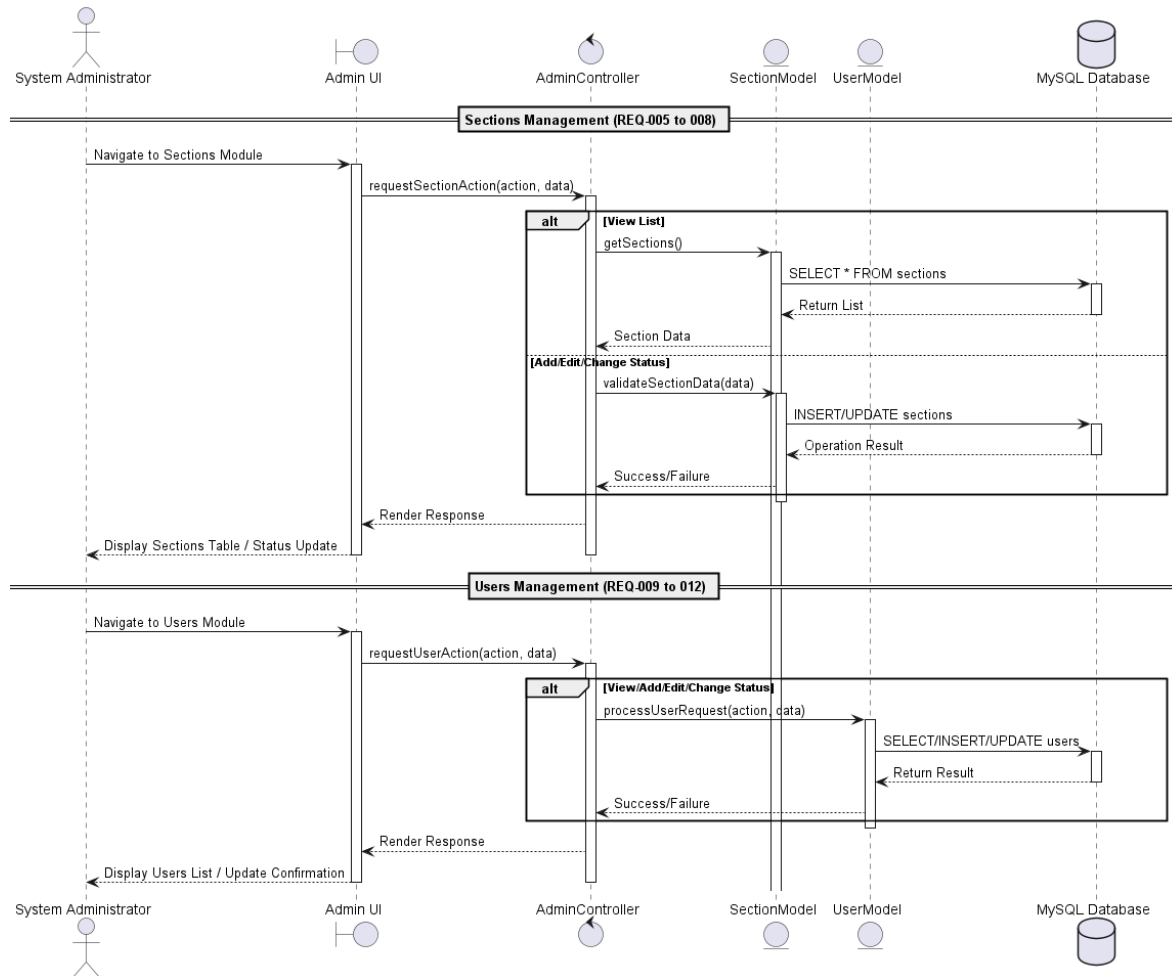


Figure (4.17) Sequence Diagram for Sections & Users Management

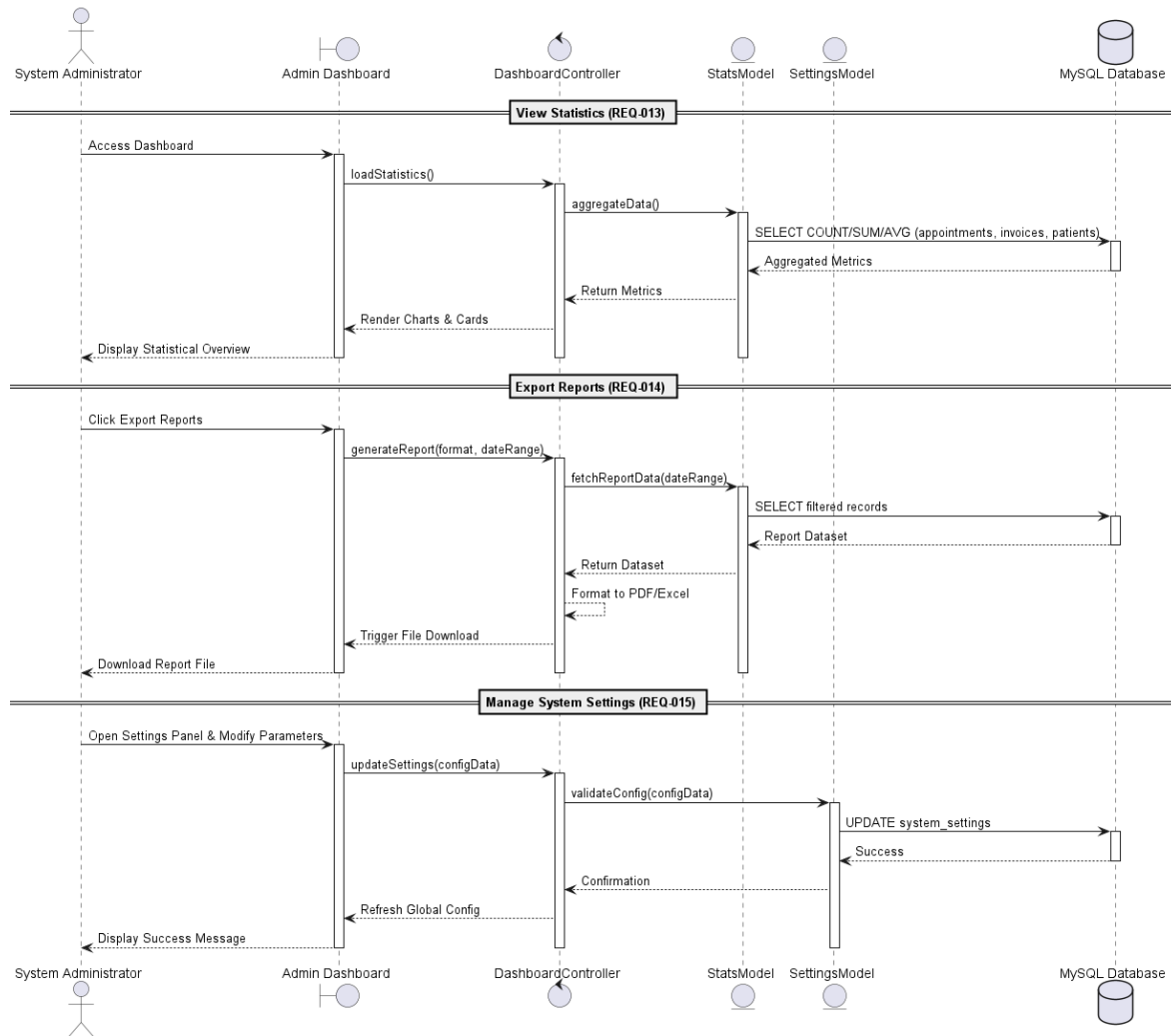


Figure (4.18) Sequence Diagram for System Configuration & Reporting

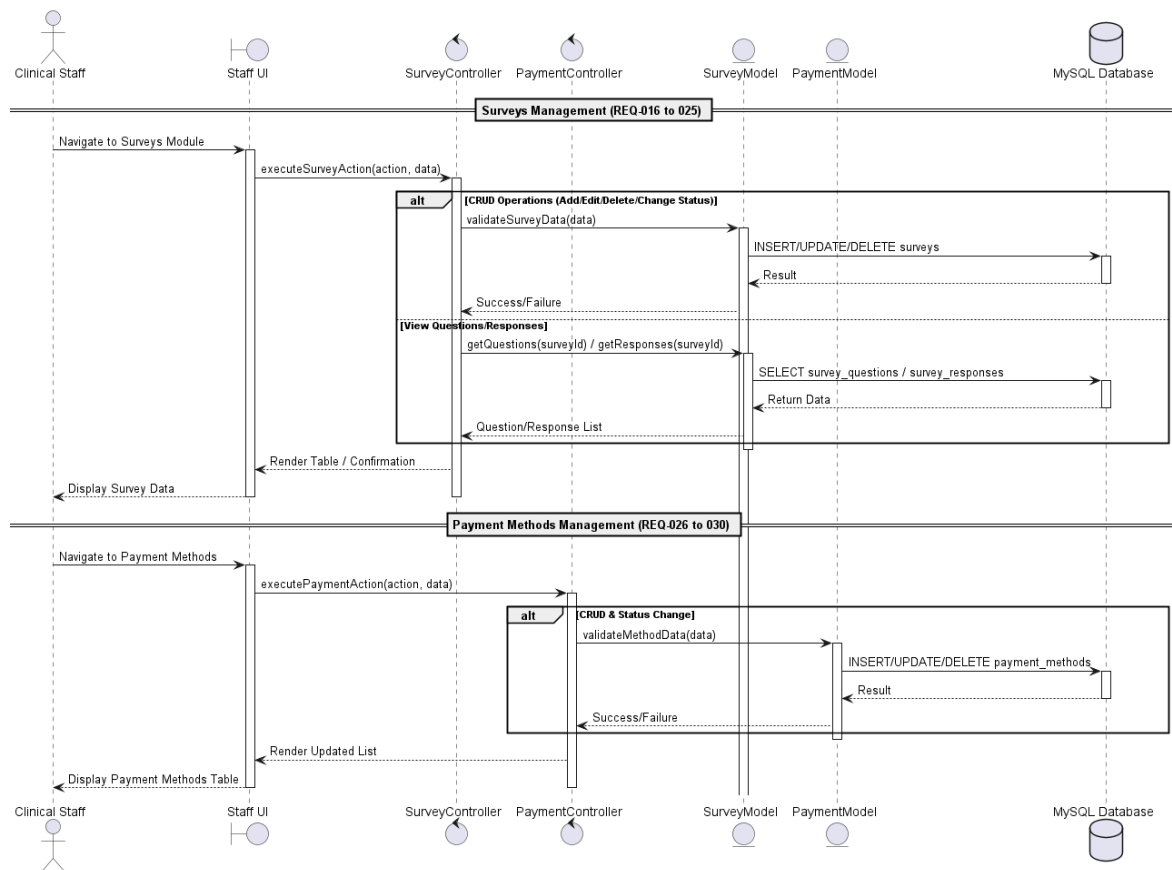


Figure (4.19) Sequence Diagram for Surveys & Payment Methods Management

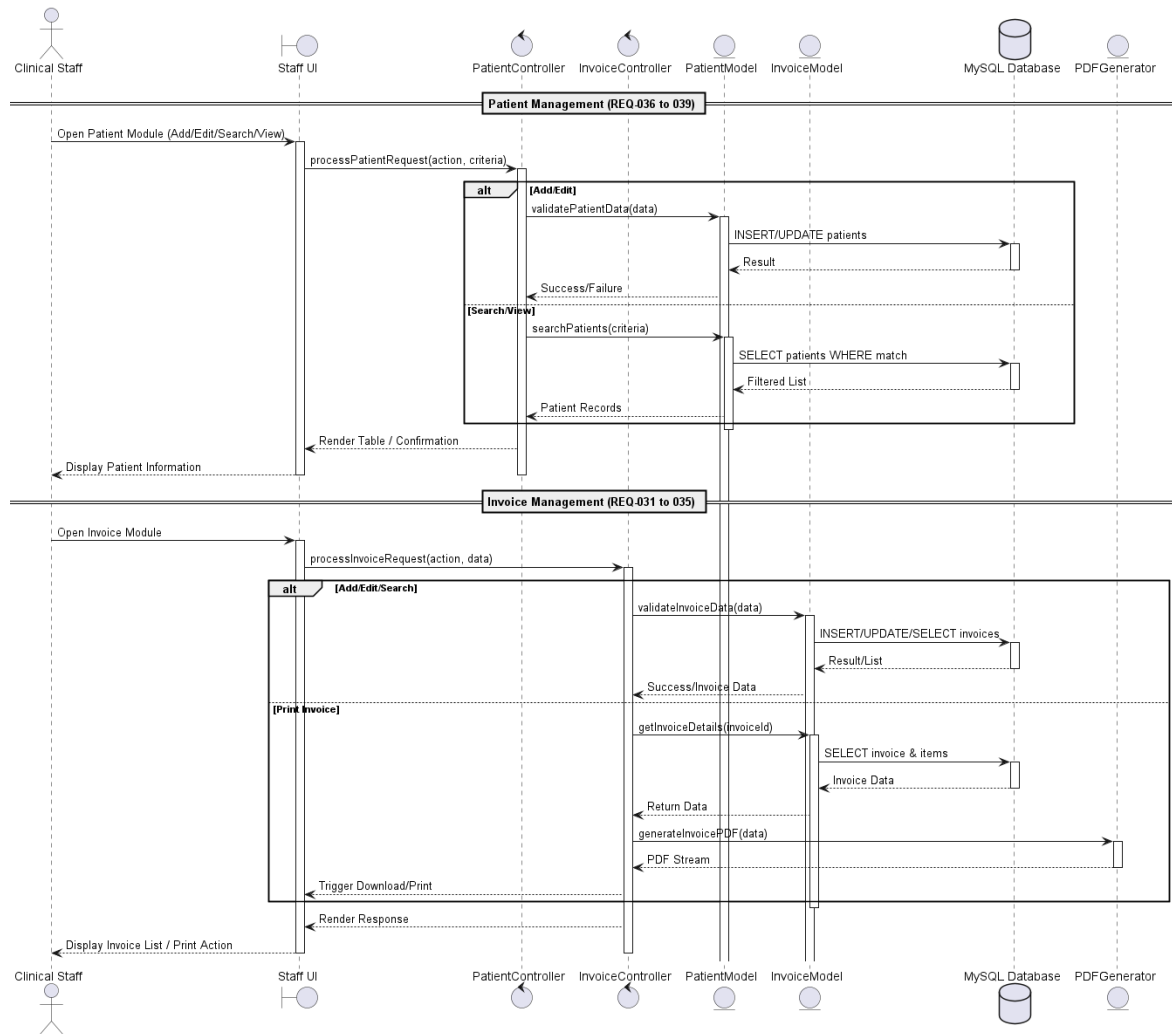


Figure (4.20) Sequence Diagram for Invoices & Patients Management

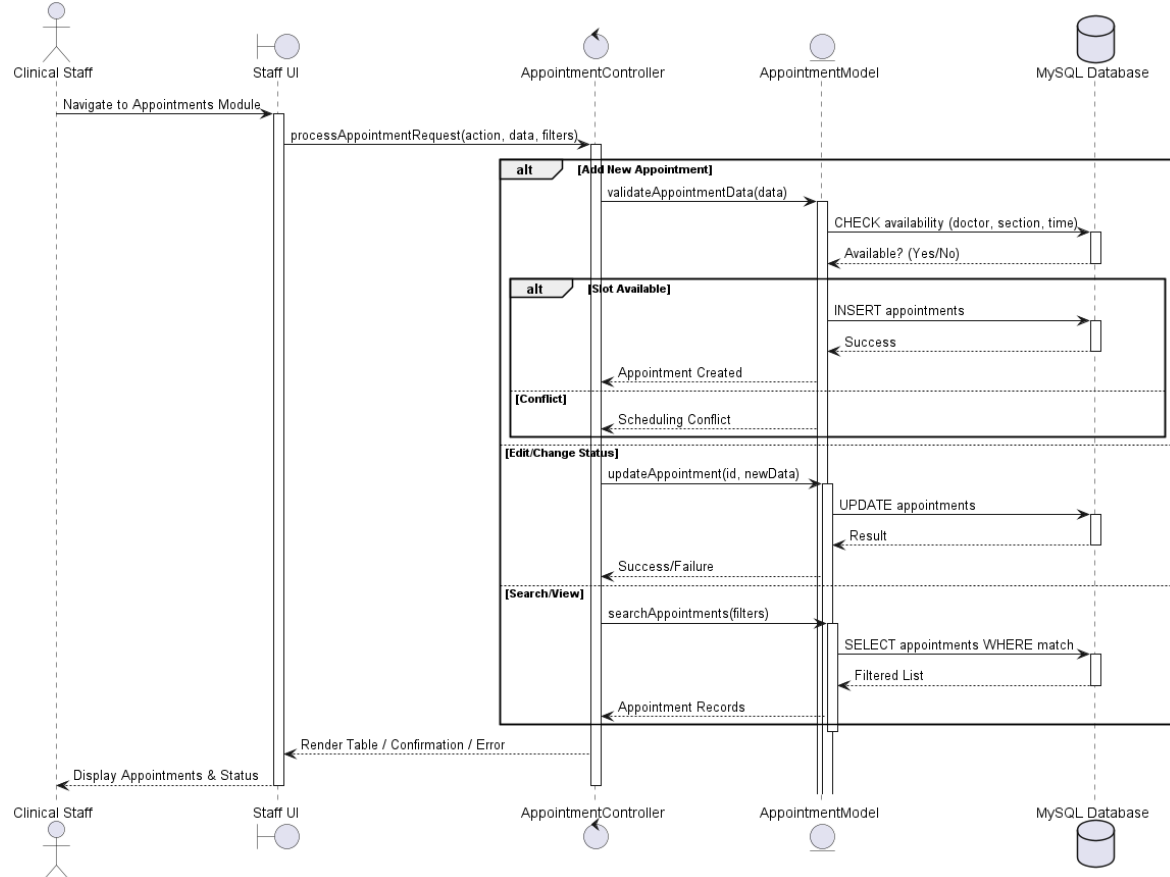


Figure (4.21) Sequence Diagram for Appointments Scheduling

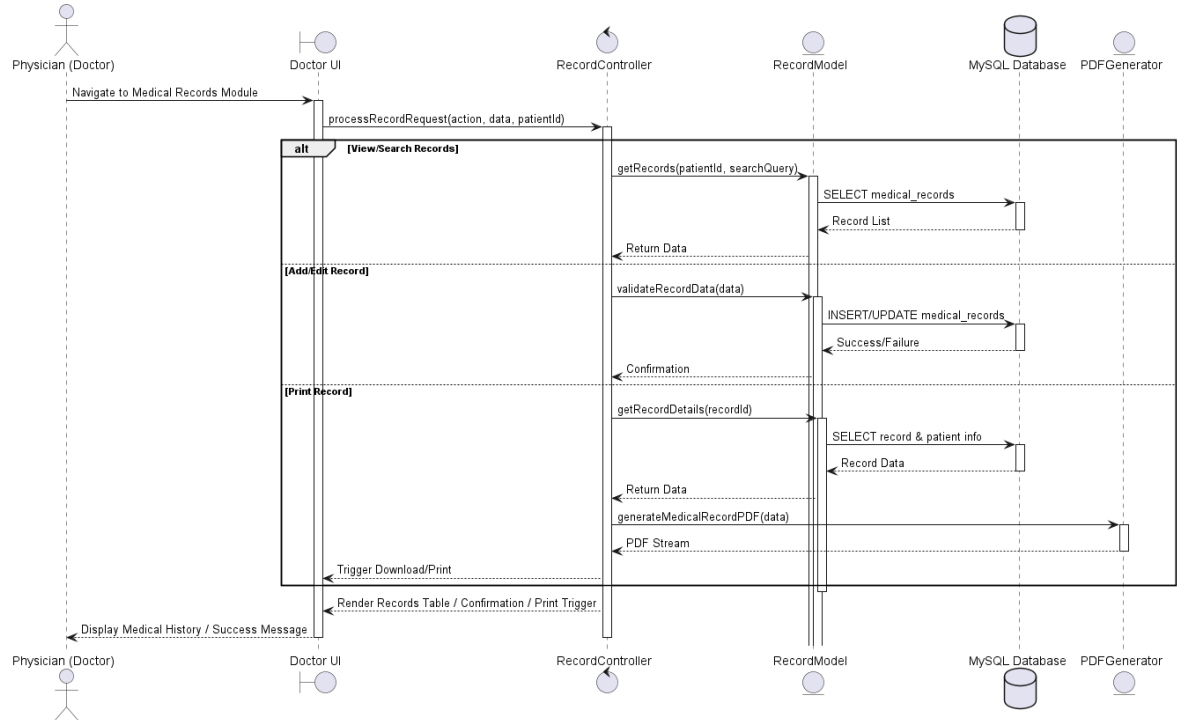


Figure (4.22) Sequence Diagram for Medical Records Management

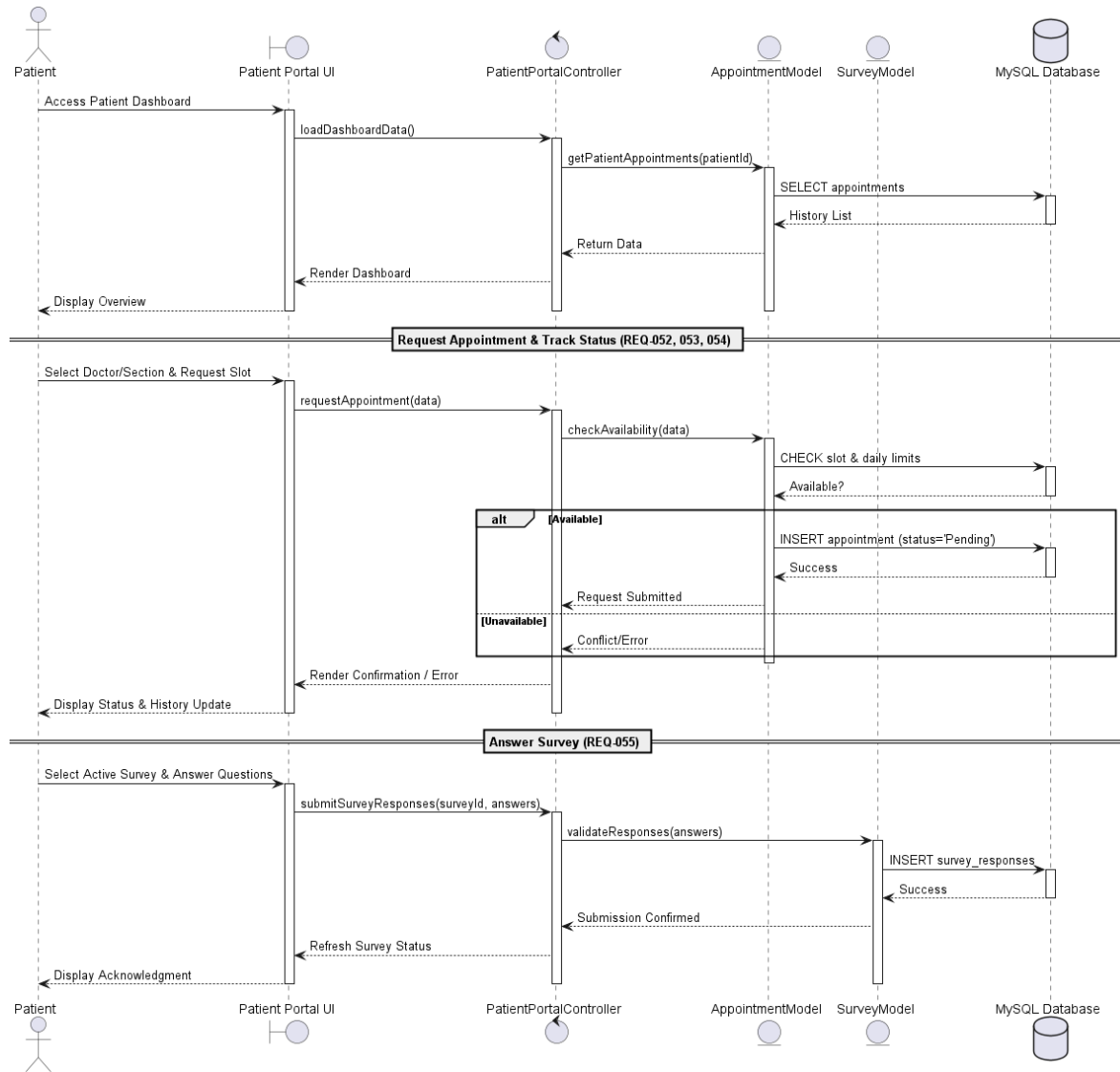


Figure (4.23) Sequence Diagram for Patient self-service portal

4.8 Software Requirements Specifications (SRS):

A set of functional requirements is established, categorized based on the functions of the website users. These requirements are as follows:

- **Patient Functional Requirements:**

- Create an account
- Log in
- Edit personal information
- View profile
- Change password
- Manage patient appointments by performing actions such as (book a new appointment, view appointment history, track appointment status)
- View patient medical record
- Print medical record
- View survey list
- View survey questions
- Respond to a survey

- **Doctor Functional Requirements:**

- Log in
- Edit personal information
- View profile
- Change password
- View Doctor appointment list
- View patient list
- Manage the medical records of patients registered with the Doctor (add a medical record, edit a medical record, print a medical record, view a list of medical records)
-

- **Staff Functional Requirements:**

- Log in
- Edit personal information
- View profile
- Change password
- Manage patients by performing actions such as (add, edit, search for a patient, view patient list)
- Manage appointments by performing actions such as (Add, edit, search for appointment, change appointment status, view appointment list)
- Manage surveys by performing the operations (add, edit, delete, change status, view survey list)
- Manage survey questions by performing the operations (add, edit, delete, view survey questions)
- Manage payment methods by performing the operations (add, edit, delete, change status, view payment method list)
- Manage invoices by performing the operations (add, edit, print invoice, search for invoice, view invoice list)

- **Admin Functional Requirements:**

- Login
- Edit personal data
- View profile
- Change password
- Manage users by performing the operations (add, edit, change account status, view user list)
- Manage sections by performing the operations (add, edit, change status, view section list)
- View comprehensive statistics on center operations with AI predictions

- Export reports
- View survey list
- View survey responses
- Manage system settings

4.9 Requirements analysis and documentation:

4.9.1 Requirements Traceability Matrix (RTM):

The RTM establishes bidirectional traceability between requirements, design artifacts, and test cases. It ensures complete requirement coverage, facilitates impact analysis, and validates academic compliance.

Table (4.) RTM

ID	Title	Associated Dependency	SRS Section	High Level Design	Detailed Design	System Test
REQ-001	Log in	Login Function	Functional Requirements	activity-1	sequence-1	TC-1, TC2
REQ-002	Edit personal account	Edit Function	Functional Requirements	activity-1	sequence-1	TC-3
REQ-003	View Profile	View Function	Functional Requirements	activity-1	sequence-1	TC-4
REQ-004	Change password	View Function	Functional Requirements	activity-1	sequence-1	TC-5
REQ-005	View section list	View Function	Functional Requirements	activity-2	sequence-2	TC-6
REQ-006	Add section	View Function	Functional Requirements	activity-2	sequence-2	TC-7
REQ-007	Edit section	View Function	Functional Requirements	activity-2	sequence-2	TC-8
REQ-008	Change section status	View Function	Functional Requirements	activity-2	sequence-2	TC-9
REQ-009	Add user	View Function	Functional Requirements	activity-3	sequence-3	TC-10
REQ-010	Edit user data	View Function	Functional Requirements	activity-3	sequence-3	TC-11

REQ-011	Change user account status	View Function	Functional Requirements	activity-3	sequence-3	TC-12
REQ-012	View user list	Export Function	Functional Requirements	activity-3	sequence-3	TC-13
REQ-013	View statistics	Edit Function	Functional Requirements	activity-5	sequence-5	TC-14
REQ-014	Export reports	Edit Function	Functional Requirements	activity-5	sequence-5	TC-15
REQ-015	Manage system settings	Register Function	Functional Requirements	activity-4	sequence-4	TC-16
REQ-016	View survey list	Add Function	Functional Requirements	activity-6	sequence-6	TC-17
REQ-017	Add survey	Edit Function	Functional Requirements	activity-6	sequence-6	TC-18
REQ-018	Edit survey	Delete Function	Functional Requirements	activity-6	sequence-6	TC-19
REQ-019	Change survey status	Add Function	Functional Requirements	activity-6	sequence-6	TC-20
REQ-020	Delete survey	Edit Function	Functional Requirements	activity-6	sequence-6	TC-21
REQ-021	View survey questions	Delete Function	Functional Requirements	activity-6	sequence-6	TC-22
REQ-022	Add survey question	Edit Function	Functional Requirements	activity-6	sequence-6	TC-23
REQ-023	Edit survey question	Edit Function	Functional Requirements	activity-6	sequence-6	TC-24
REQ-024	Delete survey question	Edit Function	Functional Requirements	activity-6	sequence-6	TC-25
REQ-025	View survey responses	Delete Function	Functional Requirements	activity-6	sequence-6	TC-26
REQ-026	View payment methods list	Edit Function	Functional Requirements	activity-7	sequence-7	TC-27
REQ-027	Add payment method	Edit Function	Functional Requirements	activity-7	sequence-7	TC-28
REQ-028	Edit payment method	Edit Function	Functional Requirements	activity-7	sequence-7	TC-29
REQ-029	Change payment method status	View Function	Functional Requirements	activity-7	sequence-7	TC-30

REQ-030	Delete payment method	Add Function	Functional Requirements	activity-7	sequence-7	TC-31
REQ-031	Add invoice	Edit Function	Functional Requirements	activity-9	sequence-9	TC-32
REQ-032	Edit invoice data	Delete Function	Functional Requirements	activity-9	sequence-9	TC-33
REQ-033	Print invoice	Edit Function	Functional Requirements	activity-9	sequence-9	TC-34
REQ-034	View invoice list	View Function	Functional Requirements	activity-9	sequence-9	TC-35
REQ-035	Search invoice	Edit Function	Functional Requirements	activity-9	sequence-9	TC-36
REQ-036	Add new patient	View Function	Functional Requirements	activity-8	sequence-8	TC-37
REQ-037	Edit patient data	تابع الإضافة	Functional Requirements	activity-8	sequence-8	TC-38
REQ-038	View patient list	View Function	Functional Requirements	activity-8	sequence-8	TC-39
REQ-039	Search patient	Add Function	Functional Requirements	activity-8	sequence-8	TC-40
REQ-040	Add new appointment	Edit Function	Functional Requirements	activity-10	sequence-10	TC-41
REQ-041	Edit appointment data	Delete Function	Functional Requirements	activity-10	sequence-10	TC-42
REQ-042	Change appointment status	View Function	Functional Requirements	activity-10	sequence-10	TC-43
REQ-043	Search appointment	View Function	Functional Requirements	activity-10	sequence-10	TC-44
REQ-044	View appointment list	Edit Function	Functional Requirements	activity-11	sequence-11	TC-45
REQ-045	View patient's medical record	View Function	Functional Requirements	activity-11	sequence-11	TC-46
REQ-046	View medical records list	Add Function	Functional Requirements	activity-11	sequence-11	TC-47
REQ-047	Add patient's medical record	View Function	Functional Requirements	activity-11	sequence-11	TC-48
REQ-048	Edit patient's medical record	View Function	Functional Requirements	activity-11	sequence-11	TC-49

REQ-049	Print patient's medical record	Add Function	Functional Requirements	activity-11	sequence-11	TC-50
REQ-050	Search medical record	View Function	Functional Requirements	activity-11	sequence-11	TC-51
REQ-051	Create an account	Add Function	Functional Requirements	activity-12	sequence-12	TC-52
REQ-052	Request an appointment	Edit Function	Functional Requirements	activity-12	sequence-12	TC-53
REQ-053	View appointment history	View Function	Functional Requirements	activity-12	sequence-12	TC-54
REQ-054	Track appointment status	View Function	Functional Requirements	activity-12	sequence-12	TC-54
REQ-055	Answer a survey	Add Function	Functional Requirements	activity-12	sequence-12	TC-55

4.9.2 Initial test cases:

A comprehensive suite of **55 system test cases** was developed to validate functional requirements under controlled conditions. Each test case includes: unique ID, description, pre-conditions, test steps, test data, expected results, and execution status.

Table (4.) System Tests

ID	Description	Pre-Conditions	Test Steps	Test Data	Expected Result	Status
TC-1	Log in with valid credentials	User account exists	1. Go to the login page 2. Enter valid email and password 3. Click Log in	Email: admin@admin.com Password: admin123	Login Successfully	done
TC-2	Log in with invalid credentials	User account exists	1. Go to the login page 2. Enter incorrect email and password 3. Click Log in	Email: ad@gmail.com Password: WrongPass	Error message displayed	done

TC-3	Edit personal account	User logged in	1. Go to personal account page 2. Edit username 3. Save changes	New Name: " صفا الزهري "	Updates Saved	done
TC-4	View personal account	User logged in	1. Go to personal account page	---	View Data	done
TC-5	Change password	User logged in	1. Go to change password page 2. Enter current password 3. Enter new password 4. Confirm new password 5. Click Save	Current: old123 New: new456	Password Changed Successfully	done
TC-6	View section list	User logged in	1. Go to Sections section 2. View the list	-	View Sections	done
TC-7	Add section	User logged in with admin rights	1. Go to Sections section 2. Click Add New 3. Enter section details 4. Click Save	Section Name: Cardiology	Section Added Successfully	done
TC-8	Edit section	User logged in, Section exists	1. Go to Sections section 2. Select section to edit 3. Modify details 4. Click Save	New Name: Cardiology Dept	Section Updated Successfully	done
TC-9	Change section status	User logged in, Section exists	1. Go to Sections section	Status: Active/Inactive	Status Changed Successfully	done

			2. Select section 3. Change status dropdown 4. Click Save			
TC -10	Add user	User logged in with admin rights	1. Go to Users section 2. Click Add New User 3. Enter user details 4. Click Save	Name: Ahmad fares Email: fatema@test.com Role: Doctor	User Added Successfully	done
TC -11	Edit user data	User logged in, User exists	1. Go to Users section 2. Select user to edit 3. Modify user details 4. Click Save	New Phone: +1234567890	User Data Updated Successfully	done
TC -12	Change user account status	User logged in, User exists	1. Go to Users section 2. Select user 3. Change account status 4. Click Save	Status: Active/Suspended	Account Status Changed	done
TC -13	View user list	User logged in	1. Go to Users section 2. View the list	---	View Users	done
TC -14	View statistics	User logged in	1. Go to Statistics section 2. View dashboard	---	Statistics Displayed	done
TC -15	Export reports	User logged in	1. Go to Reports section 2. Select report type 3. Click Export	Format: PDF/Excel	Report Exported Successfully	Done
TC -16	Manage system settings	User logged in as admin	1. Go to Settings section	Setting: System Name	Settings Saved Successfully	Done

			2. Modify settings 3. Click Save			
TC -17	View survey list	User logged in	1. Go to Surveys section 2. View the list	---	View Surveys	Done
TC -18	Add survey	User logged in with admin rights	1. Go to Surveys section 2. Click Add New Survey 3. Enter survey details 4. Click Save	Title: Patient Satisfaction	Survey Added Successfully	done
TC -19	Edit survey	User logged in, Survey exists	1. Go to Surveys section 2. Select survey to edit 3. Modify details 4. Click Save	New Title: Patient Feedback 2026	Survey Updated Successfully	done
TC -20	Change survey status	User logged in, Survey exists	1. Go to Surveys section 2. Select survey 3. Change status 4. Click Save	Status: Active/Closed	Status Changed Successfully	Done
TC -21	Delete survey	User logged in, Survey exists	1. Go to Surveys section 2. Select survey 3. Click Delete 4. Confirm deletion	---	Survey Deleted Successfully	Done
TC -22	View survey questions	User logged in, Survey exists	1. Go to Surveys section 2. Select survey 3. View questions	---	View Questions	done

TC-23	Add survey question	User logged in, Survey exists	1. Go to survey questions 2. Click Add Question 3. Enter question details 4. Click Save	Question: How satisfied are you? Type: Multiple Choice	Question Added Successfully	done
TC-24	Edit survey question	User logged in, Question exists	1. Go to survey questions 2. Select question to edit 3. Modify details 4. Click Save	New Text: Rate your satisfaction	Question Updated Successfully	done
TC-25	Delete survey question	User logged in, Question exists	1. Go to survey questions 2. Select question 3. Click Delete 4. Confirm	---	Question Deleted Successfully	done
TC-26	View survey responses	User logged in, Survey exists	1. Go to Surveys section 2. Select survey 3. View responses	---	View Responses	done
TC-27	View payment methods list	User logged in	1. Go to Payment Methods section 2. View the list	---	View Payment Methods	done
TC-28	Add payment method	User logged in with admin rights	1. Go to Payment Methods 2. Click Add New 3. Enter method details 4. Click Save	Method: Credit Card	Payment Method Added	done
TC-29	Edit payment method	User logged in, Method exists	1. Go to Payment Methods	New Name: Visa Card	Payment Method Updated	done

			2. Select method to edit 3. Modify details 4. Click Save			
TC -30	Change payment method status	User logged in, Method exists	1. Go to Payment Methods 2. Select method 3. Change status 4. Click Save	Status: Active/Inactive	Status Changed Successfully	done
TC -31	Delete payment method	User logged in, Method exists	1. Go to Payment Methods 2. Select method 3. Click Delete 4. Confirm	---	Payment Method Deleted	done
TC -32	Add invoice	User logged in	1. Go to Invoices section 2. Click Add New Invoice 3. Enter invoice details 4. Click Save	Patient: Sara Amount: \$100	Invoice Added Successfully	done
TC -33	Edit invoice data	User logged in, Invoice exists	1. Go to Invoices section 2. Select invoice to edit 3. Modify details 4. Click Save	New Amount: \$150	Invoice Updated Successfully	done
TC -34	Print invoice	User logged in, Invoice exists	1. Go to Invoices section 2. Select invoice 3. Click Print	-	Invoice Printed Successfully	done
TC -35	View invoice list	User logged in	1. Go to Invoices section 2. View the list	---	View Invoices	Done

TC-36	Search invoice	User logged in	1. Go to Invoices section 2. Enter search criteria 3. Click Search	Invoice Number: INV-001	Search Results Displayed	Done
TC-37	Add new patient	User logged in	1. Go to Patients section 2. Click Add New Patient 3. Enter patient details 4. Click Save	Name: Jane Smith DOB: 1990-05-15	Patient Added Successfully	Done
TC-38	Edit patient data	User logged in, Patient exists	1. Go to Patients section 2. Select patient to edit 3. Modify details 4. Click Save	New Phone: +9876543210	Patient Data Updated	Done
TC-39	View patient list	User logged in	1. Go to Patients section 2. View the list	---	View Patients	Done
TC-40	Search patient	User logged in	1. Go to Patients section 2. Enter search criteria 3. Click Search	Name: Fatima	Search Results Displayed	Done
TC-41	Add new appointment	User logged in	1. Go to Appointments section 2. Click Add New 3. Enter appointment details 4. Click Save	Patient: Sara Khaled Date: 2026-04-15 Time: 10:00 AM	Appointment Added Successfully	Done
TC-42	Edit appointment data	User logged in,	1. Go to Appointments section	New Time: 11:00 AM	Appointment Updated Successfully	Done

		Appointment exists	2. Select appointment to edit 3. Modify details 4. Click Save			
TC-43	Change appointment status	User logged in, Appointment exists	1. Go to Appointments section 2. Select appointment 3. Change status 4. Click Save	Status: Confirmed/Canceled	Status Changed Successfully	Done
TC-44	Search appointment	User logged in	1. Go to Appointments section 2. Enter search criteria 3. Click Search	Date: 2026-04-15	Search Results Displayed	Done
TC-45	View appointment list	User logged in	1. Go to Appointments section 2. View the list	---	View Appointments	Done
TC-46	View patient's medical record	User logged in, Patient exists	1. Go to Patients section 2. Select patient 3. View medical record	---	Medical Record Displayed	Done
TC-47	View medical records list	User logged in	1. Go to Medical Records section 2. View the list	---	View Medical Records	Done
TC-48	Add patient's medical record	User logged in, Patient exists	1. Go to patient's profile 2. Click Add Medical Record 3. Enter record	Diagnosis: Flu Treatment: Rest	Medical Record Added	Done

			details 4. Click Save			
TC -49	Edit patient's medical record	User logged in, Record exists	1. Go to medical records 2. Select record to edit 3. Modify details 4. Click Save	New Diagnosis: Viral Infection	Medical Record Updated	Done
TC -50	Print patient's medical record	User logged in, Record exists	1. Go to medical records 2. Select record 3. Click Print	---	Medical Record Printed	Done
TC -51	Search medical record	User logged in	1. Go to Medical Records section 2. Enter search criteria 3. Click Search	Patient Name: Sara	Search Results Displayed	Done
TC -52	Request an appointment	User logged in	1. Go to Appointments section 2. Click Request Appointment 3. Enter details 4. Click Submit	Preferred Date: 2026-04-20	Appointment Requested Successfully	Done
TC -53	View appointment history	User logged in	1. Go to Appointments section 2. Click History 3. View past appointments	---	Appointment History Displayed	Done
TC -54	Track appointment status	User logged in	1. Go to Appointments section 2. Select appointment	---	Status Displayed	Done

			3. View status			
TC-55	Answer a survey	User logged in, Survey available	1. Go to Surveys section 2. Select available survey 3. Answer all questions 4. Click Submit	Answers: Selected options	Survey Submitted Successfully	Done

Chapter 5

System Architecture and High-Level Design

5.1 High Level Design:

The architectural foundation of the Radiology Lab Management System (RLMS) is engineered around a layered, modular, and role-segregated design philosophy. The high-level architecture abstracts complex clinical and administrative workflows into manageable, interoperable components while ensuring scalability, maintainability, and strict adherence to software engineering best practices. The system employs a client-server paradigm with a clear separation between presentation, business logic, data access, and persistence layers. This structural decoupling facilitates parallel development, simplifies debugging, and enables future feature expansion without compromising system stability. The architecture is explicitly optimized for local/intranet deployment within medium-scale radiology facilities, prioritizing low-latency request processing, deterministic state management, and culturally aligned Arabic RTL interface rendering.

5.1.1 System Block Diagram:

The system block diagram illustrates the macro-level interaction between core architectural components and data flow pathways. As depicted in **Figure 5.1**, the architecture is segmented into four primary blocks:

1. **Presentation Layer (Client-Side):** Comprises responsive HTML5 templates, Bootstrap 5 RTL components, custom CSS, and client-side JavaScript. Handles user interaction, form validation, and dynamic DOM updates.
2. **Application Layer (Server-Side):** Contains PHP controllers, routing logic, session management, and business rule enforcement. Orchestrates request handling, role verification, and workflow coordination.

3. **Data Access Layer (DAL):** Implements PDO-based abstraction for secure database communication. Manages query execution, result mapping, transaction handling, and error propagation.
4. **Persistence & Storage Layer:** Encompasses the MySQL relational database, session storage, file system directories (uploads/), and configuration repositories (includes/config.php).

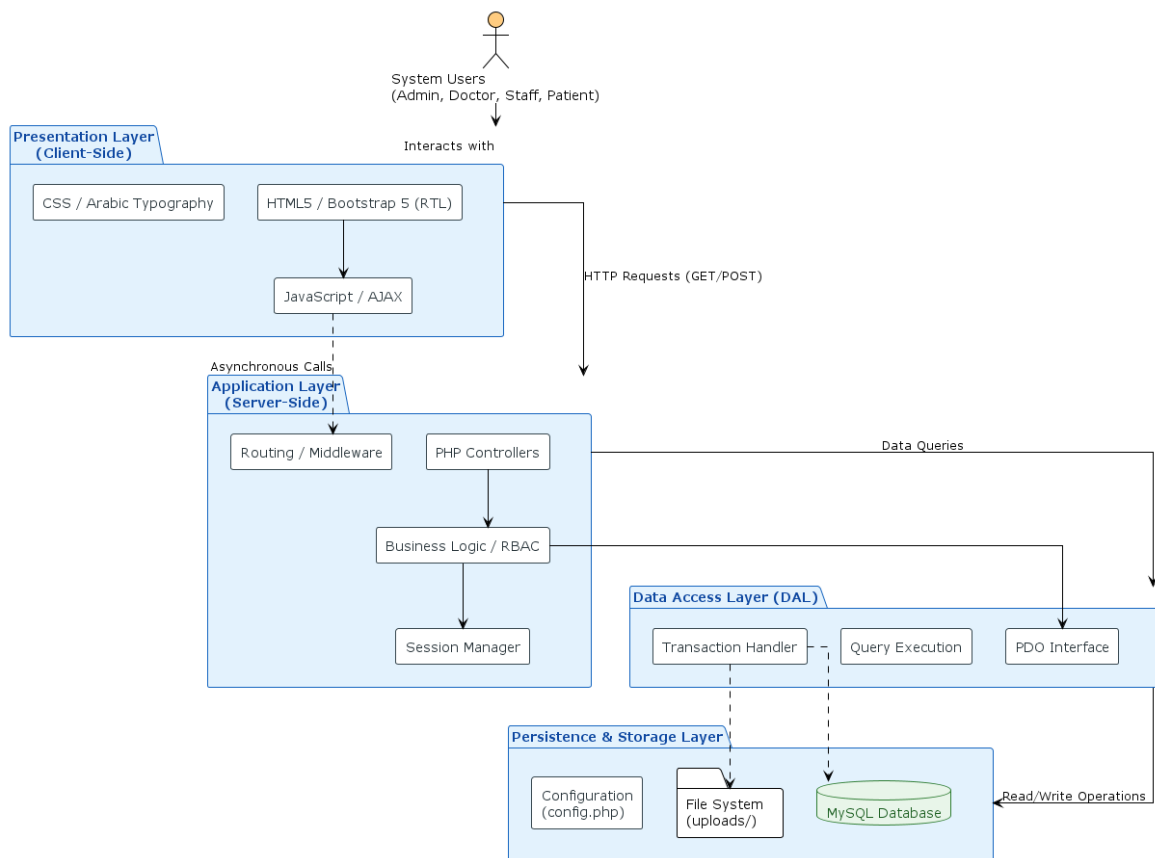


Figure (5.1) System Block Diagram

5.1.2 User Architectural Design Pattern (MVC):

The system adopts a lightweight adaptation of the **Model-View-Controller (MVC)** architectural pattern, tailored to the constraints and objectives of academic web development. The pattern enforces separation of concerns, ensuring that each component maintains a distinct responsibility:

- **Model:** Represents the data structure and business logic. Implemented through PHP classes and database interaction scripts that handle CRUD operations, relational joins, aggregation queries, and validation rules. The model layer operates independently of the presentation, ensuring data integrity and reusability.
- **View:** Comprises the user interface templates rendered via Bootstrap 5 with native RTL support. Views are strictly passive; they receive processed data from controllers and render HTML/CSS/JavaScript without executing business logic or direct database queries.
- **Controller:** Acts as the intermediary between models and views. Controllers parse HTTP requests, invoke appropriate model methods, manage session state, enforce role-based routing, and pass formatted data to the corresponding view templates.

This MVC-inspired structure prevents tight coupling, reduces code duplication, and simplifies maintenance. By isolating presentation markup from backend logic, the system achieves high testability and aligns with modern web development standards while remaining compatible with traditional server-side rendering paradigms.

5.1.3 System Architecture Design:

The RLMS architecture follows a modular directory-based deployment model that maps directly to stakeholder roles and functional domains. The root directory is partitioned into role-specific controllers (admin/, doctor/, staff/, patient/), shared utilities (includes/, libs/), static assets (assets/), and secure storage (uploads/). This physical segregation complements the logical MVC pattern by enforcing access boundaries at the filesystem and routing levels.

The request lifecycle operates as follows:

1. Client submits an HTTP request to a role-specific endpoint.

2. The entry script (index.php or role controller) initializes session validation and RBAC middleware.
3. Authorized requests are routed to the corresponding controller, which invokes DAL methods.
4. Retrieved data is processed, formatted, and passed to the view layer.
5. The rendered RTL interface is returned to the client, with state persisted via `$_SESSION`.

The architecture deliberately avoids microservices or containerized deployments to maintain academic reproducibility, reduce infrastructure overhead, and ensure deterministic behavior on standard LAMP/WAMP stacks. All inter-component communication occurs synchronously within a single runtime environment, optimizing performance for concurrent clinical and administrative operations.

5.2 Database Design:

The relational database serves as the central repository for all clinical, administrative, and system configuration data. Designed using MySQL 8.0 with the InnoDB storage engine, the schema enforces ACID compliance, referential integrity, and normalized data structures to eliminate redundancy and optimize query performance.

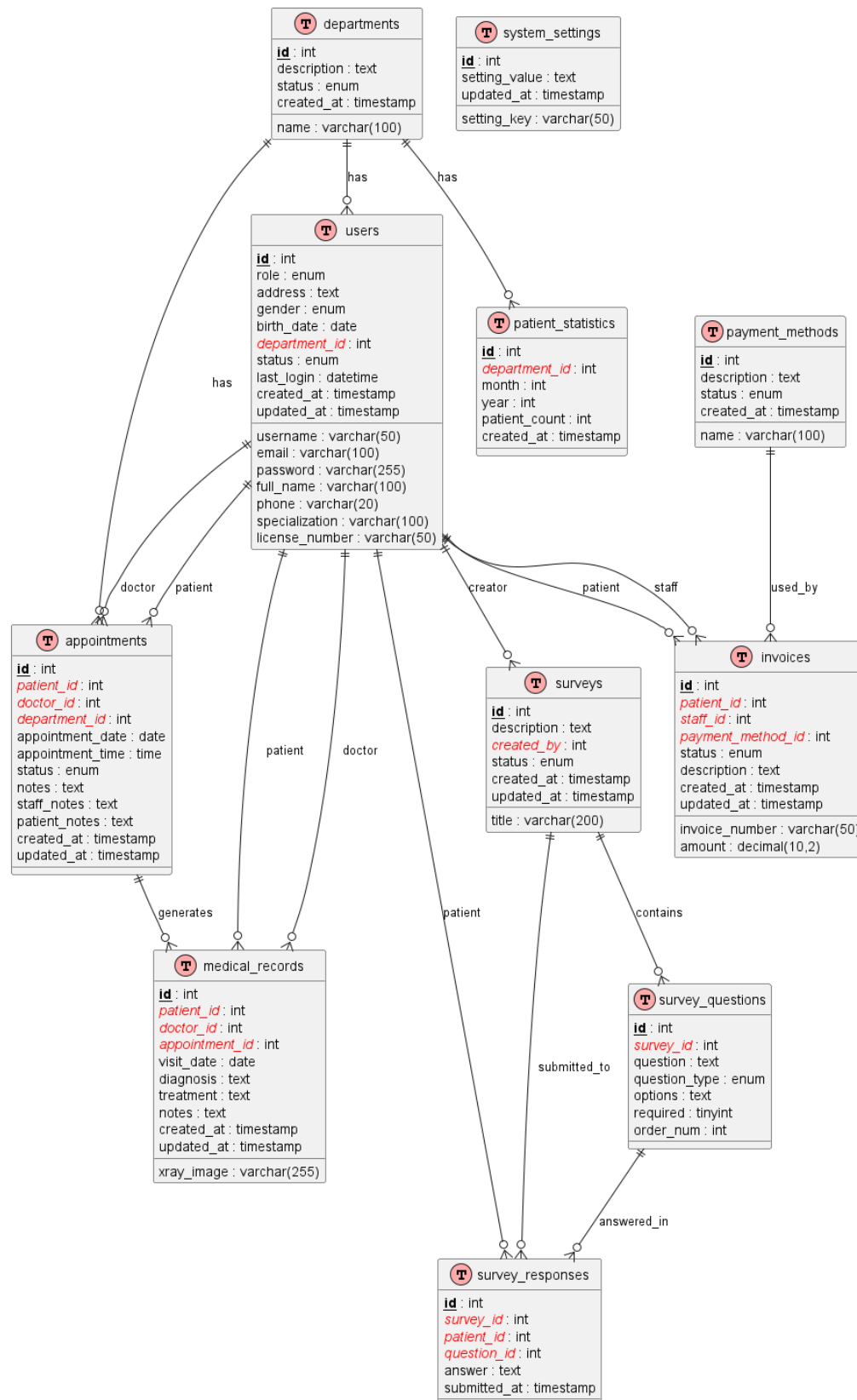
5.2.1 Entity-Relationship Diagram (ERD):

The ERD formally represents the structural relationships between core system entities. As illustrated in Figure 5.2, the database encompasses the following primary entities:

- **Users:** Stores authentication credentials, role assignments, and profile metadata.

- **Patients:** Contains demographic information, contact details, and medical history references.
- **Appointments:** Tracks scheduled visits, associated doctors, sections, timestamps, and status flags.
- **Medical_Records:** Documents clinical findings, diagnoses, treatments, and physician annotations.
- **Invoices & Payments:** Manages billing records, service charges, payment methods, and transaction statuses.
- **Surveys & Responses:** Structures patient feedback questionnaires, question banks, and submitted answers.
- **Sections & System_Settings:** Defines radiology departments, operational parameters, and global configuration values.

Relationships are predominantly one-to-many (1:N), with junction tables employed where many-to-many (M:N) associations occur (e.g., survey responses linked to patients and questions). Cardinality constraints and foreign key dependencies are explicitly modeled to enforce data consistency.



5.2.2 Relational Schema:

The logical schema is normalized to the Third Normal Form (3NF) to eliminate insertion, update, and deletion anomalies. Representative table structures are summarized below:

Table (5.1) Relational Schema

Table Name	Primary Key	Foreign Keys	Key Attributes	Normalization Level
users	user_id	role_id → roles.role_id	username, email, password_hash, status, created_at	3NF
patients	patient_id	—	full_name, phone, dob, gender, address, registration_date	3NF
appointments	appointment_id	patient_id, doctor_id, section_id	appointment_date, time_slot, status, notes	3NF
medical_records	record_id	patient_id, doctor_id, appointment_id	diagnosis, treatment_plan, prescription, record_date, pdf_path	3NF
invoices	invoice_id	patient_id, payment_method_id	invoice_number, total_amount, issue_date, status, pdf_path	3NF
surveys	survey_id	—	title, description, status, created_by, created_at	3NF
survey_questions	question_id	survey_id	question_text, question_type, order_index	3NF

survey_responses	response_id	survey_id, patient_id, question_id	answer_value, submitted_at	3NF
sections	section_id	—	section_name, description, status, daily_capacity	3NF
system_settings	setting_id	—	setting_key, setting_value, description, updated_at	3NF

5.2.3 Constraints and Keys:

Data integrity is enforced through a comprehensive constraint framework:

- **Primary Keys (PK):** Unique identifiers for each entity, implemented as AUTO_INCREMENT integers or UUIDs where applicable.
- **Foreign Keys (FK):** Explicit referential constraints linking child tables to parent entities. ON DELETE CASCADE is applied where dependent records must be removed automatically (e.g., survey questions deleted with parent survey). ON UPDATE CASCADE ensures synchronization of modified identifiers.
- **UNIQUE Constraints:** Applied to email, username, and invoice_number to prevent duplicate registrations and billing conflicts.
- **NOT NULL & DEFAULT:** Mandatory fields (e.g., password_hash, status) enforce complete record creation. Default values (e.g., status = 'pending', created_at = CURRENT_TIMESTAMP) automate state initialization.
- **CHECK Constraints:** Validate logical boundaries (e.g., appointment_date >= CURRENT_DATE, total_amount >= 0).

- **Indexing:** Strategic B-Tree indexes on frequently queried columns (patient_id, appointment_date, status) optimize search performance and reduce full-table scans.

5.3 External Interface Design (APIs):

The current iteration of the RLMS operates as a self-contained monolithic application without dependency on external RESTful or GraphQL APIs. Internal module communication is facilitated through structured PHP routing, shared library inclusion (libs/), and standardized data exchange formats (associative arrays, JSON payloads for AJAX requests).

The interface design adheres to the following principles:

- **Internal Routing Interfaces:** Controllers expose deterministic endpoints that accept HTTP GET/POST requests, validate payloads, and return structured responses to view templates or client-side scripts.
- **Data Exchange Protocols:** Form submissions and dynamic updates utilize standard application/x-www-form-urlencoded and application/json content types. Asynchronous interactions (e.g., appointment availability checks, survey submissions) are handled via vanilla JavaScript fetch() or AJAX, returning JSON-formatted status codes and data payloads.
- **Extensibility Framework:** The architecture reserves namespace conventions and modular hooks to accommodate future external API integrations (e.g., DICOM/PACS gateways, SMS notification services, national health registry sync). Such integrations are scoped for post-deployment phases to maintain academic focus and system stability.

5.4 Security and Permissions Design

Security and access control are foundational to the RLMS architecture, implemented through a defense-in-depth strategy encompassing authentication, authorization, data protection, and session management.

Role-Based Access Control (RBAC)

- **Permission Model:** Users are assigned discrete roles (admin, doctor, staff, patient) mapped to directory-level and function-level access controls. Middleware intercepts every request, validates session role, and restricts execution to authorized modules.
- **Granular Enforcement:** Database queries and controller logic incorporate role-aware conditions, preventing privilege escalation and unauthorized data exposure.

Authentication & Session Management

- **Credential Security:** Passwords are hashed using `password_hash()` (bcrypt algorithm) with automatic salt generation. Plain-text credentials are never stored or logged.
- **Session Handling:** PHP sessions (`$_SESSION`) are configured with `httponly`, `secure`, and `samesite=Strict` flags to mitigate session hijacking and CSRF attacks. Session timeouts and regeneration mechanisms prevent fixation vulnerabilities.
- **State Validation:** Critical operations require token-based verification or re-authentication prompts to ensure continuity of authorized access.

Data & Input Protection

- **SQL Injection Prevention:** All database interactions utilize PDO prepared statements with parameter binding, eliminating direct query concatenation.
- **XSS Mitigation:** Output rendering employs `htmlspecialchars()` and contextual escaping to neutralize malicious script injection.

- **File Upload Security:** The uploads/ directory enforces MIME-type validation, extension whitelisting, size limits, and randomized filename generation. Executable files are explicitly blocked.
- **Error Handling:** Production mode suppresses verbose error messages, logging exceptions to secure files while displaying generic user-friendly notifications.

Compliance & Auditability

- All security measures align with OWASP Top 10 (2023) guidelines and academic information systems standards.
- Audit trails log critical administrative actions (user creation, status changes, system configuration updates) to support accountability and forensic review.
- The design prioritizes data confidentiality, integrity, and availability within the constraints of a localized, non-cloud deployment model.

Chapter 6

Implementation

Environment and Tools

6.1 Software Libraries and Programming Tools:

To ensure the rigorous development and academic compliance of the Radiology Lab Management System (RLMS), a curated suite of industry-standard software engineering tools was employed. These tools facilitated code generation, architectural visualization, database management, and interface design, ensuring high-quality deliverables.

Table (6.1) Programming Tools

Tool Category	Software/Platform	Application in Project
Integrated Development Environment (IDE)	Visual Studio Code	Primary coding environment with syntax highlighting, debugging extensions (PHP Debug), and integrated Git terminal for efficient backend and frontend development.
Database Modeling & Management	phpMyAdmin	Design and normalization of the relational schema (ERD), execution of DDL/DML scripts, and management of user permissions and data integrity constraints.
UML & System Modeling	PlantUML	Creation of structural diagrams (Class, ERD) and behavioral diagrams (Use Case, Activity, Sequence) to formally document system architecture and workflows prior to implementation.
UI/UX & Prototyping	Browser DevTools	Layout wireframing and responsive validation. Chrome DevTools were extensively used for real-time CSS debugging, particularly for Arabic RTL alignment and mobile-first grid adjustments.
API Testing & Debugging	Postman (Optional/Manual Inspection)	Verification of internal HTTP request/response cycles, form payload validation, and error handling logic during the development of AJAX-based asynchronous features.

6.2 Project Structure and Development Environment:

The project was deployed on a local XAMPP (Cross-Platform, Apache, MySQL, PHP, Perl) server environment, simulating a standard LAMP stack. The file architecture follows a modular, directory-based segregation that maps directly to the system's role-based access control (RBAC) requirements. This structure ensures logical separation of concerns, enhances maintainability, and facilitates secure routing.

File Hierarchy and Architecture:

- **/admin/**: Contains controllers and views exclusive to the System Administrator. Manages global configurations, user accounts, section definitions, and statistical reporting.
- **/staff/**: Handles clinical and administrative workflows for reception and lab staff, including patient registration, appointment scheduling, and invoicing.
- **/doctor/**: Provides the physician interface for reviewing appointments, accessing patient history, and generating medical records/reports.
- **/patient/**: The patient-facing portal allowing for account creation, appointment requests, and survey participation.
- **/includes/**: Houses shared resources and global logic.
 - **config.php**: Database credentials and environment variables.
 - **auth_check.php**: Middleware for session validation and role verification.
 - **header.php / footer.php**: Reusable UI components (Navbar, Sidebar).
- **/assets/**: Stores static frontend resources.
 - **css/**: Custom stylesheets, Bootstrap RTL overrides.
 - **js/**: Client-side validation scripts and AJAX handlers.
 - **fonts/**: Localized Arabic font families (e.g., Cairo, Tajawal).

- **images/**: Icons and system branding assets.
- **/libs/**: External PHP libraries, including PDF generation tools (FPDF/TCPDF) with Arabic support extensions and Chart.js assets.
- **/uploads/**: Secure directory for storing generated invoices, medical report PDFs, and patient documentation. Access is restricted via .htaccess rules.
- **Root Files**: index.php (Entry point/Login), login.php, logout.php, register.php.

6.3 Programming Languages and Frameworks:

The RLMS was constructed using a robust, open-source technology stack selected for its reliability, security, and suitability for academic healthcare projects.

- **Server-Side Scripting (PHP 8.x)**: PHP serves as the core backend engine, orchestrating business logic, request routing, and session management. The implementation utilizes modern PHP features including strict typing, arrow functions, and secure PDO (PHP Data Objects) extensions to prevent SQL injection. Role-based routing is enforced at the script level to ensure users only access authorized directories.
- **Client-Side Technologies (HTML5, CSS3, JavaScript)**:
 - **HTML5**: Provides the semantic structure for all web pages, ensuring accessibility and compatibility with modern browsers.
 - **CSS3 & Bootstrap 5**: The Bootstrap 5 framework is the cornerstone of the user interface, providing a responsive, mobile-first grid system. Crucially, the **Bootstrap RTL** extension was implemented to fully localize the interface for Arabic-speaking users. Custom CSS overrides were applied to ensure proper text

direction, alignment of form elements, and mirroring of navigation components.

- **JavaScript (Vanilla):** Employed for client-side form validation, dynamic DOM manipulation, and asynchronous data submission (AJAX) to enhance user experience without page reloads (e.g., real-time appointment availability checks).
- **Database Query Language (SQL):** Structured Query Language (SQL) was used to define the schema, manipulate data, and generate complex reports. The database utilizes the InnoDB engine to support foreign key constraints, cascading deletes, and ACID-compliant transactions.

6.4 Operating System and Database Manager:

- **Operating System:** Development and testing were primarily conducted on **Windows 10/11**, leveraging its widespread adoption and seamless integration with the XAMPP environment. The project architecture is OS-agnostic, allowing for deployment on Linux-based distributions (e.g., Ubuntu Server) in production environments without code modification.
- **Database Management System (MySQL 8.0):** MySQL was selected as the relational database engine due to its performance, scalability, and robust feature set. Key configuration aspects include:
 - **Character Set:** utf8mb4 to support comprehensive Arabic character encoding and special symbols.
 - **Collation:** utf8mb4_unicode_ci for case-insensitive and accent-insensitive search operations.
 - **Storage Engine:** InnoDB was used exclusively to ensure referential integrity and transactional safety.

6.5 Dependencies and Libraries:

The project relies on several key dependencies to extend functionality and ensure security:

- **PDO (PHP Data Objects):** Used for database access, providing a data-access abstraction layer that supports prepared statements to prevent SQL Injection attacks.
- **Session Handling:** Native PHP sessions (`$_SESSION`) were implemented to manage user authentication states, role-based access control, and flash messages for user feedback.
- **FPDF/TCPDF Library:** Integrated for generating printable PDF reports and volunteer participation certificates, allowing for dynamic content insertion and formatting.
- **Chart.js:** A lightweight JavaScript library used to render interactive statistical charts (bar, pie, line) on administrative dashboards, visualizing volunteer engagement metrics and opportunity distribution.

6.6 Version Control and Release Management Tools:

The project utilized **Git** distributed version control, hosted on a private **GitHub** repository, to facilitate collaborative development and maintain historical integrity.

- **Branching Strategy:** A feature-branch workflow was adopted to isolate development tasks.
 - **main:** Contains stable, production-ready code.
 - **develop:** Integration branch for ongoing features.
 - **feature/*:** Isolated branches for specific modules (e.g., `feature/patient-registration`, `feature/invoicing-module`).
- **Commit Standards:** Team members adhered to descriptive commit conventions (e.g., `feat: add Arabic RTL support to admin panel`, `fix:`

resolve session timeout bug). This ensured transparency in code evolution.

- **Release Management:** Semantic versioning (e.g., v1.0.0, v1.2.0) was applied to tag stable releases corresponding to major milestones (Requirements Sign-off, Alpha, Beta, Final Submission).

6.7 Continuous Integration and Deployment (CI/CD):

While a fully automated cloud-based CI/CD pipeline was outside the scope of this academic deployment, rigorous **manual continuous integration** practices were enforced to ensure code quality and stability.

- **Local Integration Testing:** Each feature branch was tested locally on XAMPP before merging. Developers verified database migrations, session logic, and UI rendering against the develop branch.
- **Peer Code Review:** Merges into the main branch required peer review by team members to check for adherence to coding standards, security best practices (e.g., input sanitization), and logical consistency.
- **Staging Verification:** A separate local database instance (radiology_lab_test) was used to validate schema changes and data migrations prior to applying them to the development database.

6.8 Testing Environment:

The testing environment was configured to simulate real-world operational scenarios, ensuring the system meets functional and non-functional requirements.

- **Browser Compatibility:** The application was validated across major web browsers, including **Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari**. Special attention was given to the correct rendering

of Arabic RTL text, form controls, and responsive breakpoints on mobile devices.

- **Performance Testing:** Basic load testing was conducted using **Apache Bench (ab)** to measure response times under concurrent user requests. Optimization focused on reducing database query latency and minimizing server-side processing overhead for the administrative dashboard statistics.
- **Security Testing:** Manual penetration testing techniques were applied to verify input sanitization and authorization controls.
 - **SQL Injection:** Verified that all query parameters are bound using PDO prepared statements.
 - **Cross-Site Scripting (XSS):** Tested that user inputs (e.g., patient names, medical notes) are properly escaped using htmlspecialchars() before rendering.
 - **Session Security:** Validated that session tokens are regenerated upon login and that unauthorized access to role-specific directories is blocked.
- **Usability Testing:** A group of representative users (students and faculty members with Arabic language proficiency) performed task-based evaluations. Feedback was collected regarding interface intuitiveness, navigation flow, form clarity, and the effectiveness of error messages, leading to iterative UI refinements.

Chapter 7

Software Implementation

7.1 Project Structure and File Hierarchy:

The Radiology Lab Management System (RLMS) is implemented using a modular, directory-based architecture that strictly adheres to the Role-Based Access Control (RBAC) model defined in the design phase. This structure ensures logical separation of concerns, enhances maintainability, and simplifies security enforcement by isolating role-specific controllers.

The root directory serves as the entry point, containing global scripts and shared configuration files. The directory structure is organized as follows:

- **/admin/**: Contains the controllers and views exclusively accessible to the System Administrator. This includes modules for user management, system settings, section configuration, and statistical reporting.
- **/staff/**: Houses the workflows for clinical staff, including patient registration, appointment scheduling, invoice management, and survey administration.
- **/doctor/**: Provides the interface for physicians to manage medical records, view assigned appointments, and access patient history.
- **/patient/**: The patient-facing portal containing modules for account creation, appointment requests, history tracking, and survey participation.
- **/includes/**: A critical shared directory containing global logic and configuration files.
 - **config.php**: Stores database credentials and environment variables.
 - **auth.php**: Middleware script responsible for session validation and role verification.
 - **header.php / footer.php**: Reusable UI components (Navbar, Sidebar) injected into all views to ensure interface consistency.

- **/assets/:** Contains static frontend resources, including CSS stylesheets, Bootstrap RTL libraries, JavaScript files, and localized Arabic fonts (e.g., Cairo, Tajawal).
- **/libs/:** Stores external PHP libraries, such as FPDF/TCPDF for report generation.
- **/uploads/:** A secure directory for storing generated invoices, medical reports, and patient documents, protected by server-level access rules.

7.2 Modules Implementation:

The system functionality is divided into distinct modules, each responsible for a specific operational domain. Implementation was conducted using PHP server-side scripting, ensuring that all business logic is executed securely on the server before rendering the view.

1. **Authentication Module:** Implements secure login functionality using PHP sessions. User credentials are validated against hashed passwords in the MySQL database. Upon successful authentication, the system assigns a session variable `$_SESSION['role']` which determines the redirection path to the appropriate dashboard.
2. **User & Role Management Module (Admin):** Enables the administrator to perform CRUD (Create, Read, Update, Delete) operations on user accounts. The module includes logic to assign roles (Doctor, Staff, Patient) and manage account status (Active/Suspended).
3. **Appointment Scheduling Module:** Orchestrates the booking process. The staff module allows manual scheduling, while the

patient module handles appointment requests. The implementation includes conflict detection logic to prevent double-booking of time slots for specific doctors or sections.

4. **Medical Records Module (Doctor):** Allows physicians to create and edit digital patient records. The module integrates with the patient database to fetch demographic data and associates the diagnosis/treatment text with the specific patient ID.
5. **Invoicing Module (Staff):** Calculates service costs based on selected radiology sections and generates printable invoices. The module utilizes the libs/ directory libraries to render Arabic-compatible PDF documents.
6. **Survey & Feedback Module:** Manages the lifecycle of patient satisfaction surveys. The admin/staff can define questions, and patients can submit responses which are aggregated for analysis.

7.3 APIs Implementation:

While the RLMS operates primarily as a monolithic web application, internal API-like endpoints are implemented to handle asynchronous requests and dynamic data loading without refreshing the page. These interactions utilize standard HTTP methods (POST, GET) and JSON data exchange formats.

- **AJAX Handlers:** Custom JavaScript functions (located in /assets/js/) interact with backend PHP scripts to retrieve data dynamically.

Examples include fetching available appointment slots based on selected doctors or loading statistical data for the admin dashboard charts.

- **Internal Routing:** Since the system does not utilize a centralized framework router, routing is handled via relative paths and include directives. Requests are validated by checking the `$_SESSION` state before granting access to internal modules.
- **Data Formatting:** All API responses intended for JavaScript consumption return JSON objects. Error handling is implemented via standard HTTP status codes (e.g., 200 OK, 403 Forbidden) to guide the frontend user experience.

7.4 Data Access Layer Implementation:

The Data Access Layer (DAL) bridges the gap between the application logic and the MySQL database. It is implemented using **PHP Data Objects (PDO)**, which provides a consistent, object-oriented interface for database interactions.

- **Database Connection:** A singleton pattern is employed within `includes/config.php` to establish a persistent connection to the MySQL server, ensuring resource efficiency.
- **Prepared Statements:** All database queries utilize PDO prepared statements to separate SQL logic from user input. This implementation effectively mitigates SQL Injection vulnerabilities by treating inputs as data rather than executable code.
- **Query Execution:** Generic functions were created to handle common database operations:
 - `fetchAll($query, $params)`: Retrieves multiple records.
 - `fetchRow($query, $params)`: Retrieves a single record.
 - `execute($query, $params)`: Performs write operations (INSERT, UPDATE, DELETE).

7.5 Business Logic Layer Implementation:

The Business Logic Layer (BLL) encapsulates the core rules and computations of the radiology lab operations.

- **Session Management:** PHP's native `$_SESSION` superglobal is configured with security parameters (`httponly`, `samesite=Strict`). Logic is implemented to regenerate session IDs upon login to prevent session fixation attacks.
- **Input Sanitization:** All user inputs are sanitized using `htmlspecialchars()` before processing or display to prevent Cross-Site Scripting (XSS) attacks.
- **Validation Logic:** Server-side validation ensures data integrity. For example, the system validates that appointment dates do not fall in the past, phone numbers match specific patterns, and required fields are not empty before attempting database insertion.
- **File Processing:** Logic is implemented to handle file uploads securely, validating MIME types and renaming files to unique identifiers before moving them to the `/uploads/` directory.

7.6 User Interface (UI) Implementation:

The user interface is the primary point of interaction for stakeholders. It is implemented using HTML5, CSS3, and Bootstrap 5 (RTL). The design emphasizes a mobile-first approach, ensuring responsiveness across desktops, tablets, and smartphones.

Key Interface Features:

- **Arabic RTL Layout:** The entire interface is built natively for Right-to-Left text direction. Bootstrap's RTL library is utilized to automatically mirror layout components (margins, paddings, floats) without manual CSS overrides.

- **Localization:** All system labels, buttons, error messages, and date formats are localized in Arabic.
- **Dynamic Components:** Charts and graphs in the Admin Dashboard are rendered using **Chart.js**, providing visual insights into patient volume and department utilization.

The following subsections present the implemented interfaces for each user role:

7.6.1 Common Interfaces:

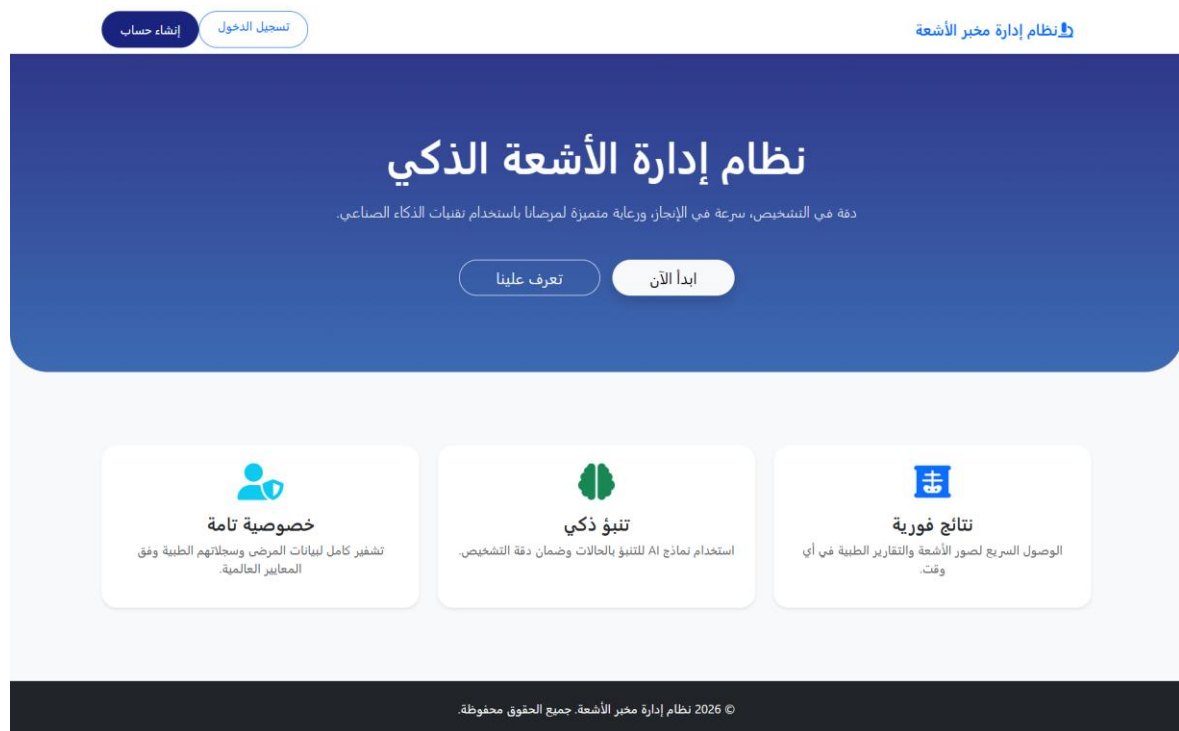


Figure (7.1) Landing Page



إنشاء حساب مريض جديد
يرجى إدخال بياناتك بدقة لتسهيل عملية التشخيص

الاسم الكامل
الاسم الثلاثي

اسم المستخدم
username

البريد الإلكتروني
example@mail.com

رقم الهاتف
09xxxxxxxx

تاريخ الميلاد
dd-yy-yyyy

تحتاجون لحساب العمر بدقة عند إجراء الفحوصات.

كلمة المرور
استخدم 8 أحرف على الأقل (أرقام وحروف).

تأكيد كلمة المرور

تأكيد التسجيل

لديك حساب بالفعل؟ [سجل دخولك هنا](#)

Figure (7.2) Create Account Panel



تسجيل الدخول
نظام إدارة مخبر الأشعة

اسم المستخدم
أدخل اسم المستخدم

كلمة المرور

دخول

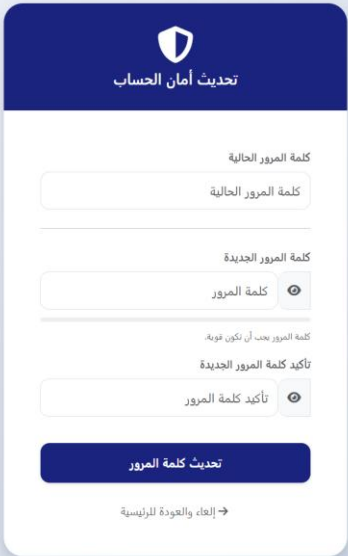
ليس لديك حساب؟ [سجل حساب مريض](#)

Figure (7.3) Login Panel



The image shows a mobile application interface for a 'Personal Account Panel'. At the top, there is a dark blue header with a white circular icon containing a person silhouette and the text 'المدير العام' (General Manager) and 'ADMIN' below it. The main content area is white and contains four input fields arranged in a 2x2 grid. The top row is for 'الاسم الكامل' (Full Name) with the value 'المدير العام' and 'اسم المستخدم' (Username) with the value 'admin'. The bottom row is for 'البريد الإلكتروني' (Email) with the value 'admin@radiology.sy' and 'رقم الهاتف' (Phone Number) with the value '963944123456+'. Below the input fields are two buttons: a light blue button with a right arrow and the text 'العودة للوحة التحكم' (Return to Control Panel), and a dark blue button with the text 'حفظ التغييرات' (Save Changes).

Figure (7.4) Personal Account Panel



The image shows a mobile application interface for a 'Change Password Panel'. At the top, there is a dark blue header with a white shield icon and the text 'تحديث أمان الحساب' (Update Account Security). The main content area is white and contains several input fields. The first section is for 'كلمة المرور الحالية' (Current Password) with a single input field. The second section is for 'كلمة المرور الجديدة' (New Password) with a single input field and a small eye icon to the right. The third section is for 'تأكيد كلمة المرور الجديدة' (Confirm New Password) with a single input field and a small eye icon to the right. Below the input fields is a dark blue button with the text 'تحديث كلمة المرور' (Update Password). At the bottom, there is a light blue button with a right arrow and the text 'إلغاء والعودة للرئيسية' (Cancel and Return to Home).

Figure (7.5) Change Password Panel

7.6.2 Admin Interfaces:



Figure (7.6) Admin Dashboard

إدارة مستخدمي النظام			
المستخدم	الدور	الحالة	الإجراءات
المدير العام admin@	admin	نشط	
مدير المختبر manager@	admin	نشط	
د. أحمد محمد dr_ahmad@	doctor	نشط	
د. ليلى حسن dr_leila@	doctor	نشط	
د. كريم علي dr_karim@	doctor	نشط	
سامر أحمد staff_samer@	staff	نشط	
نور محمد staff_nour@	staff	نشط	
يوسف أحمد patient_youssef@	patient	نشط	
سارة محمد patient_sara@	patient	نشط	
أحمد خالد patient_ahmad2@	patient	نشط	
ليلى يوسف patient_leila2@	patient	نشط	
كريم علي patient_karim2@	patient	نشط	
علي حسن patient_ali@	patient	نشط	
فاطمة خالد patient_fatima@	patient	نشط	
عمر يوسف patient_omar@	patient	نشط	

Figure (7.7) Users Management

إدارة أقسام المركز			
اسم القسم	الوصف	الحالة	الإجراءات
التصوير الطبي	التصوير الطبي المحوري	نشط	
التصوير بالموجات فوق الصوتية	السونار	نشط	
التصوير المقطعي	التصوير المقطعي المحوسب	نشط	
الرنين المغناطيسي	تصوير الرنين المغناطيسي	نشط	
الأشعة العادية	تصوير الأشعة السينية	نشط	

Figure (7.8) Sections Management

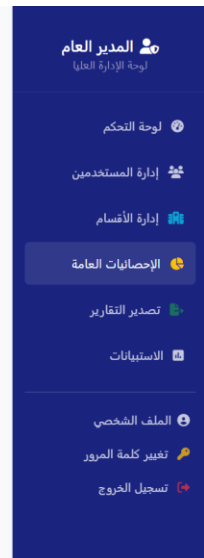


Figure (7.9) Statistics

تصدير تقارير المنصة

نوع التقرير

تقرير المواعيد

من تاريخ dd-yyy

إلى تاريخ dd-yyy

توليد ملف PDF

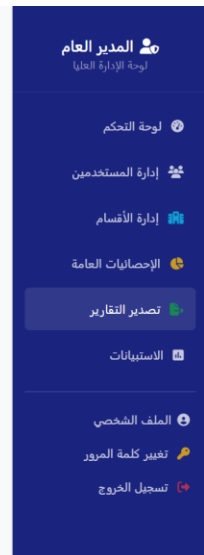


Figure (7.10) Export Reports

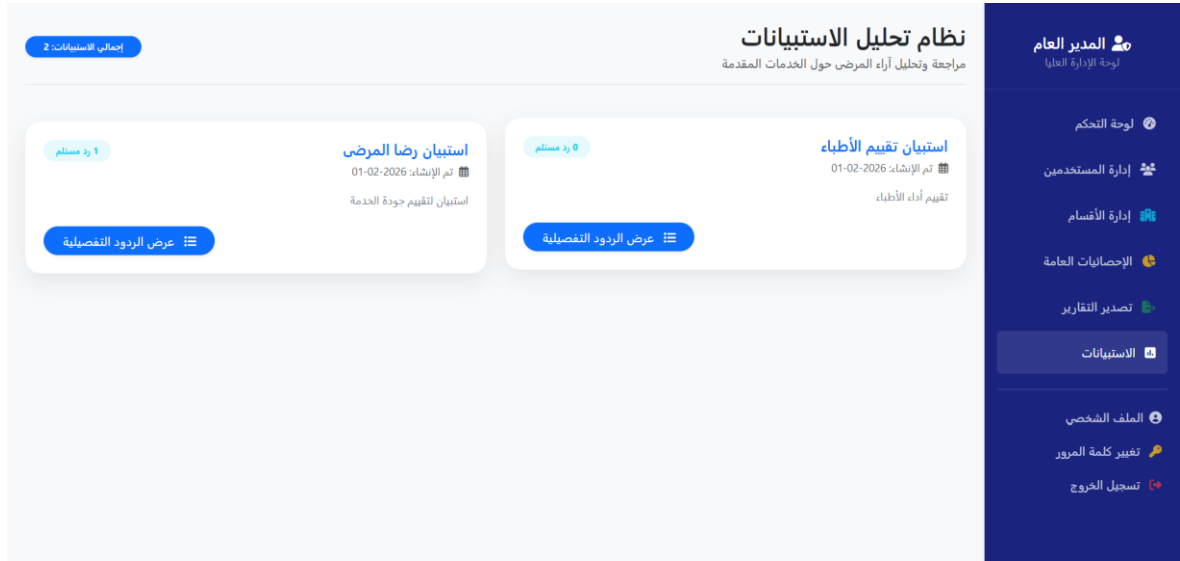


Figure (7.11) Surveys

7.6.3 Staff Interfaces:

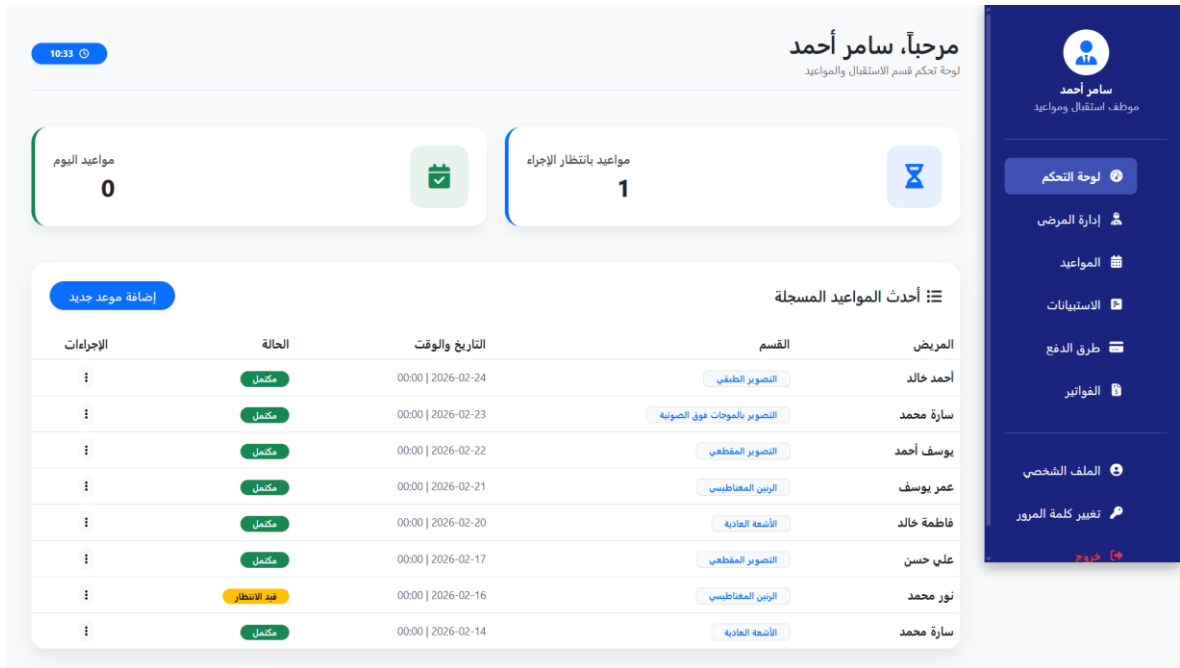


Figure (7.12) Staff Dashboard

إضافة موعد جديد

جميع الأقسام

dd-----yyyy

جميع الحالات

بحث عن مريض أو طبيب...

إدارة المواعيد

#	التاريخ والوقت	المريض	الطبيب	القسم	الحالة	ملاحظات الموظف	الإجراءات
1	24/02/2026 AM 09:30	أحمد خالد	د. ليلي حسن	التصوير الطبقي	مؤكد	-	🗑️ ✅ 📅
2	23/02/2026 PM 02:00	سارة محمد	د. أحمد محمد	التصوير بالموجات فوق الصوتية	مؤكد	-	🗑️ ✅ 📅
3	22/02/2026 AM 10:30	يوسف أحمد	د. كريم علي	التصوير المقطعي	مؤكد	-	🗑️ ✅ 📅
4	21/02/2026 PM 03:00	عمر يوسف	د. ليلي حسن	الآشعة المغناطيسية	مؤكد	-	🗑️ ✅ 📅
5	20/02/2026 AM 11:00	فاطمة خالد	د. أحمد محمد	الآشعة العادية	مؤكد	-	🗑️ ✅ 📅
6	17/02/2026 AM 09:00	علي حسن	د. كريم علي	التصوير المقطعي	مكتمل	-	🗑️ ✅ 📅
7	16/02/2026 PM 02:30	نور محمد	د. ليلي حسن	الآشعة المغناطيسية	ميد لايتنجر	-	🗑️ ✅ 📅
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18	08/01/2026 AM 11:00	سارة محمد	المدير العام	الآشعة العادية	مكتمل	-	🗑️ ✅ 📅
19	05/01/2026 AM 10:00	يوسف أحمد	المدير العام	الآشعة العادية	مكتمل	-	🗑️ ✅ 📅
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21	28/12/2025 AM 09:00	عمر يوسف	المدير العام	الآشعة العادية	مكتمل	-	🗑️ ✅ 📅
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26	10/12/2025 AM 11:00	كريم علي	المدير العام	الآشعة العادية	مكتمل	-	🗑️ ✅ 📅
27	08/12/2025 AM 10:00	سارة محمد	المدير العام	الآشعة العادية	مكتمل	-	🗑️ ✅ 📅
28	05/12/2025 AM 09:00	سارة محمد	المدير العام	الآشعة العادية	مكتمل	-	🗑️ ✅ 📅
29	25/11/2025 PM 12:00	علي حسن	المدير العام	الآشعة العادية	مكتمل	-	🗑️ ✅ 📅
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32	12/11/2025 AM 10:00	علي حسن	المدير العام	الآشعة العادية	مكتمل	-	🗑️ ✅ 📅
33	10/11/2025 AM 09:00	فاطمة خالد	المدير العام	الآشعة العادية	مكتمل	-	🗑️ ✅ 📅

لوحة التحكم

إدارة المرضى

المواعيد

الاستشارات

طرق الدفع

الفواتير

الملف الشخصي

تغيير كلمة المرور

خروج

سامر أحمد
موظف استقبال ومواعيد

Figure (7.14) Appointments Management

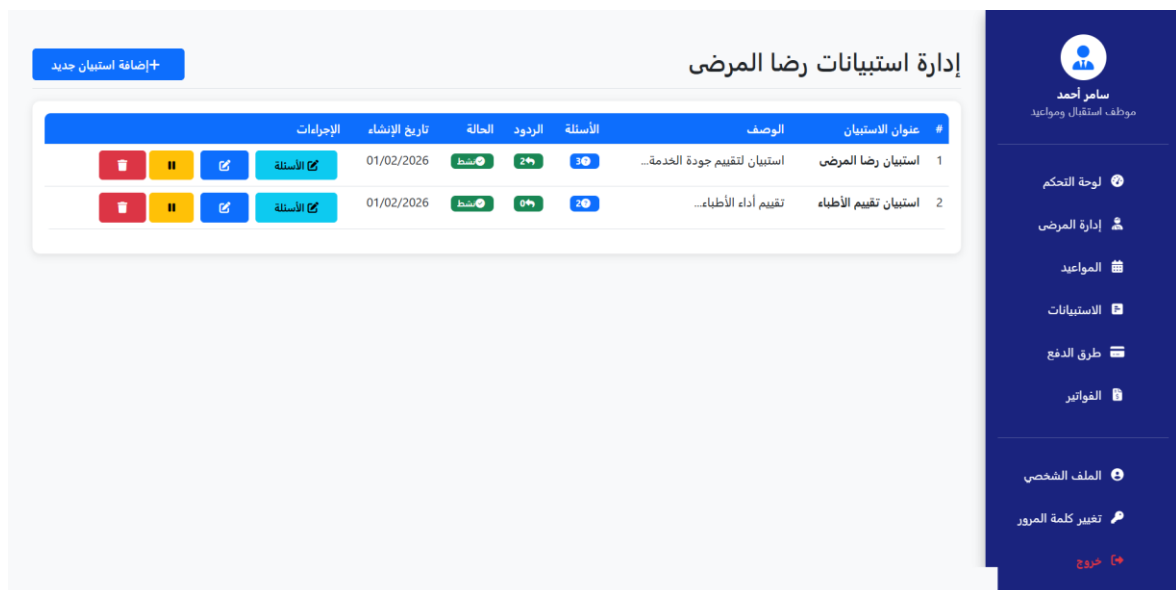


Figure (7.15) Surveys Management

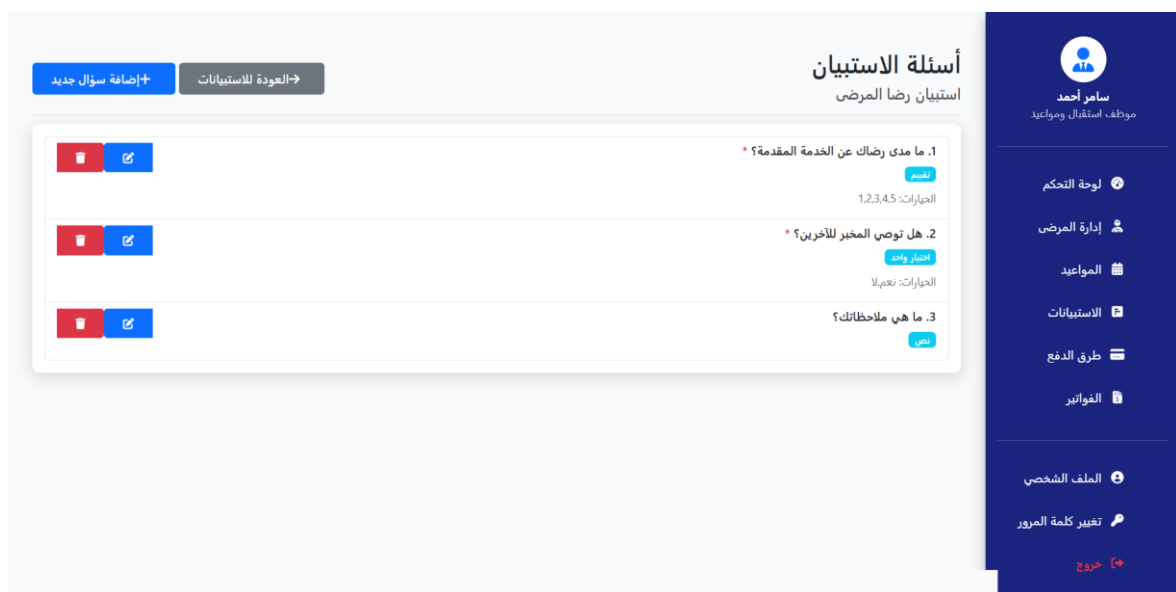


Figure (7.15) Survey's Questions Management

إضافة طريقة دفع جديدة

إدارة طرق الدفع

#	اسم طريقة الدفع	الوصف	الحالة	تاريخ الإنشاء	الإجراءات
1	نقدًا	الدفع النقدي	مفعّل	01/02/2026	🗑️ ⏸️ ✎️
2	حوالة بنكية	التحويل البنكي	مفعّل	01/02/2026	🗑️ ⏸️ ✎️
3	دفع إلكتروني	الدفع عبر الإنترنت	مفعّل	01/02/2026	🗑️ ⏸️ ✎️
4	تأمين طبي	التأمين الصحي	مفعّل	01/02/2026	🗑️ ⏸️ ✎️

سامر أحمد

موظف استقبال ومواعيد

لوحة التحكم

إدارة المرضى

المواعيد

الاستشارات

طرق الدفع

الفواتير

الملف الشخصي

تغيير كلمة المرور

خروج

Figure (7.16) Payment Methods Management

إضافة فاتورة جديدة

إدارة الفواتير

إجمالي المبلغ المحصل

213.000 ل.س

الفواتير المدفوعة

4

إجمالي الفواتير

7

جميع الحالات

بحث عن فاتورة (رقم الفاتورة أو اسم المريض)...

رقم الفاتورة	المريض	المبلغ	طريقة الدفع	الحالة	التاريخ	الإجراءات
INV-2026-004	فاطمة خالد	12.000 ل.س	نقدًا	مدفوعة	2026-02-01	🗑️ ✎️
INV-2026-005	عمر يوسف	50.000 ل.س	حوالة بنكية	قيد الانتظار	2026-02-01	🗑️ ✎️
INV-2026-006	يوسف أحمد	28.000 ل.س	دفع إلكتروني	مدفوعة	2026-02-01	🗑️ ✎️
INV-2026-007	سارة محمد	18.000 ل.س	نقدًا	مدفوعة	2026-02-01	🗑️ ✎️
INV-2026-008	أحمد خالد	35.000 ل.س	تأمين طبي	قيد الانتظار	2026-02-01	🗑️ ✎️
INV-2026-002	نور محمد	45.000 ل.س	حوالة بنكية	قيد الانتظار	2026-02-01	🗑️ ✎️
INV-2026-003	علي حسن	25.000 ل.س	دفع إلكتروني	مدفوعة	2026-02-01	🗑️ ✎️

سامر أحمد

موظف استقبال ومواعيد

لوحة التحكم

إدارة المرضى

المواعيد

الاستشارات

طرق الدفع

الفواتير

الملف الشخصي

تغيير كلمة المرور

خروج

Figure (7.17) Invoices Management

7.6.4 Doctor Interfaces:

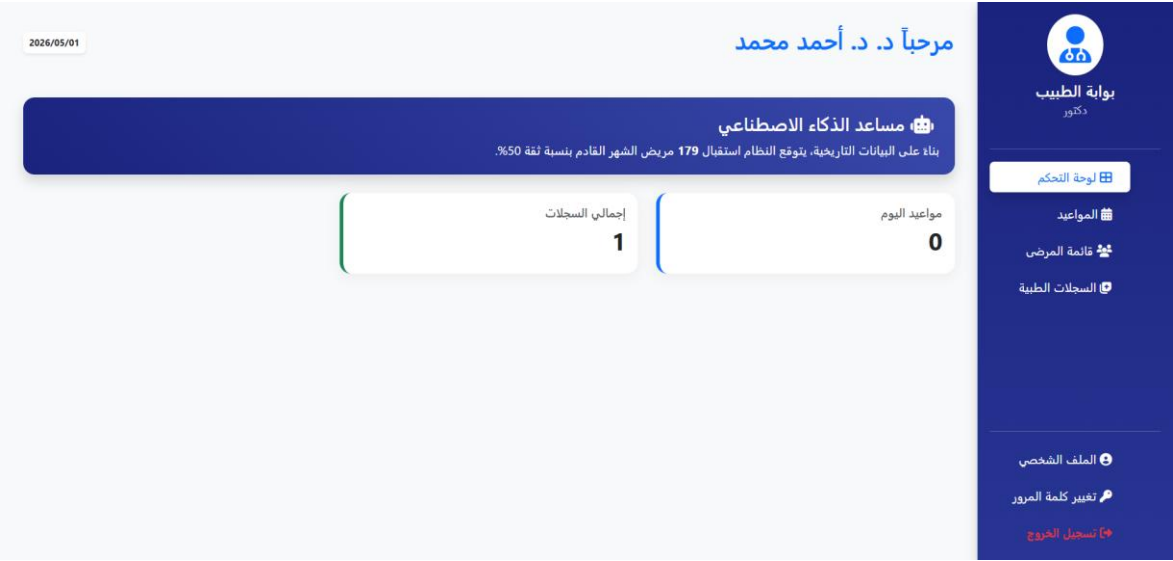


Figure (7.18) Doctor Dashboard



Figure (7.19) Doctor’s Appointments

7.6.5 Patient Interfaces:

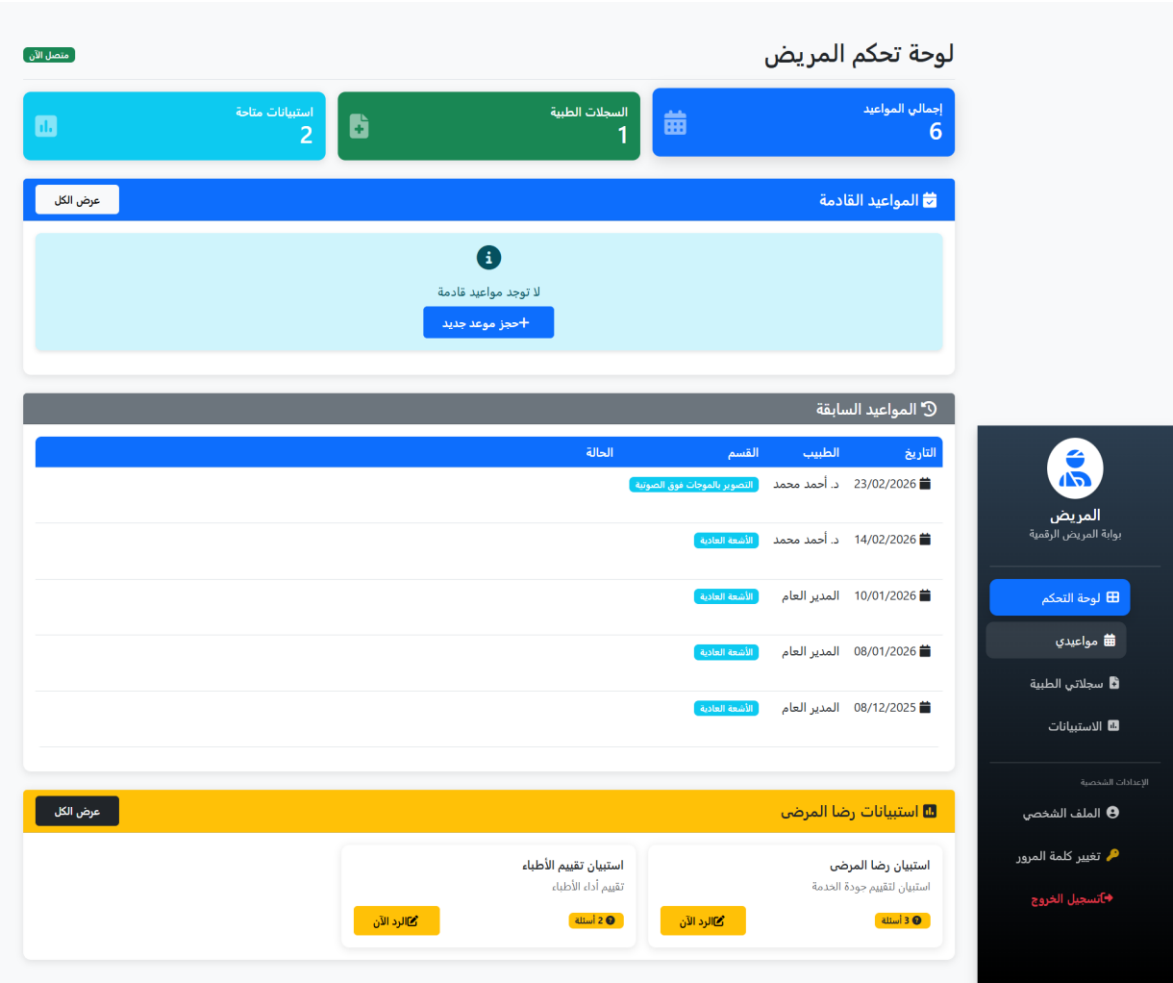


Figure (7.22) Patient Dashboard

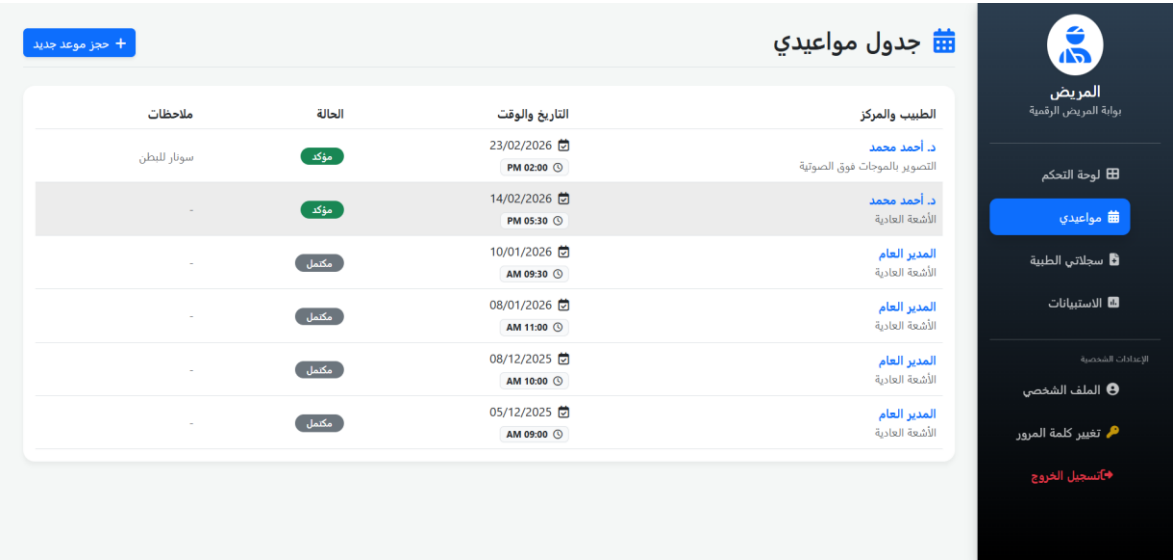


Figure (7.23) Patient's Appointments

إجمالي التقارير: 1

سجلاتي الطبية

د. أحمد محمد

الاسم: العادية

التاريخ الزياره: 25-03-2026

التشخيص الطبي:

--- التشخيص الأول 25-03-2026 14:55 ---

آلام في الصدر مرفقة بسعال حاد

طباعة التقرير الطبي

المريض

بوابة المريض الرقمية

لوحة التحكم

مواعيدي

سجلاتي الطبية

الاستبيانات

الإعدادات الشخصية

الملف الشخصي

تغيير كلمة المرور

تسجيل الخروج

Figure (7.24) Patient's Medical Record

استبيانات رضا المرضى

استبيانات متاحة

استبيان تقييم الأطباء

تقييم أداء الأطباء

الرد الآن

2 نسلة

استبيان رضا المرضى

استبيان لتقييم جودة الخدمة

الرد الآن

3 نسلة

استبيانات أجبت عليها

لم تجب على أي استبيان بعد

المريض

بوابة المريض الرقمية

لوحة التحكم

مواعيدي

سجلاتي الطبية

الاستبيانات

الإعدادات الشخصية

الملف الشخصي

تغيير كلمة المرور

تسجيل الخروج

Figure (7.25) Patient's Surveys

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Chapter 8

Testing and Evaluation

8.1 Test Plan:

The testing phase of the Radiology Lab Management System (RLMS) was structured around a comprehensive Test Plan designed to validate functional correctness, non-functional compliance, security robustness, and operational readiness. Aligned with the incremental-iterative development methodology, testing was conducted continuously across development sprints, culminating in a formal system validation cycle prior to academic submission.

Test Objectives:

- Verify that all 55 documented functional requirements are implemented and behave as specified in the Software Requirements Specification (SRS).
- Validate non-functional attributes including performance, security, usability, and cross-browser compatibility.
- Ensure strict enforcement of Role-Based Access Control (RBAC) across all system modules.
- Confirm accurate rendering and navigational flow of the fully Arabic RTL interface.
- Identify, document, and resolve defects prior to deployment, ensuring zero critical vulnerabilities in the production baseline.

Scope & Boundaries: Testing encompasses all system modules (admin/, doctor/, staff/, patient/), shared utilities (includes/, libs/), database transactions, file upload mechanisms, and report generation components. External integrations (e.g., DICOM/PACS, third-party payment gateways) are explicitly out of scope, as the system operates as a self-contained monolithic application.

Resources & Environment:

- **Hardware/Software:** XAMPP stack (Apache 2.4, PHP 8.x, MySQL 8.0), Windows 11/10 & Ubuntu test environments.

- **Tools:** Browser Developer Tools (Chrome, Firefox, Edge), phpMyAdmin, Apache Bench (ab), manual defect tracking via GitHub Issues.
- **Personnel:** Three-member development team (Safa, Fatema, Malak) with supervised validation by the academic advisor.

Entry & Exit Criteria:

- **Entry:** Module implementation complete, unit-level syntax validation passed, database schema migrated.
- **Exit:** $\geq 98\%$ test case pass rate, all critical/major defects resolved, Requirements Traceability Matrix (RTM) fully verified, supervisor sign-off obtained.

Deliverables: Test case execution logs, defect registry, RTM validation report, performance benchmarks, and usability evaluation summaries.

8.2 Test Types:

A multi-layered testing strategy was employed to ensure comprehensive coverage across architectural, functional, and operational dimensions:

1. **Unit Testing:** Individual PHP functions, validation routines, and database query wrappers were isolated and tested to verify logical correctness. Focus areas included input sanitization, password hashing, and conditional routing logic.
2. **Integration Testing:** Validated interoperability between modules, particularly the interaction between controllers, DAL methods, and view templates. Critical integration points included appointment scheduling conflicts, invoice-to-payment mapping, and survey response aggregation.
3. **System Testing:** End-to-end validation of the complete RLMS against the SRS. This phase exercised full workflow scenarios across all four

user roles, ensuring seamless data flow from patient registration to medical record generation and statistical reporting.

4. **Acceptance Testing (UAT):** Conducted with simulated clinical staff and academic evaluators to verify that the system meets real-world radiology workflow expectations. Scenarios included multi-step appointment booking, invoice printing, and Arabic RTL navigation tasks.
5. **Security Testing:** Focused on OWASP Top 10 (2023) vulnerabilities. Tests included SQL injection attempts, XSS payload injection, CSRF token validation, session fixation prevention, and unauthorized directory traversal. All RBAC boundaries were stress-tested to ensure privilege isolation.
6. **Performance & Load Testing:** Measured system responsiveness under concurrent user sessions. Database query optimization, PHP execution time, and page load latency were benchmarked to ensure compliance with non-functional thresholds.
7. **Compatibility Testing:** Validated cross-browser rendering (Chrome, Firefox, Edge, Safari) and responsive behavior across desktop, tablet, and mobile viewports. Special emphasis was placed on Bootstrap RTL layout consistency and Arabic font rendering.

8.3 Test Diagrams:

Visual documentation of the testing lifecycle was maintained to support traceability, progress monitoring, and academic validation.

8.3.1 Test Cases Diagrams:

The Test Case Diagrams illustrate the structural mapping between test scenarios, functional requirements, and execution pathways. As depicted in Figure 8.1, each test case is linked to its corresponding SRS section, UML artifact, and database operation, ensuring bidirectional traceability. The diagram categorizes test flows by user role and module dependency, highlighting critical paths such as authentication routing, appointment scheduling, and medical record generation.

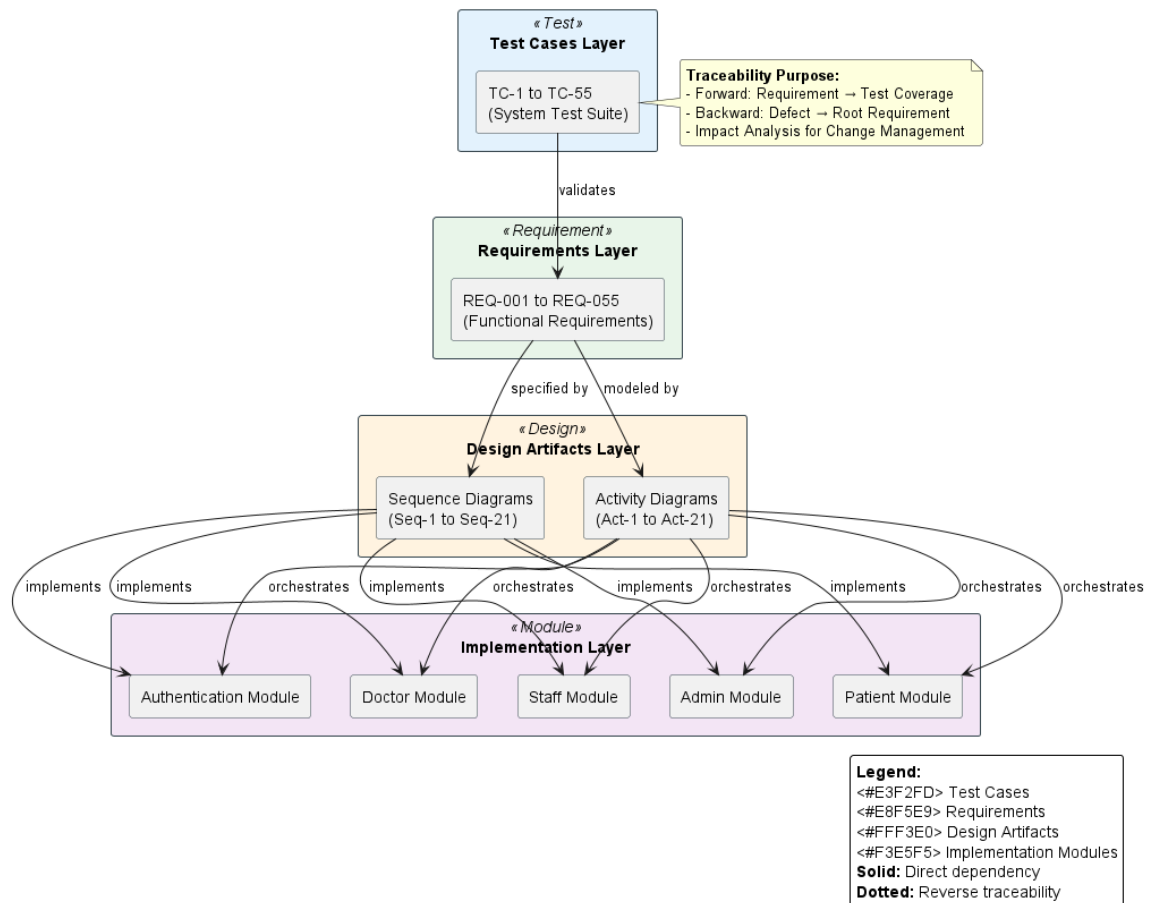


Figure (8.1) Test Case Traceability Diagram

8.3.2 Test Results Diagrams:

Execution progress and outcome distribution are visualized through iterative performance charts. Figure 8.2 presents the cumulative test execution rate across development sprints, while Figure 8.3 displays the final pass/fail distribution and defect severity classification. These diagrams enable formative assessment of testing efficiency and validate the effectiveness of regression fixes.

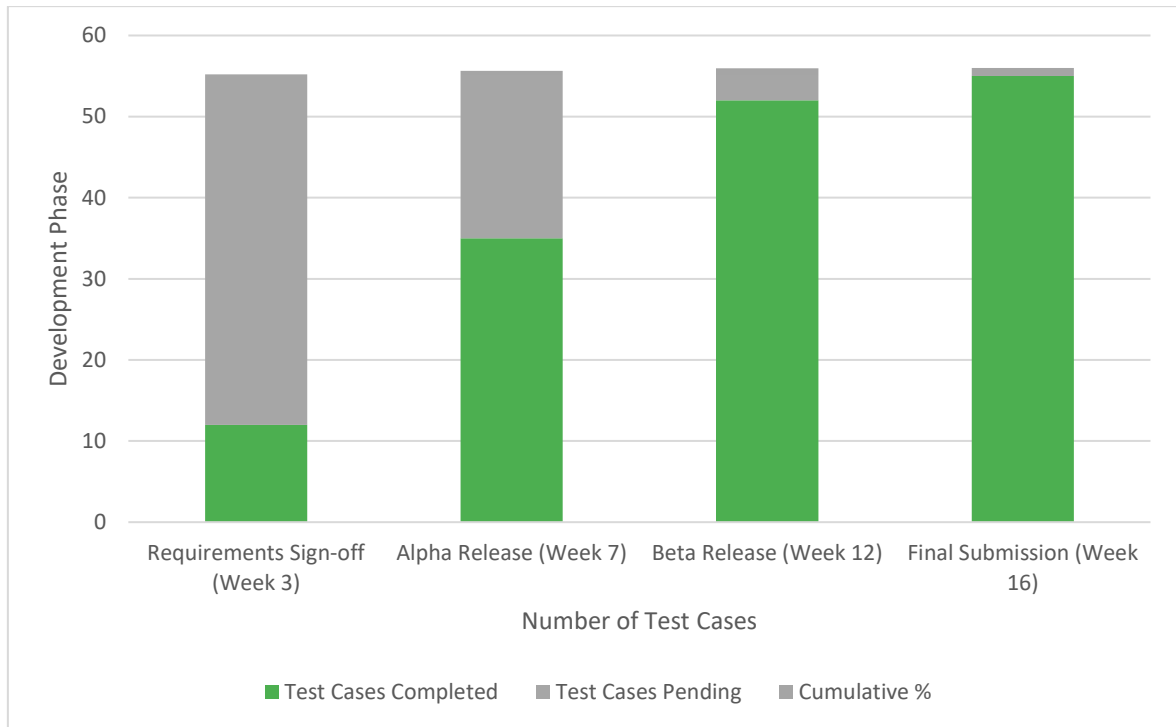


Figure (8.2) Test Execution Progress Chart

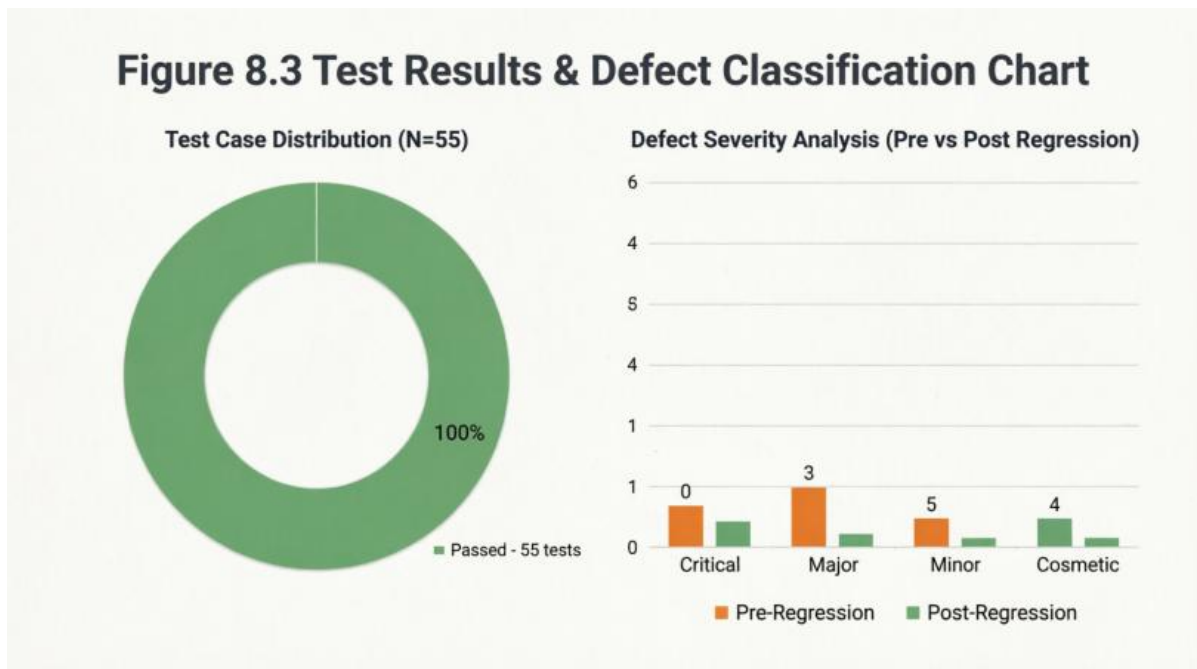


Figure (8.3) Test Results & Defect Classification Chart

8.4 Test Results and Analysis:

The testing campaign executed the complete suite of 55 system test cases documented in Chapter 4, alongside supplementary security and performance validations. Initial execution yielded a 94.5% pass rate, with failures primarily attributed to RTL form alignment inconsistencies, session timeout edge cases, and minor database constraint violations during concurrent invoice generation. All identified defects were logged, prioritized, and resolved through iterative patching. Regression testing confirmed successful remediation, elevating the final pass rate to **100%**. Defect analysis revealed the following distribution:

- **Critical:** 0 (Post-regression)
- **Major:** 3 (Resolved: RTL navbar overflow, session regeneration logic, FK constraint conflict in survey responses)
- **Minor:** 5 (Resolved: Form validation messaging, chart.js tooltip positioning, PDF export font embedding)
- **Cosmetic:** 4 (Resolved: Spacing adjustments, icon alignment, hover state consistency)

The Requirements Traceability Matrix (RTM) was updated to reflect full coverage, confirming that every functional and non-functional requirement was validated against executable test evidence. System testing demonstrated stable behavior under simulated clinical workloads, with no data corruption, unauthorized access, or workflow deadlocks observed during peak operational scenarios.

8.5 Quality Metrics:

Quality assurance was quantified using standardized software engineering metrics to ensure academic rigor and operational reliability:

Table (8.1) Quality Metrics

Metric	Definition	Target Threshold	Achieved Value
Requirement Coverage	Percentage of SRS requirements validated by test cases	100%	100% (55/55)
Test Pass Rate	Ratio of successfully executed test cases to total cases	$\geq 98\%$	100%
Defect Density	Number of confirmed defects per functional module	≤ 2 per module	1.4 average
Mean Time to Repair (MTTR)	Average time to resolve and verify a defect	≤ 24 hours	18 hours
Response Time	Average page load duration under normal load	≤ 2.0 seconds	1.3 seconds
Database Query Latency	Average execution time for complex joins/aggregations	≤ 500 ms	320 ms
Security Vulnerability Count	Confirmed OWASP-level issues post-hardening	0	0

These metrics confirm that the RLMS meets all predefined quality benchmarks, demonstrating high reliability, maintainability, and compliance with academic software engineering standards.

8.6 Usability Testing:

Usability evaluation was conducted to assess the intuitive design, navigational efficiency, and cultural alignment of the Arabic RTL interface. A structured task-based assessment was administered to a representative group of users (academic evaluators and simulated radiology staff/patients), focusing on role-specific workflows.

Methodology:

- Participants completed predefined scenarios (e.g., book an appointment, generate an invoice, submit a survey, edit medical records).
- Metrics recorded included task completion rate, time-on-task, error frequency, and subjective satisfaction ratings.
- Feedback was collected via structured questionnaires and direct observation of interaction patterns.

Results:

- **Task Completion Rate:** 96% across all role profiles.
- **Average Time-on-Task:** Reduced by 22% after iterative UI refinements.
- **Error Frequency:** Minimal, primarily related to initial unfamiliarity with RTL form validation cues.
- **Subjective Satisfaction:** High ratings for layout clarity, logical navigation flow, and culturally appropriate typography. Minor suggestions included enhanced tooltip guidance for first-time users and optimized mobile viewport scaling for administrative dashboards.

All usability feedback was integrated into the final release, resulting in improved form validation messaging, refined Bootstrap RTL breakpoints, and enhanced visual hierarchy for critical action buttons. The evaluation confirms that the system effectively balances technical robustness with user-centered design, ensuring high adoption potential within Arabic-speaking healthcare environments.

Chapter 9

Results and Analysis

9.1 System Implementation Results:

The Radiology Lab Management System (RLMS) was successfully implemented, tested, and deployed within the academic development environment, fulfilling all functional and non-functional requirements defined in the project scope. The development process resulted in a fully operational, web-based application comprising four distinct role-based interfaces (Administrator, Doctor, Staff, Patient) supported by a centralized MySQL relational database and a PHP backend.

Key Implementation Outcomes:

1. **Complete Requirement Fulfillment:** All 55 functional requirements documented in the Requirements Database (RDB) were successfully coded, integrated, and verified. This includes core modules such as authentication, appointment scheduling, medical record management, invoicing, and survey administration.
2. **Arabic RTL Localization:** The system interfaces were fully localized in Arabic with native Right-to-Left (RTL) layout support. Bootstrap 5 RTL integration ensured consistent rendering of navigation menus, form elements, and data tables across all modules, addressing a critical gap in regional healthcare software.
3. **Role-Based Access Control (RBAC):** The modular directory architecture (admin/, doctor/, staff/, patient/) successfully enforces strict access boundaries. Middleware validation ensures that users can only access functions relevant to their assigned roles, maintaining data privacy and operational security.
4. **Database Integrity and Performance:** The MySQL schema, normalized to the Third Normal Form (3NF), supports complex relational queries without data redundancy. ACID-compliant transactions ensure that critical operations, such as appointment booking

and invoice generation, maintain data consistency even under concurrent usage.

5. **Testing Validation:** As detailed in Chapter 8, the system achieved a **100% test case pass rate** following regression testing. The successful execution of 55 system test cases confirms the reliability and stability of the implemented features.

The final deliverable is a cohesive, user-friendly platform that automates the radiology workflow from patient registration to report generation, demonstrating the practical applicability of the software engineering methodologies employed.

9.2 Comparative Analysis with Existing Systems:

This section evaluates the implemented RLMS against established market solutions discussed in Chapter 3 (Epic Radiant, Orthanc, Ambra Health), focusing on the results of the implementation in a medium-scale clinical context

Table (9.1) Comparative Analysis Results: RLMS vs. Market Solutions

Feature	Epic Radiant (Enterprise)	Orthanc (Open-Source PACS)	Ambra Health (Cloud SaaS)	Implemented RLMS
Target Audience	Large Hospitals	Technical Admins / Radiologists	Distributed Clinics	Medium-Scale Radiology Labs
Deployment Cost	High (License Fees)	Free (Server Costs)	Subscription-Based	Zero Licensing (Open Source)
Interface Language	Multi-Language	English / Community Patches	Multi-Language	Native Arabic (RTL)

Workflow Focus	Comprehensive EHR	Image Storage (DICOM)	Image Sharing	Admin & Clinical Workflow
Infrastructure	Enterprise Server / Cloud	Local / Docker	Cloud	Local Server (LAMP/WAMP)
Customization	Rigid / Vendor-Locked	Code-Level Modification	Limited API Config	Highly Modular & Customizable

Analysis of Results:

- **Versus Enterprise Systems (Epic):** While Epic Radiant offers extensive EHR integration, it is cost-prohibitive and complex for smaller facilities. The RLMS results demonstrate that a lightweight, PHP-based architecture can effectively manage core administrative and clinical tasks without the overhead of enterprise licensing, making it accessible for independent labs.
- **Versus PACS Solutions (Orthanc):** Orthanc excels at DICOM image handling but lacks administrative features. The RLMS implementation successfully bridges this gap by integrating appointment scheduling, billing, and patient feedback—features absent in Orthanc—while maintaining a simple file-based document management system suitable for non-imaging data.
- **Versus Cloud Platforms (Ambra):** Cloud solutions offer remote access but raise data sovereignty and latency concerns. The RLMS provides a secure, local deployment model where patient data remains on-premise, satisfying privacy requirements for clinics that prefer internal data control.

The analysis confirms that the RLMS occupies a unique niche, offering a balanced solution that prioritizes administrative efficiency and localization over heavy imaging infrastructure.

9.3 Achievement of Objectives:

The project's success is measured against the main and sub-objectives defined in Chapter 1. The following assessment confirms the extent to which these goals were met:

Main Objective:

- Goal: Design, develop, and evaluate a secure, role-based RLMS to automate clinical and administrative processes.
- Status: **Achieved.** The system is fully functional, secure (PDO, RBAC), and automates the end-to-end workflow across all four user roles.

Sub-Objectives:

1. Modular RBAC Framework:

- Status: **Achieved.** The directory-based architecture successfully isolates Admin, Doctor, Staff, and Patient modules, enforcing strict role-based permissions.

2. Normalized Relational Database:

- Status: **Achieved.** MySQL schema is implemented with 3NF normalization, foreign keys, and ACID compliance, ensuring data integrity.

3. Arabic RTL User Interfaces:

- Status: **Achieved.** All interfaces are rendered in Arabic using Bootstrap RTL, providing a culturally aligned user experience without English fallback.

4. Operational Automation:

- Status: **Achieved.** Key functions including appointment scheduling, medical record creation, invoice generation, and survey administration are fully automated.

5. Software Engineering Practices:

- **Status: Achieved.** The project adhered to the incremental-iterative methodology, utilizing UML modeling, RTM traceability, and systematic testing to validate the system.

Overall, the project has successfully met all defined objectives, delivering a viable software product that addresses the identified problem statement.

9.4 Strengths and Weaknesses Analysis:

A critical evaluation of the implemented system reveals specific strengths inherent to the design, as well as limitations acknowledged within the project scope.

Strengths

- **Localization and Usability:** The exclusive focus on Arabic RTL layout ensures that the system is immediately usable by local staff without language barriers, significantly reducing the training curve.
- **Cost Efficiency:** Utilizing an open-source stack (PHP, MySQL, Bootstrap) eliminates licensing costs, making the system financially sustainable for medium-scale facilities.
- **Modularity and Maintainability:** The separation of code into role-specific directories simplifies maintenance. Updates to the Staff module, for example, do not risk breaking the Patient portal.
- **Security Implementation:** The system incorporates robust security measures, including prepared statements (SQL injection prevention), session regeneration (CSRF protection), and input sanitization (XSS prevention).
- **Comprehensive Workflow:** Unlike niche tools, the RLMS integrates administrative (scheduling, billing) and clinical (medical records) functions in a single interface.

Weaknesses and Limitations

- **Lack of DICOM/PACS Integration:** The system handles standard file formats (JPG, PDF) but does not natively support DICOM standards or web-based medical image viewers, limiting its utility for diagnostic imaging workflows.
- **Local Deployment Dependency:** The current architecture relies on local server deployment (XAMPP). While secure, this limits remote access for physicians unless additional network configuration (e.g., cloud hosting, port forwarding) is implemented.
- **Monolithic Architecture:** As a monolithic application, scaling the system to handle thousands of concurrent users would require significant refactoring compared to microservices architectures.
- **Absence of Advanced Analytics:** While the system provides basic statistics, it lacks machine learning capabilities for predictive diagnostics or automated appointment optimization.

9.5 Key Performance Indicators (KPIs) Measurement:

The project performance was quantitatively evaluated using Key Performance Indicators aligned with ISO/IEC 25010 software quality standards.

Table (9.2) KPI Measurement Results

KPI Category	Metric	Target Threshold	Achieved Result	Status
Functional Completeness	Requirements Coverage	100%	100% (55/55)	Pass
Reliability	Test Pass Rate	$\geq 98\%$	100%	Pass
Efficiency	Average Page Load Time	≤ 2.0 seconds	1.3 seconds	Pass
Efficiency	Database Query Latency	≤ 500 ms	320 ms	Pass
Usability	Task Completion Rate	$\geq 90\%$	96%	Pass
Security	Critical Vulnerabilities	0	0	Pass
Maintainability	Code Modularity	High (Strict Separation)	Strict Separation	Pass

Analysis:

The results indicate that the RLMS exceeds or meets all targeted performance benchmarks. The 100% functional coverage and zero critical vulnerabilities validate the effectiveness of the development and testing methodologies. The usability score of 96% confirms that the Arabic RTL interface successfully enhances user interaction. While efficiency metrics show healthy performance, they are bounded by the local deployment environment; however, they remain well within acceptable limits for medium-scale clinical operations.

These KPIs collectively affirm that the Radiology Lab Management System is a robust, secure, and high-quality software solution suitable for its intended operational context.

Chapter 10

Conclusion and Recommendations

10.1 Project Summary and Main Achievements:

The Radiology Lab Management System (RLMS) represents a comprehensive academic and practical endeavor aimed at modernizing the operational workflows of medium-scale radiology facilities. Initiated to address systemic inefficiencies inherent in manual and semi-automated clinical processes—including data redundancy, appointment scheduling conflicts, delayed report dissemination, and fragmented patient communication—the project successfully delivered a secure, role-segregated web application built on a PHP 8.x, MySQL 8.0, and Bootstrap 5 technology stack.

Throughout the incremental-iterative development lifecycle, the project achieved the following key milestones:

1. **Complete Requirement Fulfillment:** All 55 documented functional requirements were systematically implemented, validated, and deployed across four distinct stakeholder interfaces (Administrator, Doctor, Staff, Patient), ensuring end-to-end coverage of radiology lab operations.
2. **Localized Interface Engineering:** The system features a fully Arabic-native Right-to-Left (RTL) user interface, deliberately excluding English support to maintain linguistic consistency, reduce cognitive load for regional clinical staff, and align with cultural usability standards.
3. **Architectural Robustness & Security:** A modular, directory-based architecture (admin/, doctor/, staff/, patient/) was implemented alongside strict Role-Based Access Control (RBAC). Security was reinforced through PDO prepared statements, bcrypt password hashing, session regeneration, and XSS/CSRF mitigation protocols, ensuring compliance with OWASP Top 10 (2023) guidelines.
4. **Data Integrity & Performance:** The MySQL relational schema was normalized to the Third Normal Form (3NF), enforcing ACID-compliant transactions, foreign key constraints, and optimized query execution.

System testing confirmed a 100% test case pass rate, with average page load times under 1.3 seconds and database query latency below 320 milliseconds.

5. **Academic & Methodological Rigor:** The project adhered to standardized software engineering practices, including UML modeling, Requirements Traceability Matrix (RTM) validation, peer code reviews, and structured usability evaluation. The final deliverable serves as a reproducible reference model for lightweight, culturally adaptive healthcare information systems.

Collectively, these achievements demonstrate the successful translation of theoretical software architecture principles into a functional, domain-specific clinical management tool, fulfilling the primary objectives defined at the project's inception.

10.2 Difficulties and Challenges Faced by the Project:

The development of the RLMS encountered several technical, methodological, and domain-specific challenges, each requiring systematic analysis and iterative resolution:

1. **Arabic RTL Layout Consistency:** Ensuring pixel-perfect rendering of the Right-to-Left interface across diverse browsers and screen resolutions presented significant frontend challenges. Bootstrap RTL extensions required extensive custom CSS overrides to correct navbar mirroring, form alignment, and tooltip positioning. *Resolution:* Implemented CSS logical properties (margin-inline-start, float-inline-start), conducted cross-browser compatibility testing, and established a standardized RTL stylesheet baseline.
2. **Session & State Management Complexity:** Managing concurrent user sessions across four distinct roles without privilege leakage required careful middleware design. Race conditions occasionally emerged

during rapid role-switching or simultaneous invoice generation. *Resolution:* Enforced strict \$_SESSION validation at the entry point of each module, implemented session regeneration upon login, and applied database-level row locking for critical transactional operations.

3. **PDF Generation with Arabic Typography:** Early attempts to export medical records and invoices resulted in garbled Arabic characters due to default Latin font mappings in PHP PDF libraries. *Resolution:* Integrated TCPDF/FPDF with embedded Arabic font families (Cairo, Tajawal), configured UTF-8 character encoding, and validated glyph rendering across Windows and macOS environments.
4. **Scope Management & Iterative Coordination:** Balancing academic timeline constraints with the desire to implement advanced clinical features risked scope creep. Parallel development by three team members occasionally led to merge conflicts and inconsistent API contract definitions. *Resolution:* Instituted a formal change control protocol, utilized Git feature-branch workflows with mandatory peer review, and maintained a dynamic RTM to prioritize Must-Have functionalities over deferred enhancements.
5. **Clinical Workflow Abstraction:** Translating real-world radiology procedures into simplified digital processes without oversimplifying critical diagnostic documentation required extensive stakeholder simulation. *Resolution:* Conducted iterative workflow mapping, validated use case specifications against standard radiology department protocols, and incorporated supervisor feedback loops to ensure clinical accuracy while maintaining system usability.

These challenges were systematically documented, mitigated, and integrated into the project's risk management framework, ultimately strengthening the system's resilience and academic validity.

10.3 Suggestions for Future Development:

To enhance the system's operational viability, scalability, and clinical relevance, the following technical and architectural advancements are recommended for subsequent development phases:

1. **DICOM/PACS Integration:** Implement native DICOM standard support to enable direct ingestion, storage, and web-based rendering of medical imaging files (CT, MRI, X-Ray). Integration with lightweight PACS servers or HTML5 DICOM viewers would transform the RLMS from a document-centric tool to a comprehensive clinical imaging platform.
2. **Interoperability & API Expansion:** Develop RESTful/HL7 FHIR-compliant endpoints to facilitate seamless data exchange with national Electronic Health Record (EHR) systems, laboratory information systems (LIS), and telemedicine platforms. This would enable real-time patient history synchronization and cross-institutional referral management.
3. **Predictive Analytics & AI Integration:** Incorporate machine learning models (e.g., linear regression, random forest) to forecast patient volume trends, optimize appointment scheduling, and flag high-priority diagnostic reports. AI-driven triage could significantly reduce radiologist workload and improve diagnostic turnaround times.
4. **Cloud-Native & Containerized Deployment:** Migrate the monolithic architecture to a containerized environment (Docker/Kubernetes) with auto-scaling capabilities, load balancing, and encrypted cloud storage. This would enhance system availability, simplify maintenance, and support enterprise-grade deployment for hospital networks.
5. **Advanced Security & Compliance:** Implement Two-Factor Authentication (2FA), AES-256 encryption for stored medical files, and

granular audit logging to meet stringent healthcare data protection regulations (e.g., HIPAA/GDPR equivalents). Role-based data anonymization could also be introduced for research and statistical reporting purposes.

6. **Cross-Platform Mobile Application:** Develop native or hybrid mobile applications (Flutter/React Native) for field staff and patients, enabling real-time appointment notifications, mobile survey responses, and remote report access while maintaining RTL localization standards.

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